Time series analysis

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Chapter 1

Stationary time series

1.1 Data and model

The term *time series* is used to indicate two objects:

1. A collection of data (observations) all collected at particular time point t: $\{x_t | t \in T\}$, where T is an *index set*. Typically $T = [a, b]$ with continuous time observations and $T = \{t_1, t_2, \ldots\}$ in discrete time observations. In this lecture we will consider $T \subseteq \mathbb{Z}$. Some examples are shown in Figure 1.1.

2. The model describing data. For this purpose we will use a stochastic process (see definition below) $\{X_t | t \in T\}$. General idea: each x_t is the realization of a random variable X_t.

Note that for the data we usually have $\{x_1, \ldots, x_n\}$ (we start observing at time 1 and stop at time n), while we use as a model $\{X_t, t \in \mathbb{Z}\}$, i.e. process runs from infinite past to infinite future. *The interpretation is*: the process will usually already run before we start observing the output (before time 1) and it will continue after we stop observing at time n. *Technical reason:* mathematical convenience.

What makes time series analysis different from classical statistics? Data are dependent and correlated, whereas in statistics usually independence between the observations is assumed. Roughly speaking, in statistics the experimenter can repeat the experiment under the same conditions and *independently*. In time series we cannot redo the experiment, we only observe from a running process. On the one hand that requires more complex models but on the other hand, dependence is the crucial ingredient for *forecasting*, one of the central topics in time series analysis.

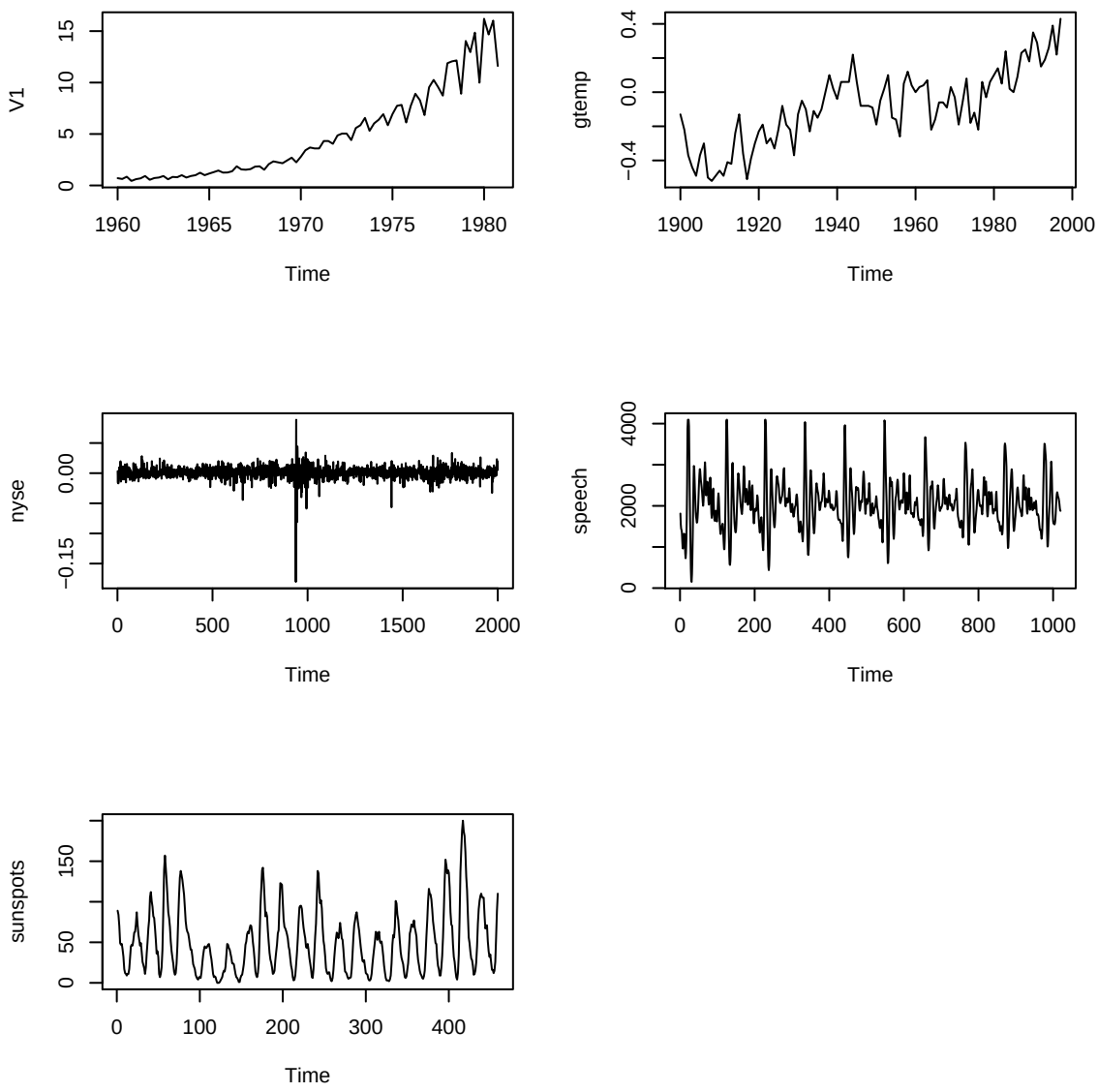


Figure 1.1: Examples of time series data: Johnson & Johnson quarterly earning per share (upper left), Yearly average global temperature deviations (upper right), NYSE stock return data from 1984–1991 (middle left), speech recording of the syllable *aaa...hhh* (middle right) and sunspot numbers measured biannually form 1749–1978 (lower left). [R-code]

Definition 1.1.1 (Stochastic Process)**.** A stochastic process is a collection of random variables $\{X_t|t \in T\}$ defined on a common probability space $(\Omega, \mathscr{A}, P)$. The functions $\{X_t(\omega)|t \in T\}$, with $\omega \in \Omega$ are called *realizations* or *trajectories* or also *sample paths*. We use $\{x_t|t \in T\}$ to indicate the values observed.

Recall: A random variable X is a measurable mapping defined on a probability space $(\Omega, \mathscr{A}, P)$:

$$X : (\Omega, \mathscr{A}) \to (\mathbb{R}, \mathcal{B}(\mathbb{R})).$$

Here $\mathcal{B}(\mathbb{R})$ is the Borel σ-algebra. In other words

$$\omega \mapsto X(\omega) \in \mathbb{R}$$

and

$$X^{-1}(B) \in \mathscr{A}, \forall B \in \mathcal{B}(\mathbb{R}).$$

The distribution function of X is given by

$$F_X(x) = P(X \leq x) = P(\{\omega|X(\omega) \leq x\}) = P(X^{-1}(-\infty, x]).$$

Simple examples:

1. **IID sequences**. $\{X_t, t = 1, 2, \ldots\}$ is a sequence of independent and identically distributed (iid) random variables. We will write $\{X_t\} \sim$ IID or $\{X_t\} \sim \text{IID}(\mu, \sigma^2)$ to indicate $EX_t = \mu$ and $\text{Var}(X_t) = \sigma^2$.

2. **Random walk**. $\{Z_t, t = 1, 2, \ldots\}$ is a sequence of iid random variables and $\{X_t, t = 1, 2, \ldots\}$ is defined as

$$\begin{cases} X_t = 0 & \text{if } t = 0, \\ X_t = \sum_{k=1}^{t} Z_k & \text{if } t \geq 1. \end{cases}$$

3. **Branching process**. This stochastic model is used to describe the evolution of a population size. Let's call X_t the size of the population at time t described by:

$$\begin{cases} X_t = x & \text{if } t = 0, \\ X_{t+1} = \sum_{k=1}^{X_t} Z_{k,t} & \text{if } t \geq 1, \end{cases}$$

   where $Z_{j,t}$ is a random variable indicating the number of offsprings generated by individual j for $j = 1, 2, \ldots$ at time t for $t = 0, 1, \ldots$.

4. **Moving average**. Assume $\{Z_t\} \sim \text{IID}(\mu, \sigma^2)$. A process of the form $X_t = a_1 Z_t + a_2 Z_{t-1} + \cdots + a_q Z_{t-q+1}$ is called a *moving average*.

5. **AR(1)**. Assume $\{Z_t\} \sim \text{IID}(0, \sigma^2)$. Then we call a process of the form $X_t = \rho X_{t-1} + Z_t$ an *auto-regressive process of order 1 (AR(1))*.

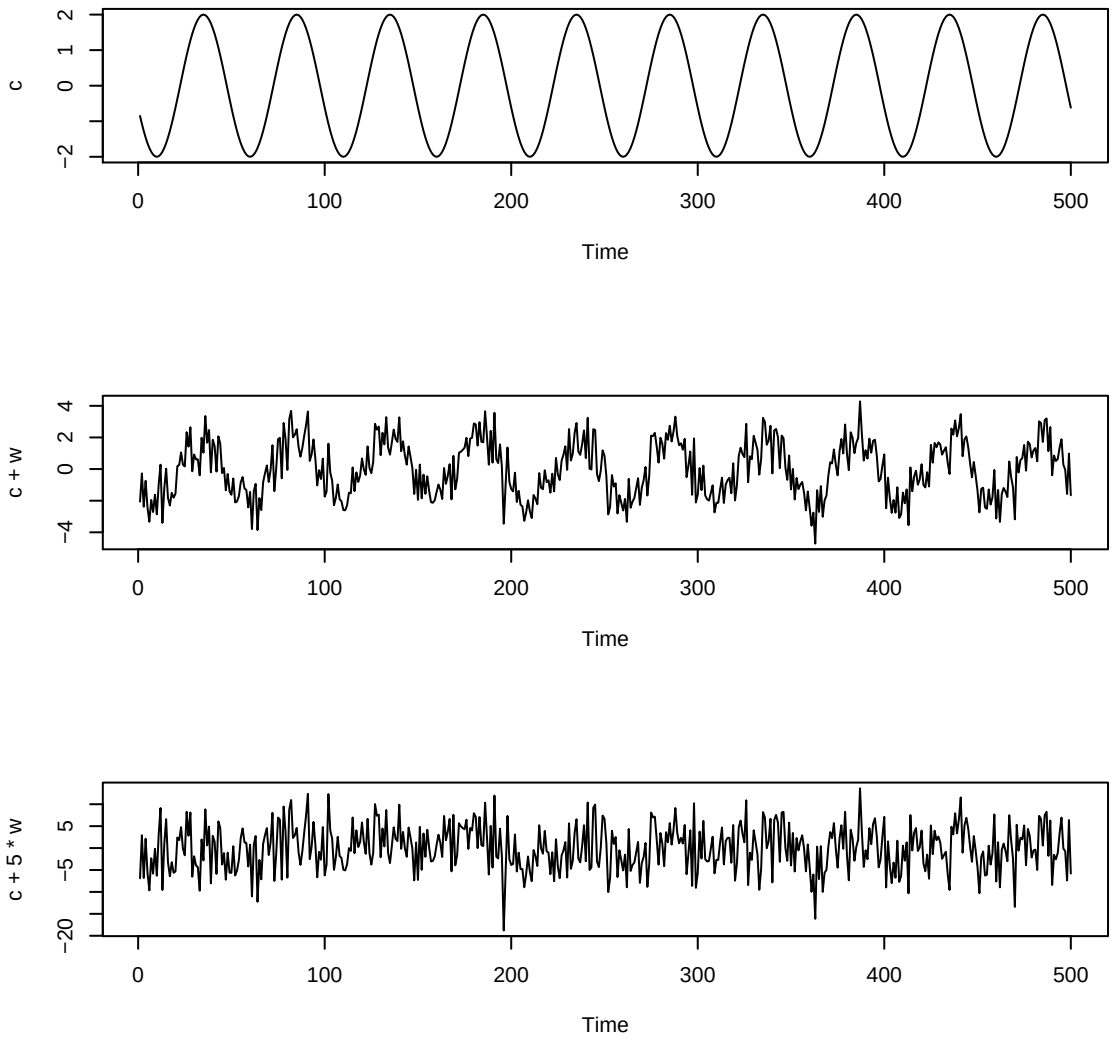


Figure 1.2: Cosine wave with period 50 (top) compared with cosine wave contaminated with additive Gaussian noise ($\sigma = 1$, middle) and ($\sigma = 5$, bottom). [R-code]

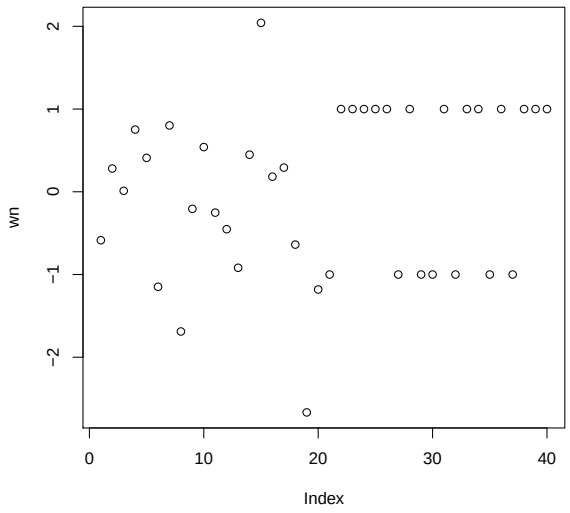


Figure 1.3: A white noise sequence. Data are independent and $N(0,1)$ for $t \in \{1,\ldots,20\}$ and Rademacher (± 1 with probability $1/2$) for $t \geq 21$.

6. **Signal plus noise**. Assume that $\{Z_t\} \sim \text{IID}(0,\sigma^2)$ and $X_t = \mu(t) + Z_t$, where $\mu(t)$ is a deterministic process. (See Figure 1.2).

More general than i.i.d. variables is *white noise*.

Definition 1.1.2 (White noise)**.** A process $\{Z_t, t \in \mathbb{Z}\}$ is called white noise if the following holds:

- $\text{Var}(Z_t) = \sigma_Z^2 < \infty, \ \forall t,$
- $EZ_t = \mu, \ \forall t,$
- $\text{Cov}\,(Z_t, Z_{t+h}) = 0, \ \forall t, \ |h| \geq 1.$

Short: $Z_t \sim \text{WN}(\mu, \sigma^2)$.

Note that white noise can be (i) *dependent*! E.g. the famous *GARCH* model used in econometric applications is only white noise. (ii) The distribution of the marginals can be entirely different for different time points. (See Figure 1.3)

It is also important to note that for a general stochastic process the marginal distributions

$$F_{X_t}(\cdot) = P(X_t \leq \cdot)$$

don't say anything about the dynamics of the process. For example, consider the processes

$$X_t = X, \quad \forall t \geq 1,$$

and

$$X_t \sim_{iid} X, \quad \forall t \geq 1.$$

They have the same marginal distribution but they behave completely different as a process. For the first, knowledge of X_1 provides perfect knowledge of X_t for any $t \geq 1$. For the second example, knowledge of X_1 doesn't provide information of X_t for $t > 1$.

Another example is shown in Figure 1.4. The four processes shown their all came from a process where the marginal distributions are the same.

Hence, we need more information on joint distributions.

Definition 1.1.3 (Finite dimensional distributions, f.d.d.'s)**.** Let $\{X_t, t \in T\}$ be a stochastic process and let $\mathcal{T} = \{\mathbf{t} = (t_1, \ldots, t_n), t_1 < t_2 < \ldots < t_n, n = 1, 2, \ldots\}$. Then the finite-dimensional distribution functions of the stochastic process are the functions $\{F_{\mathbf{t}}(\cdot),\ \mathbf{t} = (t_1, \ldots, t_n) \in \mathcal{T}\}$ with

$$F_{\mathbf{t}}(\mathbf{x}) = F_{\mathbf{t}}(x_1, \ldots, x_n) = P\left(X_{t_1} \leq x_1, \ldots, X_{t_n} \leq x_n\right).$$

Example 1.1.1. If the random variables X_t are independent, then the finite-dimensional distribution functions are

$$F_{\mathbf{t}}(\mathbf{x}) = F_{\mathbf{t}}(x_1, \ldots, x_n) = \prod_{i=1}^{n} F_{X_{t_i}}(x_i) = P\left(X_{t_1} \leq x_1\right) \cdot \ldots \cdot P\left(X_{t_n} \leq x_n\right).$$

It seems that one possible way to define a stochastic process (i.e. define a time series model) is to make assumptions on the finite dimensional distributions.

There is (at least) one problem: how do we know that to a given family of f.d.d.'s a stochastic process will exist? NOTE: it is actually not even trivial to claim the existence of an iid sequence (see probability theory)! This theorem helps:

Theorem 1.1.1 (Existence theorem of Kolmogorov)**.** *Denote* $\mathbf{x} = (x_1, \ldots, x_n)$ *and* $\mathbf{t} = (t_1, \ldots, t_n)$ *and* $\mathbf{x}^{(i)}$ *and* $\mathbf{t}^{(i)}$ *the same vectors leaving out the i-th coordinate. A family of finite-dimensional distribution functions* $\{F_{\mathbf{t}}(\cdot), \mathbf{t} \in \mathcal{T}\}$ *defines a stochastic process if and only if*

$$\lim_{x_i \to \infty} F_{\mathbf{t}}(\mathbf{x}) = F_{\mathbf{t}^{(i)}}\left(\mathbf{x}^{(i)}\right), \tag{1.1}$$

$\forall n \geq 1, \forall \mathbf{t} \in \mathcal{T}, \forall i \in \{1, \ldots, n\}$.
Fourier version: Let $\varphi_{\mathbf{t}}(\mathbf{u}) = \int_{\mathbb{R}^n} \exp(i\mathbf{u}^T\mathbf{x}) dF_{\mathbf{t}}(\mathbf{x})$ *be the characteristic function of* $F_{\mathbf{t}}(\mathbf{x})$*. Then the statement of the theorem holds true, if* $lim_{u_i \to 0} \varphi_{\mathbf{t}}(u_1, \ldots, u_n) = \varphi_{\mathbf{t}^{(i)}}(\mathbf{u}^{(i)})$.

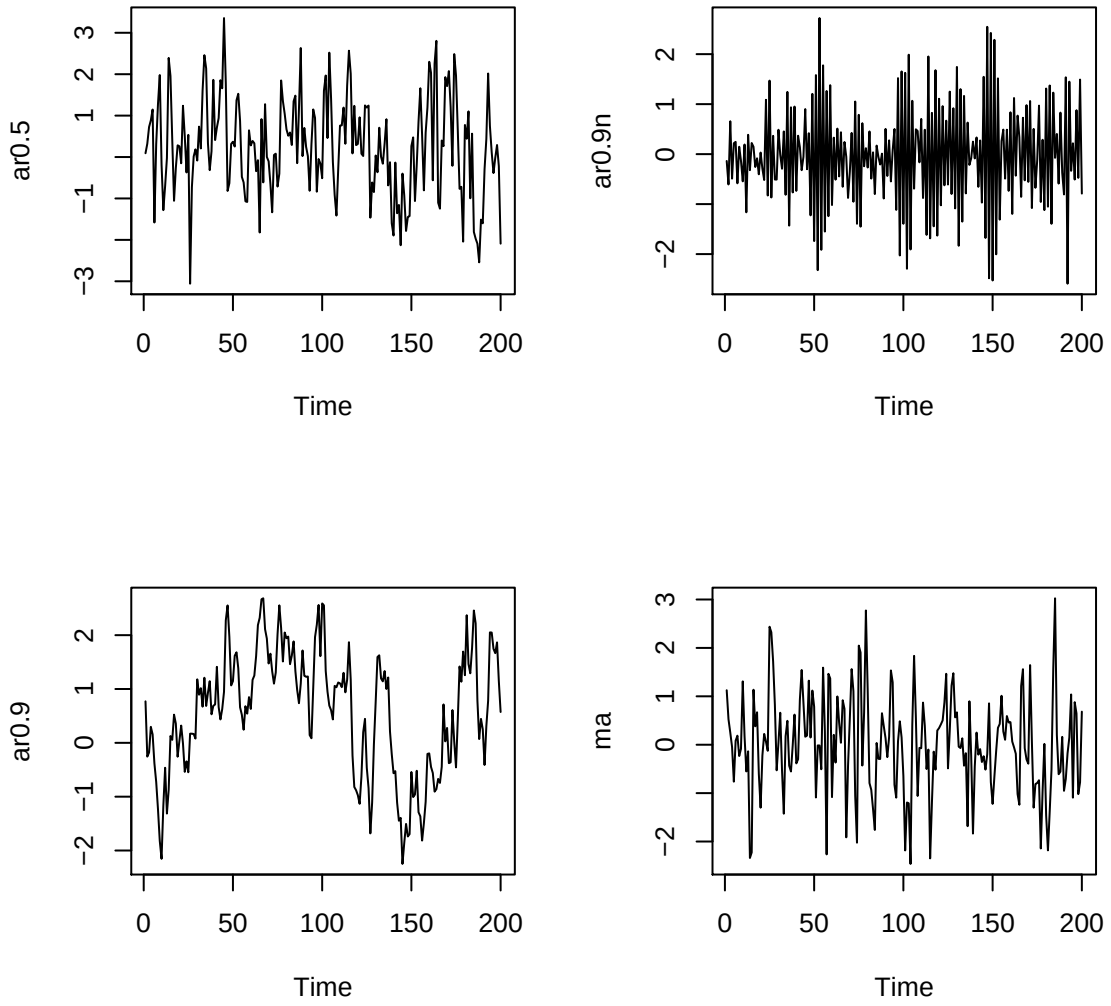


Figure 1.4: Four different time series with i.i.d. Gaussian innovations: AR(1) with $\rho = 0.5$ and $\sigma^2 = 1$ (upper left), $\rho = -0.9$ and $\sigma^2 = 0.253$ (upper right), $\rho = 0.9$ and $\sigma^2 = 0.253$ (lower left) and an MA(1) process of the form $X_t = Z_t + bZ_{t-1}$ where $b = 0.58$ and $\sigma^2 = 1$ (lower right). [R-code]

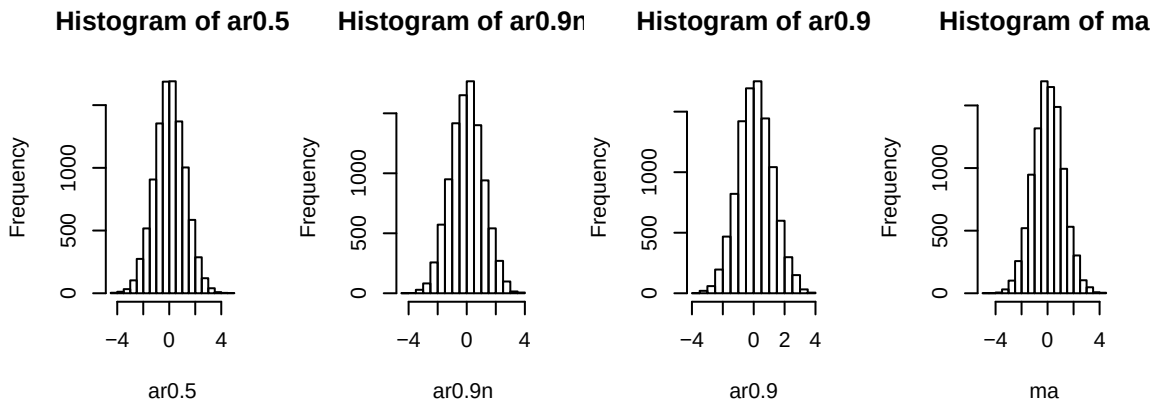


Figure 1.5: Histograms from 10000 observations of respective processes shown in Figure 1.4. [R-code]

Proof. Not here. □

What does it say? If we let x_i in (1.1) go to ∞, then this means that we allow ultimately arbitrary values for X_i. The resulting limit should be the same as if we kick out the i-th variable a priory. So this is a very plausible consistency requirement.

Exercise: Use Theorem 1.1.1 to prove the existence of an iid sequence. (Exercise 1.1.)

We will later use Kolmogorov's extension theorem to prove the existence of processes, belonging to a very important family (see proof of Theorem 1.3.1):

Example 1.1.2 (Gaussian processes)**.** A process $\{X_t\}$ is called Gaussian, if all its finite dimensional distributions are multivariate normal.

Recall: A vector $\mathbf{X} = (X_1, \ldots, X_n)^T$ is multivariate normally distributed if we can write $\mathbf{X}$ in the following way: $\mathbf{X} = \boldsymbol{\mu} + B\mathbf{N}$ where $\boldsymbol{\mu} \in \mathbb{R}^n$, $B \in \mathbb{R}^{n\times m}$, $\mathbf{N} = (N_1, \ldots, N_m)^T$ and $N_i \sim_{iid} N(0,1)$. Then

- $E[\mathbf{X}] = \boldsymbol{\mu}$, $\mathrm{Cov}(\mathbf{X}) = BB^T \in \mathbb{R}^{n\times n}$.
- If $\mathbf{X} = \boldsymbol{\mu} + B\mathbf{N}$ and $\mathbf{Y} = \boldsymbol{\mu} + A\tilde{\mathbf{N}}$ with $BB^T = AA^T$ and thus have the same mean and same covariance, then $\mathbf{X} \stackrel{d}{=} \mathbf{Y}$ (i.e. $\mathbf{X}$ has the same distribution as $\mathbf{Y}$).

Proof. Look at the characteristic function and assume without loss of generality that

$\boldsymbol{\mu} = \mathbf{0}$. Then

$$\begin{aligned}\varphi_{\mathbf{X}}(\mathbf{u}) &= E[\exp(i\mathbf{u}^T\mathbf{X})] \\ &= E[\exp(i(B^T\mathbf{u})^T\mathbf{N})] \\ &= \varphi_{\mathbf{N}}(B^T\mathbf{u}) = \exp(-\frac{1}{2}\mathbf{u}^T BB^T\mathbf{u}) \\ &= \exp(-\frac{1}{2}\mathbf{u}^T AA^T\mathbf{u}) \\ &= \varphi_{\mathbf{Y}}(\mathbf{u}).\end{aligned}$$

□

- $\Sigma := BB^T$ is non-negative definite:
  $\mathbf{a}^T BB^T\mathbf{a} \geq 0,\ \forall \mathbf{a} \in \mathbb{R}^n$ since $\mathbf{a}^T BB^T\mathbf{a} = (B^T\mathbf{a})^T(B^T\mathbf{a}) = \| B^T\mathbf{a} \|^2$.
- BB^T is symmetric: $(BB^T)^T = BB^T$.
- every non negative definite matrix Σ can be written in the form $\Sigma = BB^T$ (by the spectral theorem).

Consequences.

- The multivariate normal distribution depends only on $\boldsymbol{\mu}$ and $\Sigma = BB^T$.
- For all Σ symmetric and non-negative a multivariate normal vector exists, whose covariance matrix is Σ.
- Notation: $\mathbf{X} \sim \mathbf{N}(\boldsymbol{\mu}, \Sigma)$.

Theorem 1.1.1 is very important, but it has still one drawback. How would we in general come up with a reasonable family of finite dimensional distributions?

Another very important method to construct a stochastic process is by constructing them from independent variables or some other existing process:

Example 1.1.3 (Sinusoid wave with random phase and amplitude)**.**

$$X_t = A \cdot \cos(\rho t + B)$$

where A is a non-negative random variable independent of $B \sim U([0, 2\pi])$ (uniform distribution on $[0, 2\pi]$).

Example 1.1.4 (Moving average process of order q (MA(q))**.** $X_t = Z_t + \theta_1 Z_{t-1} + \cdots + \theta_q Z_{t-q}$, with $\{Z_t\} \sim \mathrm{WN}(0, \sigma^2)$.

Example 1.1.5 (Linear process)**.**

$$X_t = \sum_{k \geq 0} a_k \varepsilon_{t-k}, \qquad \{\varepsilon_t\} \sim \mathrm{WN}(0, \sigma^2), \qquad a_0, a_1, \ldots \in \mathbb{R}.$$

For these examples it might be difficult to calculate the f.d.d.'s, but existence is no problem!

1.2 Stationarity

Within the context of a finite number random variables we often use the covariance function in order to understand the dependence between the variables. Now we need a device that can extend the concept of covariance matrix to deal with infinite number of random variables. This is the *autocovariance function.*

Definition 1.2.1 (Autocovariance function, ACF)**.** For a process $\{X_t, t \in T\}$ for which $\text{Var}\,(X_t) < \infty$ for each $t \in T$, the autocovariance function $\gamma_X(\cdot,\cdot)$ of $\{X_t\}$ is given by:

$$\gamma_X(r,s) = \text{Cov}(X_r, X_s) = E[(X_r - E[X_r])(X_s - E[X_s])],\ r,s \in T.$$

Remark: Note that $\text{Var}(X) < \infty \Leftrightarrow EX^2 < \infty$. The latter implies (by Cauchy-Schwarz inequality, Proposition 2.1.1) that $E|X| < \infty$.

An appealing aspect of iid sequences is that there stochastic behavior remains stable. We introduce now a framework which is very often used to describe a certain structural stability in terms of the first two moments of the process and of its dependence.

Definition 1.2.2 (Weak Stationarity)**.** The time series $\{X_t, t \in \mathbb{Z}\}$ is said to be stationary if:

(i) $EX_t^2 < \infty, \quad \forall t \in \mathbb{Z}$

(ii) $EX_t = m, \quad \forall t \in \mathbb{Z}$ (i.e. the expected value is constant)

(iii)
$$\gamma_X(r,s) = \text{Cov}(X_r, X_s) = \gamma_X(r+t, s+t),\ \forall t \in \mathbb{Z}, \forall r,s \in \mathbb{Z}.$$

Example 1.2.1. An iid sequence is weakly stationary (provided it has second moments), whereas the random walk model is not weakly stationary.

Remark. Apparently $\gamma_X(r,s) = \gamma_X(s,r)$. Hence by (iii)

$$\gamma_X(0, s-r) = \gamma_X(r,s) = \gamma_X(r-s, 0) = \gamma_X(0, r-s)$$

This means that the ACF of a stationary time series can be expressed as a one parameter functions $\gamma_X(h)$, with $h = |r-s|$. In this case one often uses the *autocorrelation function* (acf) to asses dependence:

$$\rho_X(h) = \frac{\gamma_X(h)}{\gamma_X(0)} = \frac{\text{Cov}(X_{t+h}, X_t)}{\text{Var}(X)} = \text{Corr}(X_{t+h}, X_t).$$

Remark. $\rho_X(h)$ is a good and simple measure for dependence. But $\rho_X(h) = 0$ does generally not imply that X_t independent of X_{t+h}. For Gaussian processes, uncorrelated implies independent.

Here is another, more stringent notion of stationarity.

Definition 1.2.3 (Strict Stationarity)**.** A process $\{X_t | t \in \mathbb{Z}\}$ is *strictly stationary* if

$$F_{\mathbf{t}}(\cdot) = F_{\mathbf{t}+h}(\cdot), \quad \forall h \in \mathbb{Z}, \mathbf{t} \in \mathcal{T}$$

or equivalently

$$P(X_{t_1} \leq x_1, \ldots, X_{t_n} \leq x_n) = P(X_{t_1+h} \leq x_1, \ldots, X_{t_n+h} \leq x_n).$$

Example 1.2.2. An iid sequence is strictly stationary.

Example 1.2.3. $X_t = X \ \forall t$ is strictly stationary.

The following proposition is quite useful:

Proposition 1.2.1. *Assume $\{X_t | t \in \mathbb{Z}\}$ is strictly stationary and $f : \mathbb{R}^\infty \to \mathbb{R}$ is $\mathcal{B}(\mathbb{R}^\infty)$-$\mathcal{B}(\mathbb{R})$-measurable. Then $Y_t = f(X_t, X_{t-1}, \ldots)$ is again strictly stationary.*

Proof. Not here. □

Example 1.2.4. Assume $\{\varepsilon_t\} \sim$ IID. Then it is a strictly stationary process. Define the linear process

$$X_t = \sum_{k \geq 0} a_k \varepsilon_{t-k} = a_0 \varepsilon_t + a_1 \varepsilon_{t-1} + a_2 \varepsilon_{t-1} + \ldots = f(\varepsilon_t, \varepsilon_{t-1}, \ldots)$$

If the series converges (in some sense to be specified), $\{X_t, t \in \mathbb{Z}\}$ is strictly stationary.

Relation between weak and strict stationarity. If $\{X_t\}$ is strictly stationarity, the distribution F_t of X_t is the same for each $t \in \mathbb{Z}$. Moreover, any pair (X_t, X_{t+h}) has a joint distribution $F_{t,t+h}$ that doesn't depend on t. Hence

$$EX_t = \int_{\mathbb{R}} x dF_t(x) = m$$

is independent of t and

$$\text{Cov}(X_t, X_{t+h}) = \int_{\mathbb{R}} \int_{\mathbb{R}} (x - m)(y - m) dF_{t,t+h}(x, y)$$

is independent of t. Hence:

Strict stationarity $\Longrightarrow$ weak stationarity *provided the sequence has finite second moments.*

The reverse is in general not true. (Find a simple counter example.) There is an important case where weak stationarity implies strict stationarity. It is the case of the *Gaussian process.* In this case weak stationarity of $\{X_t\}$ implies strict stationarity since $(X_{t_1}, \ldots, X_{t_n})'$ and $(X_{t_1+h}, \ldots, X_{t_n+h})'$ have the same mean and covariance matrix and hence the same distribution for all $n = \{1, 2, \ldots\}$ and for all $h, t_1, \ldots, t_n \in \mathbb{Z}$.

NOTE: We will henceforth call a time series stationary, if it is weakly stationary.

1.3 Properties of the ACF

Proposition 1.3.1 (Elementary properties of the ACF)**.** *If $\gamma_X(\cdot)$ is the ACF of a stationary process $\{X_t, t \in \mathbb{Z}\}$, then*

(i) $\gamma_X(0) \geq 0$,

(ii) $|\gamma_X(h)| \leq \gamma(0), \ \forall h \in \mathbb{Z}$,

(iii) $\gamma_X(h) = \gamma_X(-h), \ \forall h \in \mathbb{Z}$.

Proof. (i) follows from $\gamma(0) = \mathrm{Var}(X_t) \geq 0$, (ii) follows from

$$|\gamma_X(h)| = |\mathrm{Cov}(X_{t+h}, X_t)| \leq (\mathrm{Var}(X_{t+h}))^{\frac{1}{2}}(\mathrm{Var}(X_t))^{\frac{1}{2}} = \gamma_X(0)$$

(Cauchy-Schwarz inequality, Proposition 2.1.1) and (iii) is trivial. □

Definition 1.3.1 (Non-negative definiteness, n.n.d.)**.** A real-valued function on the integers $\kappa : \mathbb{Z} \to \mathbb{R}$ is called non negative definite (n.n.d.) if

$$\sum_{i=1}^{n}\sum_{j=1}^{n} a_i \kappa(t_i - t_j) a_j \geq 0, \qquad a_i \in \mathbb{R}, \quad t_i \in \mathbb{Z}, \quad n \geq 1.$$

In other words: the matrices $K = ((\kappa(t_i - t_j)))_{i,j=1}^n$ are n.n.d.

Theorem 1.3.1 (Characterization of ACFs)**.** *A function $\gamma : \mathbb{Z} \to \mathbb{R}$ is the autocovariance function of a stationary process $\{X_t, t \in \mathbb{Z}\}$, iff γ is even and n.n.d.*

Proof. Assume γ is an ACF of a stationary process. Then γ is even (Proposition 1.3.1, (iii)). Further

$$\begin{aligned} 0 &\leq \mathrm{Var}\left(\sum_{i=1}^{n} a_i X_{t_i}\right) \\ &= \sum_{i=1}^{n}\sum_{j=1}^{n} a_i a_j \mathrm{Cov}(X_{t_i}, X_{t_j}) \\ &= \sum_{i=1}^{n}\sum_{j=1}^{n} a_i a_j \gamma(t_i - t_j). \end{aligned}$$

Assume γ is even and n.n.d. We will show that a Gaussian process with ACF γ exists. We define $\Gamma_{\mathbf{t}} = (\gamma(t_i - t_j))_{i,j=1}^n$. Note that since $\Gamma_{\mathbf{t}}$ is nnd, we can find a multivariate normal vector $\mathbf{X}$ with $\Gamma_{\mathbf{t}} = \mathrm{Cov}(\mathbf{X})$. We will use now the "Fourier version" of Kolmogorov's extension theorem: Let $\varphi_{\mathbf{t}}(\mathbf{u})$ be the characteristic function of a $\mathbf{N}(\mathbf{0}, \Gamma_{\mathbf{t}})$ random vector.

$$\begin{aligned}
\lim_{u_k \to 0} \varphi_{\mathbf{t}}(\mathbf{u}) &= \lim_{u_k \to 0} \int_{\mathbb{R}^n} \exp(i\mathbf{u}^T\mathbf{x}) dF_{\mathbf{t}}(\mathbf{x}) \\
&= \lim_{u_k \to 0} \exp(-\frac{1}{2}\mathbf{u}^T\Gamma_{\mathbf{t}}\mathbf{u}) \\
&= \lim_{u_k \to 0} \exp(-\frac{1}{2}\sum_{i,j=1}^{n} u_i\gamma(t_i - t_j)u_j) \\
&= \exp(-\frac{1}{2}\sum_{\substack{i,j=1 \\ i,j\neq k}} u_i\gamma(t_i - t_j)u_j) \\
&= \varphi_{\mathbf{t}^{(k)}}(\mathbf{u}^{(k)}).
\end{aligned}$$

Hence: Kolmogorov's extention theorem applies. □

Example 1.3.1. Define:

$$\begin{aligned}
X_t &= Z_t + \theta Z_{t-1}, \text{ where } Z_t \sim \text{IID}(0, \sigma^2),\ \theta \in \mathbb{R} \text{ (MA(1) process)} \\
X_{t-1} &= Z_{t-1} + \theta Z_{t-2} \\
X_{t-2} &= Z_{t-2} + \theta Z_{t-3}, \text{ independent from the first equation.}
\end{aligned}$$

Hence:

$$\begin{aligned}
\gamma_X(0) &= \sigma^2(1 + \theta^2) \\
\gamma_X(1) &= \sigma^2\theta = \gamma_X(-1) \\
\gamma_X(k) &= 0,\ \forall |k| > 1
\end{aligned}$$

Assume now that:

$$\begin{aligned}
\kappa(0) &= a \\
\kappa(1) &= b = \kappa(-1) \\
\kappa(k) &= 0,\ \forall |k| > 1
\end{aligned}$$

What requirements on a and b do we need to prove the existence of a stationary process with ACF κ? See Exercise 1.8.

1.4 Sample mean and sample ACF

Assume we have a stationary process $\{X_t\}$ with $E[X_t] = m$ and ACF $\gamma_X(h)$. In practice we don't know m and γ_X, hence they must be estimated from the data.

Natural estimators are the *sample mean*

$$\hat{m} = \bar{X}_n = \frac{1}{n}(X_1 + \ldots + X_n),$$

and the *sample ACF*

$$\hat{\gamma}_X(h) = \frac{1}{n}\sum_{k=1}^{n-|h|}(X_{k+|h|} - \bar{X}_n)(X_k - \bar{X}_n)$$

Figure 1.6 shows the sample auto-correlation function of 2000 observations from the processes in Figure 1.4.

One can show under quite general conditions that these estimators are *consistent* (and *asymptotically normal*, not now.)

(a) $P(|\bar{X}_n - m| > \varepsilon) \overset{n\to\infty}{\longrightarrow} 0$, $\forall \varepsilon > 0$ (weak consistency $\leftrightarrow$ convergence in probability)

(b) $P(\max_{0\le h\le k} |\hat{\gamma}_X(h) - \gamma_X(h)| > \varepsilon) \overset{n\to\infty}{\longrightarrow} 0$, $\forall \varepsilon > 0$ and $k \ge 0$ fixed (weak consistency).

For the moment we treat only (a).

Proposition 1.4.1. *Assume $\{X_t, t \in \mathbb{Z}\}$ is stationary and let $\gamma_X(\cdot)$ be its ACF. Then*

(i) $\mathrm{Var}(\bar{X}_n) = E[(\bar{X}_n - m)^2] \overset{n\to\infty}{\longrightarrow} 0$ *if* $\gamma_X(h) \overset{h\to\infty}{\longrightarrow} 0$;

(ii) $n\mathrm{Var}(\bar{X}_n) = nE[(\bar{X}_n - m)^2] \overset{n\to\infty}{\longrightarrow} \sum_{h\in\mathbb{Z}} \gamma_X(h)$ *if* $\sum_{h\in\mathbb{Z}} |\gamma_X(h)| < \infty$.

Remark. (i) implies consistency: by the Markov inequality

$$P(|\bar{X}_n - m| > \varepsilon) \le \frac{\mathrm{Var}(\bar{X}_n)}{\varepsilon^2} \overset{n\to\infty}{\longrightarrow} 0, \ \forall \varepsilon > 0.$$

Remark. When $\{X_t, t \in \mathbb{Z}\}$ are iid, $E[X_t] = m$ and $\mathrm{Var}(X_t) = \sigma^2 < \infty$, the central limit theorem tells us that

$$Z_n := \sqrt{n}\left(\frac{\bar{X}_n - m}{\sigma}\right) \overset{d}{\longrightarrow} N(0,1).$$

Here $E[Z_n] = 0$ and $\mathrm{Var}(Z_n) = E[Z_n^2] = 1$, while in the general stationary case we have $E[Z_n] = 0$, $E[Z_n^2] = \frac{n}{\sigma^2}E[(\bar{X} - m)^2] \to \frac{\sum_{h\in\mathbb{Z}}\gamma_X(h)}{\sigma^2}$ follows from (ii). Hence if the central limit theorem applies, we should scale with τ instead of σ where

$$\tau^2 = \sum_{h\in\mathbb{Z}} \gamma_X(h) = \sum_{h\in\mathbb{Z}} \mathrm{Cov}(X_0, X_h).$$

The parameter τ^2 is called *long-run variance*.

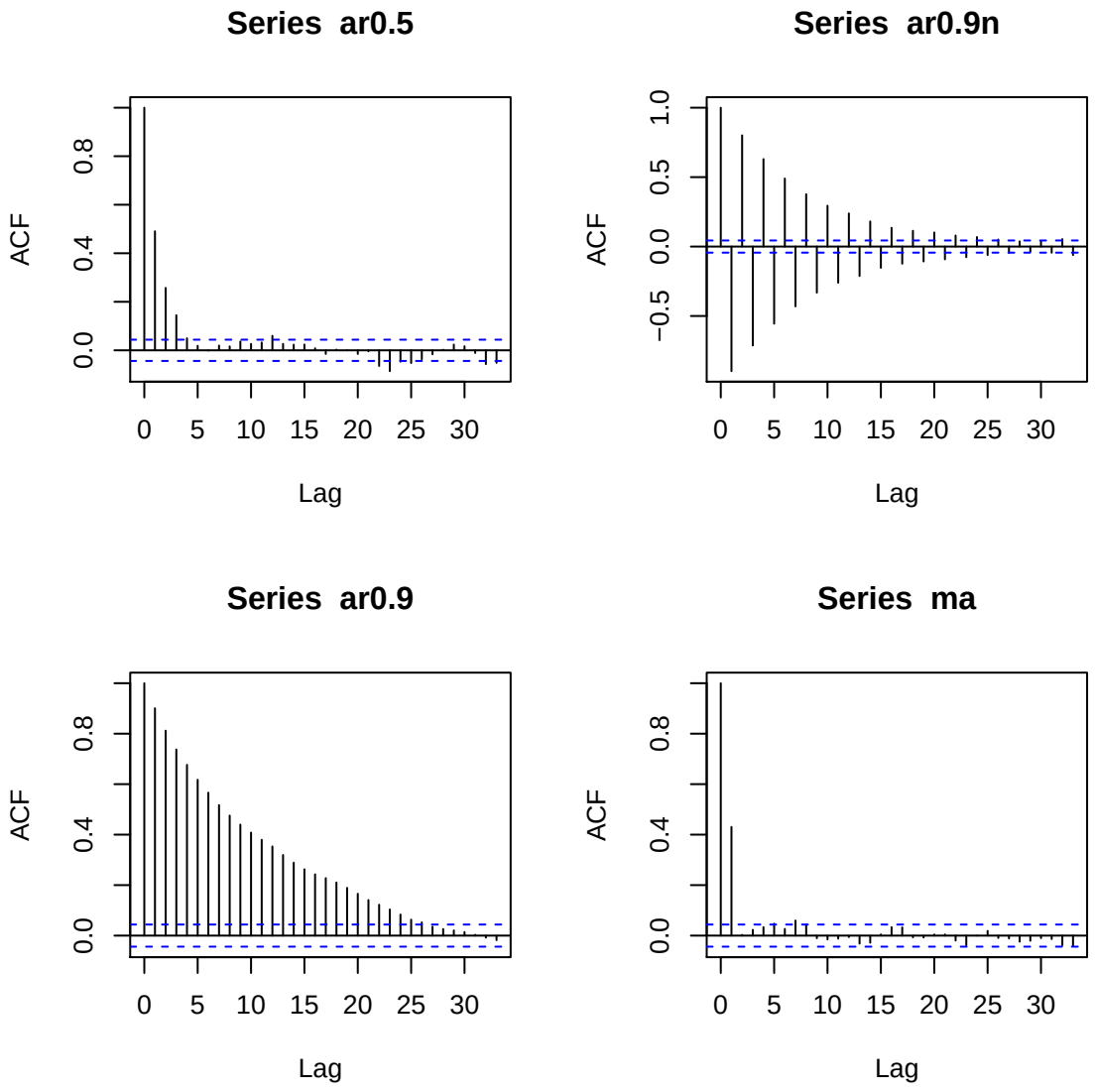


Figure 1.6: The sample auto-correlation function of 2000 observations from the processes in Figure 1.4. [R-code]

Proof of Proposition 1.4.1.

$$\begin{aligned} n\mathrm{Var}(\bar{X}_n) &= \frac{1}{n}\mathrm{Var}\left(\sum_{k=1}^{n}(X_k - m)\right) \\ &= \frac{1}{n}\sum_{k=1}^{n}\sum_{l=1}^{n}\mathrm{Cov}(X_k, X_l) \\ &= \frac{1}{n}\big(n\gamma_X(0) + [(n-1)\gamma_X(1) + (n-1)\gamma_X(-1)] \\ &\quad + [(n-2)\gamma_X(2) + (n-2)\gamma_X(-2)] + \ldots + [\gamma_X(n-1) + \gamma_X(-n+1)]\big) \\ &= \sum_{|h|<n}\left(1 - \frac{|h|}{n}\right)\gamma_X(h). \end{aligned}$$

Hence if $|\gamma_X(h)| \to 0$ then

$$\begin{aligned} \mathrm{Var}(\bar{X}_n) &= \frac{1}{n}\sum_{|h|<n}\underbrace{\left(1 - \frac{|h|}{n}\right)}_{\in[0,1]}\gamma_X(h) \\ &\leq \frac{1}{n}\sum_{|h|<n}|\gamma_X(h)| \overset{n\to\infty}{\longrightarrow} 0, \end{aligned}$$

and if $\sum_{h\in\mathbb{Z}}|\gamma_X(h)| < \infty$ then by dominated convergence

$$\begin{aligned} n\mathrm{Var}(\bar{X}_n) &= \sum_{h\in\mathbb{Z}}\left(1 - \frac{|h|}{n}\right)\gamma_X(h)\cdot I\{|h| < n\} \\ &\overset{n\to\infty}{\longrightarrow} \sum_{h\in\mathbb{Z}}\gamma_X(h). \end{aligned}$$

□

The following result is very important in time series analysis. It is a special case of *Birkhoff's ergodic theorem* and gives a strong law of large numbers for an important class of strictly stationary processes.

Proposition 1.4.2 (Version of the ergodic theorem)**.** *Assume that X_t has the following representation*

$$X_t = f(\varepsilon_t, \varepsilon_{t-1}, \ldots),$$

where $\{\varepsilon_t\}$ is an iid sequence and

$$f : (\mathbb{R}^\infty, \mathcal{B}(\mathbb{R}^\infty)) \to (\mathbb{R}, \mathcal{B}(\mathbb{R})).$$

Then if $E|X_1| < \infty$, we have

$$\frac{1}{n}\sum_{k=1}^{n}X_k \overset{\text{a.s.}}{\longrightarrow} EX_1 \quad \textit{for } n \to \infty.$$

1.5 Estimation and elimination of trend and seasonal components

Suppose we have a non-stationary time series $\{X_t\}$ of the form

$$X_t = m_t + s_t + Y_t,$$

where

- m_t is a trend,
- s_t is a cyclical (seasonal) component ($s_t = s_{t+d}$ for some $d > 1$),
- Y_t is the stationary component, with $EY_t = 0$.

Most of the classical methods (and the ones we will develop here) are designed for stationary time series. The idea is to (a) either estimate m_t and s_t and treat the stationary component Y_t separately or to (b) transform $\{X_t\}$ such that trend and seasonal component disappear.

If we ignore such trends, our resulting analysis can be misleading. For example, we have printed in Figure 1.7 the empirical autocorrelation function belonging to signal plus noise data in Figure 1.2. By definition $\mathrm{Cov}(X_t, X_{t+h}) = \mathrm{Cov}(Z_t, Z_{t+h}) = 0$. However, our estimator for the `acf` assumes a constant mean, and hence the empirical `acf` doesn't look closely as it should. This doesn't mean we shouldn't apply the `acf` in data analysis, it gives us a good indication of the periodicity!

Method I (Linear Filter): Assume that $s_t = 0$. We define for some real valued coefficients $\ldots, a_{-1}, a_0, a_1, \ldots$

$$\begin{aligned}\widehat{m}_t &= \sum_{j=-\infty}^{\infty} a_j X_{t-j} \\ &= \sum_{j=-\infty}^{\infty} a_j m_{t-j} + \sum_{j=-\infty}^{\infty} a_j Y_{t-j} \\ &=: \mu_t + R_t.\end{aligned}$$

By a clever choice we can have $\mu_t \approx m_t$, while $R_t \approx 0$.

Example 1.5.1. Assume $m_t = b_0 + b_1 t$ (linear trend). Define

$$a_j = \frac{1}{2q+1} I\left\{j \in \{-q, \ldots, 0, \ldots, q\}\right\}, \quad j \in \mathbb{Z}.$$

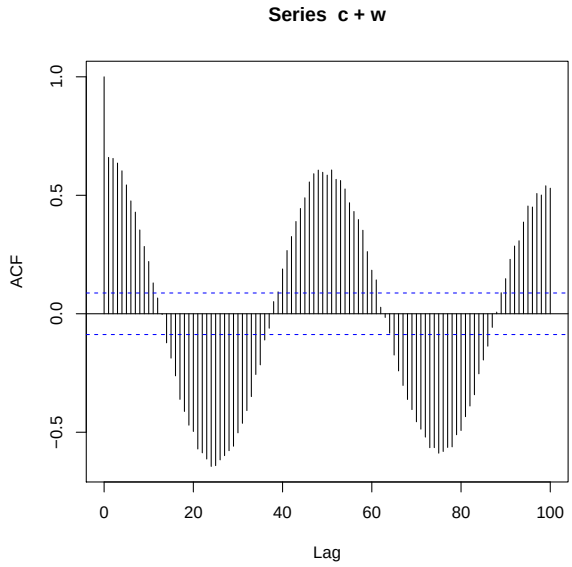


Figure 1.7: Empirical `acf` of the signal plus noise sequence ($\sigma^2 = 1$) in Figure 1.2.

Then,

$$\begin{aligned}
\widehat{m}_t &= \sum_{j=-q}^{q} \frac{1}{2q+1}(m_{t-j} + Y_{t-j}) \\
&= \sum_{j=-q}^{q} \frac{1}{2q+1}((b_0 + b_1(t-j) + Y_{t-j}) \\
&= b_0 + b_1 t + \frac{b_1}{2q+1}\sum_{j=-q}^{q}(-j) + \frac{1}{2q+1}\sum_{j=-q}^{q} Y_{t-j} \\
&= m_t + \frac{1}{2q+1}\sum_{j=-q}^{q} Y_{t-j}.
\end{aligned}$$

The linear trend *passes the filter without distortion*. If q is sufficiently large, then by a "law of large numbers effect" we can hope that

$$\frac{1}{2q+1}\sum_{j=-q}^{q} Y_{t-j} \approx 0.$$

Proposition 1.5.1. *A linear filter a_i passes any polynomial trend of order k without distortion, i.e.*

$$m_t = \sum_j a_j m_{t-j}, \quad \textit{for all trends of the from} \quad m_t = c_0 + c_1 t + \ldots + c_k t^k,$$

if and only if:

- $\sum_j a_j = 1$;
- $\sum_j j^r a_j = 0, \ \forall r = \{1, \ldots, k\}$.

Proof. Exercise 1.9. □

Example 1.5.2 (Detrending global temperature)**.** There seems to be a linear increase in the global temperature data in Figure 1.1. We assume that $X_t = \mu_t + Y_t$, where $\beta_0 + \beta_1 \times t$. We use $q = 3$ and compute $\hat{\mu}_t$ and $\hat{Y}_t = X_t - \hat{\mu}_t$, Figure 1.9. The empirical `acf` changes drastically. Conclusion: it is difficult to distinguish between change in the mean and strong dependence.

Method II (Regression approach). Another convenient method to estimate a trend m_t is to use a least squares approach. The ideas is to write $m_t = b_0 + b_1 h_1(t) + \cdots + b_p h_p(t)$, where $h_k(t)$ are certain *basis functions*, e.g. $h_k(t) = t^k$. Then we estimated b_k by least squares:

$$(\hat{b}_0, \ldots, \hat{b}_p) = \text{argmin}_{(b_0,\ldots,b_p)\in\mathbb{R}^{p+1}} \sum_{t=1}^{n} \left(X_t - [b_0 + b_1 h_1(t) + \cdots + b_p h_p(t)]\right)^2.$$

Method III (Differentiation). We define the back-shift operator B:

$$\begin{aligned} BX_t &= X_{t-1}; \\ B^2(X_t) = B(BX_t) &= BX_{t-1} = X_{t-2}; \\ B^k X_t &= X_{t-k}. \end{aligned}$$

More generally, let $P(z) = c_0 + c_1 z + \ldots + c_k z^k$ (a polynomial of degree k),

$$P(B)X_t = c_0 X_t + c_1 X_{t-1} + \ldots + c_k X_{t-k}.$$

In this case, we have simple algebraic relations. If $P(z)$ and $Q(z)$ are two polynomials, then

$$\begin{aligned} &(i) \quad P(B)(Q(B)X_t) = (P(B)Q(B))X_t = Q(B)(P(B)X_t), \\ &(ii) \quad P(B)[X_t + Y_t] = P(B)X_t + P(B)Y_t, \end{aligned}$$

where $P(B)Q(B)$ is the formal multiplication of $P(B)$ and $Q(B)$. (Exercise 1.12.)

The difference operator is given by $\nabla = (I - B)$. So $\nabla X_t = X_t - X_{t-1}$.

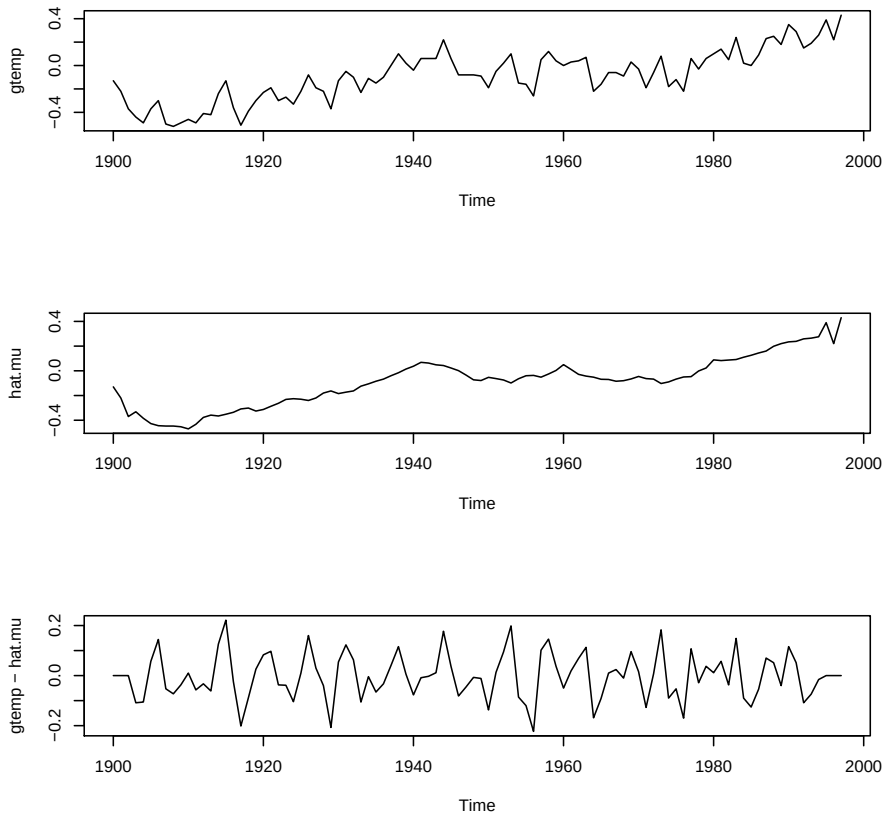


Figure 1.8: Global temperature data (top), estimated trend with uniform filter (middle, $q = 3$) and detrended data (bottom). [R-code]

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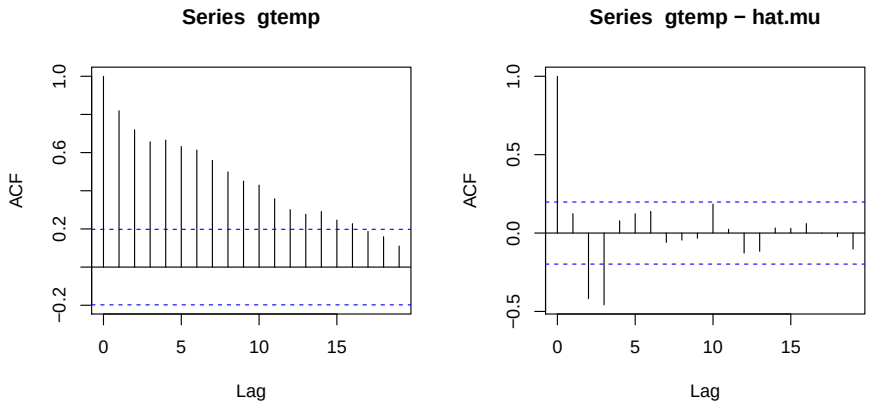


Figure 1.9: Empirical autocorrelation function before (left) and after (right) detrending. [R-code]

Proposition 1.5.2. *The following statements hold:*

- *If m_t is a degree k polynomial of the form*

$$m_t = c_0 + c_1 t + \ldots + c_k t^k,$$

  then $\nabla^k(m_t) = k!\, c_k$;

- $(I - B^d)s_t = s_t - s_{t-d} = 0$;

- *if $\{Y_t, t \in \mathbb{Z}\}$ is stationary then $\{P(B)Y_t, t \in \mathbb{Z}\}$ is stationary.*

Proof. Exercises 1.10 and 1.11. □

Example 1.5.3. If we have $m_t = b_0 + b_1 t$ and $X_t = m_t + Y_t$, then

$$\begin{aligned}\nabla X_t &= \nabla(m_t + Y_t) \\ &= m_t - m_{t-1} + (Y_t - Y_{t-1}) \\ &= b_0 + b_1 t - (b_0 + b_1(t-1)) + (Y_t - Y_{t-1}) \\ &= b_1 + (Y_t - Y_{t-1})\end{aligned}$$

where b_1 is constant: we have eliminated the trend.

Example 1.5.4 (Johnson & Johnson data)**.** The Johnson & Johnson data (see Figure 1.1) show an apparent increase. For financial price data, it is common that

we have exponential growth, and hence we will first a apply a `log`–transformation to stabilize the process:

$$Y_t = \log X_t.$$

Other common variance stabilizing transformations are of the form $Y_t = (X_t^\lambda - 1)/\lambda$ (*Box-Cox family*). The newly derived process is shown in Figure 1.10 (middle). Besides some linear trend we observe a cyclic pattern. Since these are quarterly data, a cycle of length 4 seems natural. We get rid of trend and seasonal pattern by defining

$$D_t = Y_t - Y_{t-4}.$$

Figure 1.10 (bottom).

Remark: In the model $X_t = m_t + s_t + Y_t$ where $\{Y_t\}$ is stationary we can assume that $EY_t = 0$. If $EY_t = \mu$ we can relace Y_t by $U_t = Y_t - \mu$ and m_t by $v_t = m_t + \mu$. Hence the mean function EX_t is captured in $m_t + s_t$, which we can estimate or eliminate by the methods described above. It is hence often assumed in theoretical results that the mean of the time series is 0.

1.6 Exercises

Ex 1.1 A sequence $\{X_t | t \in \mathbb{Z}\}$ is called independent, if each finite subcollection is independent, i.e. if

$$P(X_{t_1} \le x_1 \dots X_{t_n} \le x_n) = \prod_{i=1}^{n} P(X_{t_i} \le x_i).$$

Let $\{F_t | t \in \mathbb{Z}\}$ be a given sequence of distribution functions. Use Kolmogorov's extension theorem to prove the existence of an independent sequence $\{X_t | t \in \mathbb{Z}\}$ with given marginal distributions $F_t(x)$.

Ex 1.2 Let $\{X_t | t \ge 0\}$ be the random walk model with iid innovations ε_t and $E\,\varepsilon_t = \mu$, $\operatorname{Var}\varepsilon_t = \sigma^2 < \infty$. Calculate the mean function and the ACF. Show that ∇X_t is stationary.

Ex 1.3 Give an example for a process $\{X_t | t \in \mathbb{Z}\}$ where $\operatorname{Cov}(X_t, X_{t+h}) = \gamma(h)$ [depends only on lag] but which is not stationary.

Ex 1.4 Let $\{Z_t | t \in \mathbb{Z}\}$ be iid $N(0,1)$ [normal distribution with mean 0 and variance 1]. Let a, b and c be constants. Which, if any, of the processes are stationary?

- $X_t = a + bZ_t + cZ_{t-1}$;

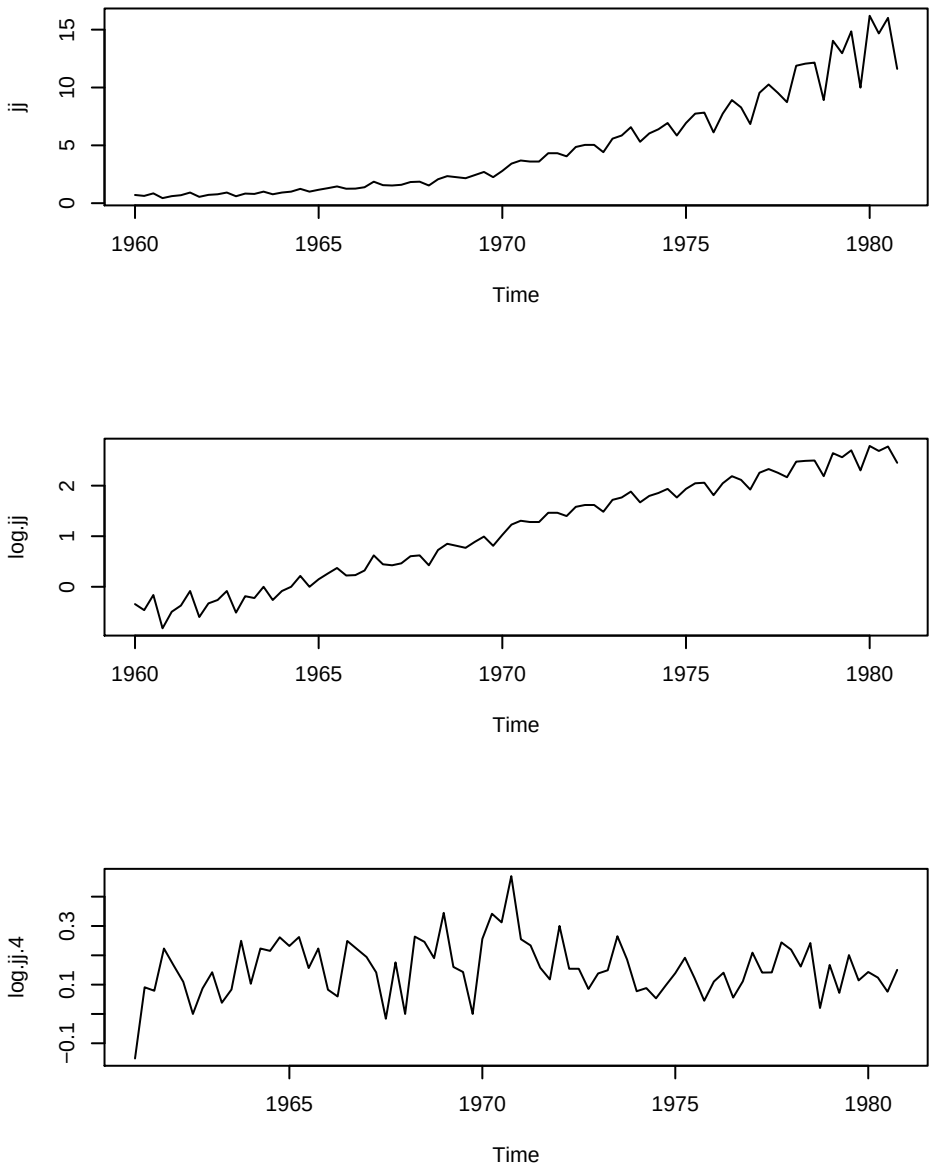


Figure 1.10: Johnson & Johnson data (top), the `log`-transformation (middle) and the `lag`–4 differences (bottom). [R-code]

- $X_t = Z_1 \cos(ct) + Z_2 \sin(ct)$;
- $X_t = Z_t \cos(ct) + Z_{t-1} \sin(ct)$;
- $X_t = a + bZ_0$;
- $X_t = Z_0 \cos(ct)$;
- $X_t = Z_t Z_{t-1}$.

For the stationary processes compute the ACF.

Ex 1.5 Assume X_i have a finite variance and $a_i \in \mathbb{R}$, $1 \leq i \leq n$. Show that

$$\mathrm{Var}(\sum_{i=1}^{n} a_i X_i) = \sum_{i,j=1}^{n} a_i a_j \mathrm{Cov}(X_i, X_j).$$

Ex 1.6 Let Θ be uniformly distributed on $[0, 2\pi)$ and let Z be independent of Θ have a standard exponential distribution [the density of Z is given by $f_Z(t) = e^{-t}$ if $t > 0$ and 0 else]. Compute the ACF of

$$X_t = Z \cos(t + \Theta).$$

Ex 1.7 Give an example for a process that is weakly stationary but not strictly stationary.

Ex 1.8 Which, if any, of the following functions defined on the integers is the ACF of a stationary time series?

- $f(h) = 1$ if $h = 0$ and $f(h) = 1/h$ if $h \neq 0$.
- $f(h) = (-1)^{|h|}$.
- $f(h) = 1$ if $h = 0$, $f(h) = 0.4$ if $h = \pm 1$ and $f(h) = 0$ otherwise.
- $f(h) = 1$ if $h = 0$, $f(h) = 0.6$ if $h = \pm 1$ and $f(h) = 0$ otherwise.

Ex 1.9 Show that a linear filter (a_j) passes an arbitrary polynomial of degree k without distortion, i.e.

$$m_t = \sum_j a_j m_{t-j}$$

for all k-th degree polynomials $m_t = c_0 + c_1 t + \cdots + c_k t^k$, if and only if (a) $\sum_j a_j = 1$ and (b) $\sum_j j^r a_j = 0$, for $r = 1, \ldots, k$.

Ex 1.10 Let $m_t = \sum_{k=0}^{p} c_k t^k$, $t \in \mathbb{Z}$. Show that ∇m_t is a polynomial of degree $(p-1)$ in t and hence that $\nabla^{p+1} m_t = 0$. [Here $\nabla x_t = (1-B)x_t = x_t - x_{t-1}$.]

Ex 1.11 Assume that $\{Y_t | t \in \mathbb{Z}\}$ is weakly stationary. Define

$$Q_k(B) = c_0 + c_1 B + \cdots + c_k B^k$$

as "backshift polynomial". Show that

$$X_t = Q_k(B)Y_t = c_0 Y_t + c_1 Y_{t-1} + \cdots + c_k Y_{t-k}$$

is again weakly stationary.

Ex 1.12 Show that if $P_n(B)$ is another backshift polynomial of degree n, then

$$Q_k(B)\Big[P_n(B)Y_t\Big] = R_{n+k}(B)Y_t$$

with $R_{n+k}(t) = Q_k(t)P_n(t)$.

Ex 1.13 Assume that $\{Z_t | \, t \in \mathbb{Z}\}$ is an i.i.d. sequence with $EZ_t = 0$ and $\mathrm{Var}(Z_t) = \sigma_Z^2 < \infty$ and that there is a stationary process satisfying

$$X_t = \alpha X_{t-1} + Z_t,$$

such that $\mathrm{Cov}(Z_t, X_s) = 0$ for all $s < t$. Compute the ACF of this process [it is called the AR(1) process]. Show that such a solution cannot exist if $|\alpha| \geq 1$.

Ex 1.14 Assume that $\{Z_t | \, t \in \mathbb{Z}\}$ is an i.i.d. sequence with $EZ_t = 0$ and $\mathrm{Var}(Z_t) = \sigma_Z^2 < \infty$. The process

$$X_t = P_q(B)Z_t$$

with $P_q(z) = 1 + \theta_1 z + \cdots + \theta_q z^q$ is called $MA(q)$ process. What is its ACF?

Ex 1.15 Obtain a closed form for $(\hat{b}_0, \ldots, \hat{b}_p)$ in the regression approach. Detrend the global temperature data using the regression approach assuming $m_t = b_0 + b_1 t$.

Ex 1.16 Assume $\{X_t\}$ is a stationary time series with ACF $\gamma_X(\cdot)$. Let $\hat{\gamma}_X(h)$ be the estimator of $\hat{\gamma}_X(h)$ described in Section 1.4. If $\gamma_X(h) \to 0$ for $h \to \infty$, then $\hat{\gamma}_X(h)$ is asymptotically unbiased, i.e. $E\hat{\gamma}_X(h) \to \gamma_X(h)$.

Ex 1.17 Assume that $\kappa(\cdot)$ is non-negative definite and even. Show (directly, i.e. without using that it is an ACF) that $\kappa(0) \geq 0$ and $\kappa(0) \geq |\kappa(h)|$.

Ex 1.18 Suppose we predict X_{t+h} by ϕX_t, $\phi \in \mathbb{R}$. (a) Show that the mean square prediction error $E(X_{t+h} - \phi X_t)^2$ is minimized when $\phi = \rho_X(h)$. (b) Show that the minimum mean square prediction error is $\gamma_X(0)(1 - \rho_X^2(h))$.

Ex 1.19 * If $\{X_t\}$ is a stationary time series with mean μ then the usual estimator for μ is the sample mean $\bar{X} = \frac{1}{n}(X_1 + \cdots + X_n)$. Assume we have $X_t = \mu + Z_t + \theta Z_{t-1}$, where $\{Z_t\} \sim \mathrm{WN}(0, \sigma^2)$. Can you find an unbiased estimator for μ of the form $\hat{\mu} = a_1 X_1 + \cdots + a_n X_n$ such that $\mathrm{Var}(\hat{\mu}) < \mathrm{Var}(\bar{X})$?

Ex 1.20 Compute the longrun variance of the MA(1) process $X_t = Z_t + \theta Z_{t-1}$ where $\{Z_t\}$ is i.i.d. with $EZ_t = 0$ and $EZ_t^2 = 1$. What happens if $\theta = -1$?

Ex 1.21 Compute the longrun variance of the AR(1) process defined in Ex 13. of the previous exercises.

Chapter 2

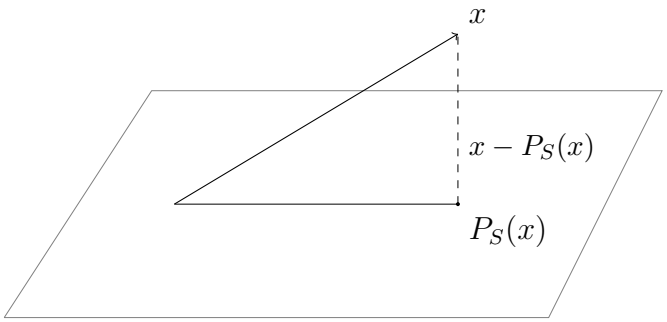
Hilbert spaces

A Hilbert space has a similar geometry like the Euclidean space $(\mathbb{R}^2, \mathbb{R}^3, \ldots)$. In particular we have the concept of *orthogonality* and *projection*. These concepts are very useful to obtain for a given element x in the Hilbert space $\mathcal{H}$ a closest element $\hat{x}$ in a subspace $\mathcal{S}$, say. In terms of time series, we will see that this closest element has something to do with optimal forecasting. In our setup

$$\mathcal{S} = \quad \text{information carried by} \quad X_1, \ldots, X_n,$$

whereas $x = X_{n+1}$ is the future element we wish to predict. In order to formalize

Figure 2.1: $P_S(x)$ is the element in S that is closest to x.



this we have to do some preparatory work.

2.1 Inner product spaces

Definition 2.1.1 (Inner product spaces)**.** A (complex) vector space $\mathcal{H}$ is called inner product space if there is a mapping

$$I : \mathcal{H} \times \mathcal{H} \to \mathbb{R} \quad (\mathbb{C}),$$

such that

(i) $I(x,y) = \overline{I(y,x)}$, (if $z = a + ib$ and $\bar{z} = a - ib$) (symmetry);
[Note: If we are considering a real vector space then $I(x,y) = I(y,x)$.]

(ii) $I(x+v,y) = I(x,y) + I(v,y),\ \forall x,y,v \in \mathcal{H}$ (linearity);

(iii) $I(\alpha x,y) = \alpha I(x,y),\ \forall \alpha \in \mathbb{R}\ (\mathbb{C}),\ \forall x,y,v \in \mathcal{H}$ (linearity);

(iv) $I(x,x) \geq 0$ and $I(x,x) = 0 \Leftrightarrow x = 0$ (non-negative definiteness).

Examples:

1. $\mathcal{H} = \mathbb{R}^n,$ with $I(x,y) = \sum_{i=1}^n x_i y_i,$
where $x = (x_1,\ldots,x_n)^T, y = (y_1,\ldots,y_n)^T$;
2. $\mathcal{H} = \mathbb{C}^n,$ with $I(x,y) = \sum_{i=1}^n x_i \bar{y}_i$;
3. $\mathcal{H} = \{X : (\Omega,\mathcal{A}) \to (\mathbb{R},\mathcal{B}(\mathbb{R})) \mid EX^2 < \infty\}$ (random variables with finite variance), with $I(X,Y) = EXY$.

Usually one uses, instead of $I(x,y)$ the notation $\langle x,y\rangle$, we call it the inner product between x and y.

Proposition 2.1.1 (Cauchy-Schwarz inequality)**.** *Let $\mathcal{H}$ be an inner product space. Then for all x and y in $\mathcal{H}$*

$$|\langle x,y\rangle| \leq \sqrt{\langle x,x\rangle} \cdot \sqrt{\langle y,y\rangle}.$$

Proof. We give the proof for real inner product spaces. First note that

$$\begin{aligned} 0 &\overset{\text{(iv)}}{\leq} \langle x - ty, x - ty\rangle,\ \forall x,y \in \mathcal{H},\ t \in \mathbb{R} \\ &\overset{\text{(i)-(iii)}}{=} \langle x,x\rangle - 2t\langle x,y\rangle + t^2\langle y,y\rangle. \end{aligned}$$

Setting $t = \frac{\langle x,y\rangle}{\langle y,y\rangle}$ gives the result. □

A nice consequence is that the inner product gives a *norm* on $\mathcal{H}$ by setting

$$\|x\| := \sqrt{\langle x,x\rangle}.$$

Recall: A norm has the following properties:

(N1) $\|x+y\| \leq \|x\| + \|y\|$;

(N2) $\|\alpha x\| = |\alpha| \cdot \|x\|, \ \forall \alpha \in \mathbb{R}$;

(N3) $\|x\| \geq 0$ and $\|x\| = 0 \ \Leftrightarrow x = 0$.

(N2) and (N3) follow directly from the definition of inner product. To prove (N1) note that by the Cauchy-Schwarz inequality

$$\begin{aligned}\|x+y\|^2 &= \langle x+y, x+y\rangle \\ &= \langle x,x\rangle + \underbrace{2\langle x,y\rangle}_{\leq \sqrt{\langle x,x\rangle}\cdot\sqrt{\langle y,y\rangle}} + \langle y,y\rangle \\ &\leq \left(\sqrt{\langle x,x\rangle} + \sqrt{\langle y,y\rangle}\right)^2 \\ &= \left(\|x\| + \|y\|\right)^2.\end{aligned}$$

Definition 2.1.2. If $\{x_n\}$ is a sequence in $\mathcal{H}$ then we say that x_n *converges in norm* to $x \in \mathcal{H}$ if $\|x_n - x\| \overset{n\to\infty}{\longrightarrow} 0$. Short: $x_n \overset{n\to\infty}{\longrightarrow} x$.

Proposition 2.1.2 (Continuity of the inner product)**.** *If* $\{x_n\}$ *and* $\{y_n\} \in \mathcal{H}$ *and* $x_n \to x$ *and* $y_n \to y$*, then*

(i) $\|x_n\| \to \|x\| \quad (n \to \infty)$*;*

(ii) $\langle x_n, y_m\rangle \to \langle x, y\rangle \quad (n, m \to \infty)$.

Proof.

(i) By the triangular inequality

$$\begin{aligned}\|x_n\| &= \|x_n - x + x\| \\ &\leq \|x_n - x\| + \|x\| \overset{n\to\infty}{\longrightarrow} \|x\|.\end{aligned}$$

On the other hand

$$\|x_n\| \geq \|x\| - \|x_n - x\| \overset{n\to\infty}{\longrightarrow} \|x\|.$$

(ii) Exercise 2.1.

□

Definition 2.1.3 (Cauchy sequence)**.** Let $\{x_n\} \in \mathcal{H}$, then $\{x_n\}$ is called Cauchy sequence if:

$$\|x_n - x_m\| \overset{n,m\to\infty}{\longrightarrow} 0.$$

Definition 2.1.4 (Completeness)**.** The space $\mathcal{H}$ is said to be complete if every Cauchy sequence has a limit in $\mathcal{H}$, i.e. there is an $x \in \mathcal{H}$ such that $\|x_n - x\| \overset{n\to\infty}{\longrightarrow} 0$.

Definition 2.1.5 (Hilbert space)**.** A complete inner product space is called Hilbert space.

Examples for Hilbert spaces.

1. $\mathbb{R}^n$ with $\langle x, y\rangle = \sum_{i=1}^n x_i y_i$;
2. $\mathbb{C}^n$ with $\langle x, y\rangle = \sum_{i=1}^n x_i \bar{y}_i$;
3. $L^2(\Omega, \mathscr{A}, P) = \{X \text{ on } (\Omega, \mathcal{A}, P) \text{ with finite variance}\}$ with $\langle X, Y\rangle = E(XY)$.

Remark. If $\mathcal{H} = L^2(\Omega, \mathscr{A}, P)$ and $E[X] = E[Y] = 0$ then

- $\langle X, Y\rangle = EXY = \mathrm{Cov}(X, Y)$;
- $\|X\|^2 = \langle X, X\rangle = EX^2 = \mathrm{Var}(X)$.

Remark. If $\mathcal{H} = L^2(\Omega, \mathscr{A}, P)$ then

$$X_n \overset{n\to\infty}{\longrightarrow} X \Leftrightarrow E[(X_n - X)^2] \overset{n\to\infty}{\longrightarrow} 0$$

(mean squared convergence).

Proposition 2.1.3 (Cauchy criterion)**.** *Assume $\{x_n\}$ is a sequence in $\mathcal{H}$, where $\mathcal{H}$ is a Hilbert space. Then*

$$x_n \to x \in \mathcal{H} \;\Leftrightarrow\; \{x_n\} \text{ is a Cauchy sequence.}$$

Proof. $(\Rightarrow)$ $||x_n - x_m|| \le ||x_n - x|| + ||x_m - x|| \to 0$, if $m, n \to \infty$.

$(\Leftarrow)$ By definition of completeness.

□

2.2 A first application

Many time series we study have the form $X_t = \sum_{k\ge 0} a_k Z_{t-k}$ with $\{Z_k\} \sim \mathrm{WN}(0, \sigma^2)$ (e.g. ARMA(p, q) processes which we study in the next section). We ask, when such a sequence is convergent (in some sense to be specified). Consider

$$X_t^{(n)} = \sum_{k=0}^{n} a_k Z_{t-k}$$

By assumption $Z_{t-k} \in L^2 = L^2(\Omega, \mathscr{A}, \mathbb{P})$ and hence $X_t^{(n)} \in L^2$, $\forall t, n$. We check under which assumptions on the a_k's $\left\{X_t^{(n)}, n \ge 1\right\}$ is a Cauchy sequence. Assume without loss of generality that $m \le n$, then

$$\left\| X_t^{(n)} - X_t^{(m)} \right\|^2 = E\left[\sum_{k=m+1}^{n} a_k Z_{t-k} \right]^2$$
$$= \sum_{k=m+1}^{n} \sum_{l=m+1}^{n} a_k a_l \operatorname{Cov}(Z_{t-k}, Z_{t-l})$$
$$= \sigma^2 \left(\sum_{k=m+1}^{n} a_k^2 \right).$$

Now

$$\sum_{k=m+1}^{n} a_k^2 \to 0 \quad \text{for } m, n \to \infty \Leftrightarrow \sum_{k \geq 0} a_k^2 < \infty.$$

I.e. there exists an element $X_t \in L^2$ such that

$$X_t^{(n)} \to X_t \quad \text{(mean square convergence)} \Leftrightarrow \sum_{k \geq 0} a_k^2 < \infty.$$

We will write $X_t = \sum_{k=0}^{\infty} a_k Z_{t-k}$.

It also follows that

- $\operatorname{Var}(X_t) < \infty$;
- and by continuity of the norm that

$$\operatorname{Var}(X_t) = \|X_t\|^2 = \lim_{n \to \infty} \left\| X_t^{(n)} \right\|^2 = \lim_{n \to \infty} \operatorname{Var}\left(X_t^{(n)} \right) = \sigma_Z^2 \sum_{k \geq 0} a_k^2.$$

2.3 The projection theorem

Definition 2.3.1 (Orthogonality)**.** If $x, y \in \mathcal{H}$ (inner product space) then $x \perp y$ (x orthogonal to y) $\Leftrightarrow \langle x, y \rangle = 0$.

Example 2.3.1. In L^2 if we have $E[X] = E[Y] = 0$, then $X \perp Y \Leftrightarrow \langle X, Y \rangle = 0 \Leftrightarrow E[XY] = 0 \Leftrightarrow \operatorname{Cov}(X, Y) = 0$. Orthogonal thus means uncorrelated.

Example 2.3.2. A very useful result is *Pythagoras' Theorem* which states that $\|x+y\|^2 = \|x\|^2 + \|y\|^2$ in case $x \perp y$. (Exercise 2.4). In case of orthogonal random variables it means that the variance of a sum, is the sum of the variances.

A look at Figure 2.1 shows that $x - P_s(x) \perp y$ for all $y \in S$. This characterizes the projection in $\mathbb{R}^k$. The projection theorem states, that the same holds in all Hilbert spaces.

Example 2.3.3. Assume $\mathcal{H} = L^2$ and $S = \{\alpha\mathbf{1}, \alpha \in \mathbb{R}\}$, where $\mathbf{1}$ is the random variable that is equal to 1. Then S is the space of all constant random variables; it is a closed linear subspace of L^2. For any $X \in L^2$ we have

$$\begin{aligned} P_S(X) &= \alpha_0\mathbf{1} \\ &\Leftrightarrow \langle X - \alpha_0\mathbf{1}, \underbrace{\alpha\mathbf{1}}_{\text{any other element in the space}, \forall\alpha \in \mathbb{R}} \rangle = 0 \\ &\Leftrightarrow \langle X - \alpha_0\mathbf{1}, \mathbf{1}\rangle = 0 \\ &\Leftrightarrow \underbrace{\langle X, \mathbf{1}\rangle}_{EX} = \alpha_0 \underbrace{\langle \mathbf{1}, \mathbf{1}\rangle}_{1}. \end{aligned}$$

It follows that (i) $\alpha_0 = EX$ and (ii) $E\left|X - EX\right|^2 \leq E\left|X - \alpha\right|^2$, $\forall\alpha \in \mathbb{R}$.

Example 2.3.4. Assume $\{X_t, t \in \mathbb{Z}\}$ is stationary with zero mean. What is the optimal (linear) forecast for X_{n+1} of the form $\hat{X}_{n+1} = \alpha X_n$?

$$\begin{aligned} &\langle X_{n+1} - \underbrace{\hat{X}_{n+1}}_{\alpha_0 X_n}, \alpha X_n\rangle = 0 \quad \forall\alpha \in \mathbb{R} \\ &\Leftrightarrow \underbrace{\langle X_{n+1}, X_n\rangle}_{\gamma_X(1)} = \alpha_0 \underbrace{\langle X_n, X_n\rangle}_{\gamma_X(0)} \\ &\Rightarrow \alpha_0 = \frac{\gamma_X(1)}{\gamma_X(0)} = \rho_X(1). \end{aligned}$$

The forecast is optimal in the sense that

$$E\left[X_{n+1} - \alpha X_n\right]^2 \geq E\left[X_{n+1} - \rho_X(1)X_n\right]^2, \ \forall\alpha \in \mathbb{R}.$$

Here is the main result.

Theorem 2.3.1 (Projection theorem)**.** *Let $\mathcal{H}$ be a Hilbert space and let S be a closed linear subspace. Then*

(i) $\forall x \in \mathcal{H}$ there is a unique $\hat{x} \in S$ such that $\|x - \hat{x}\| = \inf_{y\in S}\|y - x\|$.

(ii) $\hat{x}$ is characterized by $x - \hat{x} \perp y$, $\forall y \in S$. That is

(a) $x - \hat{x} \perp y$, $\forall y \in S$

(b) if $z \in S$ such that $x - z \perp y$ for all $y \in S$, then $z = \hat{x}$.

Example 2.3.5. In Example 2.3.4 we considered the optimal *linear* forecast. We could ask more generally, what is the best forecast of the form $f_0(X_n)$, i.e.

$$E[X_{n+1} - f_0(X_n)]^2 \leq E[X_{n+1} - f(X_n)]^2,$$

for all f, such that $f(X_n)$ is a square integrable random variable? The set

$$\mathcal{M} = \{T = f(X_n) | ET^2 < \infty\}$$

is the space $L^2(\Omega, \mathcal{A}_0, P)$, where $\mathcal{A}_0 = X_n^{-1}(\mathcal{B}(\mathbb{R}))$ and is a closed subspace of $L^2(\Omega, \mathcal{A}_0, P)$ (it is again a Hilbert space). We thus have that $f_0(X_n) = P_{\mathcal{M}}(X_{n+1})$. In fact, one can show that it equals the *conditional expectation* $E[X_{n+1}|X_n]$.

Proof of Theorem 2.3.1. Existence. Set $d := \inf_{y \in S} \|x - y\|$. Then $\exists \{y_n\} \subseteq S$ such that $\|x - y_n\| \overset{n\to\infty}{\longrightarrow} d$. We show that $\{y_n\}$ is a Cauchy sequence. Then $y_n \to y$ and by continuity of the norm $\lim_{n\to\infty} \|x - y_n\| = \|x - y\| = d$. So we can set $\hat{x} = y$.

Now we show that $\{y_n\}$ is a Cauchy sequence. By Exercise 2.5 and the fact that $(y_n + y_m)/2 \in S$ we have

$$\begin{aligned}
\|y_n - y_m\|^2 &= \|y_n - x - (y_m - x)\|^2 \\
&= 2\|y_n - x\|^2 + 2\|y_m - x\|^2 - \|y_n + y_m - 2x\|^2 \\
&= 2\|y_n - x\|^2 + 2\|y_m - x\|^2 - 4 \underbrace{\left\| \frac{y_n + y_m}{2} - x \right\|^2}_{\geq d^2 \text{ by our construction}} \\
&\leq 2\|y_n - x\|^2 + 2\|y_m - x\|^2 - 4d^2, \quad \forall n, m \geq 1.
\end{aligned}$$

By construction of the y_n the last expression tends to 0 for $m, n \to \infty$.

Uniqueness. Assume there exists $\hat{y} \in S$ such that $\|\hat{x} - x\| = \|\hat{y} - x\|$. By similar arguments as before

$$\begin{aligned}
\|\hat{x} - \hat{y}\|^2 &= \|(\hat{x} - x) - (\hat{y} - x)\|^2 \\
&= 2\|\hat{x} - x\|^2 + 2\|\hat{y} - \hat{x}\|^2 - 4\left\| \frac{\hat{x} + \hat{y}}{2} - \hat{x} \right\|^2 \\
&\leq 2d^2 + 2d^2 - 4d^2 = 0
\end{aligned}$$

Then $\|\hat{x} - \hat{y}\| = 0 \Leftrightarrow \hat{x} = \hat{y}$.

Characterization.

(b) We want to prove that if $x - z \perp y, \forall y \in S \Rightarrow z = \hat{x}$. By Pythagoras' theorem we have for all $y \in S$

$$\|x - y\|^2 = \|x - z\|^2 + \|z - y\|^2,$$

and hence $\|x - z\|^2 \leq \|x - y\|^2, \forall y \in S$. But then $z = \hat{x}$ by uniqueness of the minimizer.

(a) We want to show that $\forall y \in S, x - \hat{x} \perp y$. By contradiction, assume $\exists\, y \in S$ such that $\langle x - \hat{x}, y\rangle \neq 0$. Apparently this implies that $y \neq 0$. So we can define the following element in S:

$$\tilde{x} = \hat{x} + \frac{\langle x - \hat{x}, y\rangle}{\|y\|^2} y.$$

Then

$$\hat{x} - \tilde{x} = -\frac{\langle x - \hat{x}, y\rangle}{\|y\|^2} y$$

and we can write

$$\begin{aligned}
\|x - \tilde{x}\|^2 &= \langle (x - \hat{x}) + (\hat{x} - \tilde{x}), (x - \hat{x}) + (\hat{x} - \tilde{x})\rangle \\
&= \langle (x - \hat{x}), (x - \hat{x})\rangle + \langle (\tilde{x} - \hat{x}), (\tilde{x} - \hat{x})\rangle + 2\langle x - \hat{x}, \hat{x} - \tilde{x}\rangle \\
&= \|x - \hat{x}\|^2 + \|\hat{x} - \tilde{x}\|^2 + 2\left\langle x - \hat{x}, -\frac{\langle x - \hat{x}, y\rangle}{\|y\|^2} y \right\rangle \\
&= \|x - \hat{x}\|^2 + \frac{\langle x - \hat{x}, y\rangle^2}{\|y\|^2} - 2\frac{\langle x - \hat{x}, y\rangle^2}{\|y\|^2} \\
&= \|x - \hat{x}\|^2 - \underbrace{\frac{\langle x - \hat{x}, y\rangle^2}{\|y\|^2}}_{>0 \text{ by our assumption}} \\
&\Rightarrow \|x - \tilde{x}\|^2 < \|x - \hat{x}\|^2
\end{aligned}$$

which is clearly a contradiction.

□

2.4 Exercises

Ex 2.1 Let $\mathcal{H} = \mathbb{R}^n$. Show that for $x = (x_1, \ldots, x_n)^T$ and $y = (y_1, \ldots, y_n)^T$ the mapping $I(x, y) = \sum_{i=1}^n x_i y_i$ defines an inner product. What is the norm this inner product generates?

Ex 2.2 Let $\mathcal{H} = \{x : [0, 1] \to \mathbb{R}\}$. Show that for $x = x(t)$ and $y = y(t)$ the mapping $I(x, y) = \int_0^1 x(t)y(t)dt$ defines an inner product on $\mathcal{H}$. (This space is called $L^2 = L^2([0, 1])$.)

Ex 2.3 Show that the inner product is continuous, i.e. $x_n \to x$ and $y_n \to y$ imply that $\langle x_n, y_n\rangle \to \langle x, y\rangle$.

Ex 2.4 Prove *Pythagoras Theorem*: if x, y are orthogonal elements of some inner-product space $\mathcal{H}$ then $\|x+y\|^2 = \|x\|^2 + \|y\|^2$.

Ex 2.5 Prove the *parallelogram rule*: if x, y are elements of some inner-product space $\mathcal{H}$ then $\|x-y\|^2 = 2\|x\|^2 + 2\|y\|^2 - \|x+y\|^2$.

Ex 2.6 Show that if $X_n \xrightarrow{\text{m.s.}} X$, then $\text{Var}(X_n) \to \text{Var}(X)$.

Ex 2.7 Let $x_1, \ldots, x_n \in \mathcal{H}$ which is assumed to be a real Hilbert space. Show that $\overline{\text{sp}}\{x_1, \ldots, x_n\} = \{\sum_{i=1}^n \alpha_i x_i : \alpha_i \in \mathbb{R}\}$. Hint: You can assume that x_i are orthogonal.

Ex 2.8 Let $\mathcal{H}$ be a Hilbertspace and let S be a closed subspace of $\mathcal{H}$. Denote by $P_S(x)$ the projection of x onto S. Then

- $P_S(\alpha x + \beta y) = \alpha P_S(x) + \beta P_S(y) \quad \forall x, y \in \mathcal{H},\ \alpha, \beta \in \mathbb{R}$.
- $\|x\|^2 = \|P_S(x)\|^2 + \|(1-P_S)(x)\|^2 \quad \forall x \in \mathcal{H}$.
- Any $x \in \mathcal{H}$ has a unique decomposition $x = s + s'$ where $s \in S$ and $s' \perp S$.
- $P_S(x_n) \to P_S(x)$ if $x_n \to x$.
- $x \in S$ if and only if $P_S(x) = x$.
- $x \in S^\perp$ if and only if $P_S(x) = 0$.
- $S_1 \subset S_2$ if and only if $P_{S_1}(P_{S_2}(x)) = P_{S_1}(x)$ for all $x \in \mathcal{H}$.

Ex 2.9 Assume that $e_1, \ldots, e_m$ are *orthonormal* in some Hilbert space $\mathcal{H}$, i.e.

$$\langle e_i, e_j \rangle = \begin{cases} 0 & \text{if} \quad i \neq j; \\ 1 & \text{else.} \end{cases}$$

Let $x \in \mathcal{H}$ be arbitrary.

(a) What is $P_S(x)$ for $S = \overline{\text{sp}}\{e_1, \ldots, e_m\}$?

(b) Show that $\|P_S(x)\|^2 = \sum_{j=0}^m |\langle x|e_i\rangle|^2$

(c) Conclude that $\|x\|^2 \geq \sum_{j=0}^m |\langle x|e_i\rangle|^2$

(d) Show that $\|x - P_S(x)\|^2 \leq \|x - \sum_{i=1}^m c_i e_i\|^2$. When does equality hold?

Ex 2.10 Assume that S is a closed subspace of $\mathcal{H}$. Show that $(S^\perp)^\perp = S$.

Ex 2.11 Let $\mathcal{M}$ be any subset of $\mathcal{H}$. Show that $\mathcal{M}^\perp$ is a closed subspace.

Ex 2.12 Let $\{X_t : t \in \mathbb{Z}\}$ be weakly stationary with ACF γ and $EX_t = 0$. Show that

$$Y_n = \sum_{k=1}^{n} a_k X_k$$

converges in mean square if $\sum \sum a_j a_i \gamma(j-i)$ is finite.

Ex 2.13 In the previous example assume that X_t is white noise. Show that Y_n converges in mean square if and only if $\sum a_j^2 < \infty$.

Ex 2.14 Suppose that $\{X_t : t \in \mathbb{Z}\}$ is a stationary process with mean zero and ACF $\gamma(\cdot)$ for which

$$\sum_{h \geq 0} |\gamma(h)| < \infty.$$

Define

$$f(\lambda) = \frac{1}{2\pi} \sum_{h=-\infty}^{\infty} \gamma(h) \exp(-ih\lambda), \quad -\pi \leq \lambda \leq \pi,$$

and show that $\gamma(h) = \int_{-\pi}^{\pi} \exp(ih\lambda) f(\lambda) d\lambda$. The function $f(\lambda)$ is called the spectral density of the process $\{X_t : t \in \mathbb{Z}\}$.

Chapter 3

ARMA models

3.1 Introduction and definitions

We already defined a special case of linear processes in Chapter 1. Here we give a more general definition.

Definition 3.1.1 (Linear Process)**.** Let $\{Z_t\}$ be any random process and consider the time series $\{X_t, t \in \mathbb{Z}\}$ given by

$$X_t = \sum_{k \in \mathbb{Z}} \psi_k Z_{t-k} = \ldots + \psi_{-1} Z_{t+1} + \psi_0 Z_t + \psi_1 Z_{t-1} + \ldots \tag{3.1}$$

with ψ_k numerical constants, such that the expression in (3.1) makes sense (converges almost surely or in L^2). Such a process is called *linear process with innovations* $\{Z_t\}$.

Remark.

- Linear processes can in principle also depend on future innovations.
- Throughout these notes we will study linear processes with innovations $\{Z_t\} \sim \mathrm{WN}(0, \sigma^2)$.
- A different way to write (3.1) is

  $$X_t = \psi(B) Z_t,$$

  where $\psi(B)$ is the *linear filter* defined by $\psi(z) = \sum_{k \in \mathbb{Z}} \psi_k z^k$.
- Linear processes are convenient since they are comparably easy to handle. In terms of modeling covariance structure they are quite flexible.

- Problem: we have infinitely many parameters and so they are not easy to interpret. A way out is to define $\psi_k = g_k(\phi_1, \ldots, \phi_p, \theta_1, \ldots, \theta_q)$ for some functions g_k. Naturally such parametrizations result from *stochastic difference equations*. One of the most prominent linear processes is the ARMA(p, q), given by such a difference equation.

Definition 3.1.2 (ARMA(p, q))**.** A stationary process $\{X_t\}$ is called ARMA(p, q) process if it satisfies the following equation:

$$X_t - \phi_1 X_{t-1} - \phi_2 X_{t-2} - \cdots - \phi_p X_{t-p} = Z_t + \theta_1 Z_{t-1} + \cdots + \theta_q Z_{t-q}, \tag{3.2}$$

where $\{Z_t\} \sim \text{WN}(0, \sigma^2)$, ϕ_i $(1 \leq i \leq p)$ and θ_j $(1 \leq j \leq q)$ are real coefficients. If $\{X_t - \mu\}$ is an ARMA(p, q) process, then $\{X_t\}$ is called ARMA(p, q) with mean μ.

Remarks.

- ARMA stands for Auto Regressive Moving Average.
- Let $\{X_t\}$ be a stationary solution of (3.2) and assume $EX_t = \mu$. Taking expectations on both sides of (3.2) gives:

  $$\mu(1 - \phi_1 - \phi_2 - \ldots - \phi_p) = 0.$$

  This is true if and only if $\mu = 0$ or $(1 - \phi_1 - \phi_2 - \cdots - \phi_p) = 0$. In the latter case, there is no stationary solution as we will see later. Hence in general stationary solutions of (3.2) have zero mean.
- We will see that if (3.2) has a stationary solution, it will be a linear process.
- Define the *autoregressive polynomial*

  $$\phi(z) = 1 - \phi_1 z - \phi_2 z^2 - \cdots - \phi_p z^p$$

  and the *moving average polynomial*

  $$\theta(z) = 1 + \theta_1 z + \theta_2 z^2 + \cdots + \theta_q z^q$$

  and let B be the back shift operator. Then (3.2) can be rewritten as:

  $$\phi(B) X_t = \theta(B) Z_t. \tag{3.3}$$

Example 3.1.1 (MA(q) processes)**.** Assume that $\phi(z) = 1$ and $\theta(z) = 1 + \theta_1 z + \cdots + \theta_q z^q$. Then (3.3) is

$$X_t = \theta(B) Z_t = Z_t + \theta_1 Z_{t-1} + \cdots + \theta_q Z_{t-q}.$$

This is called the *moving average process of order* q.

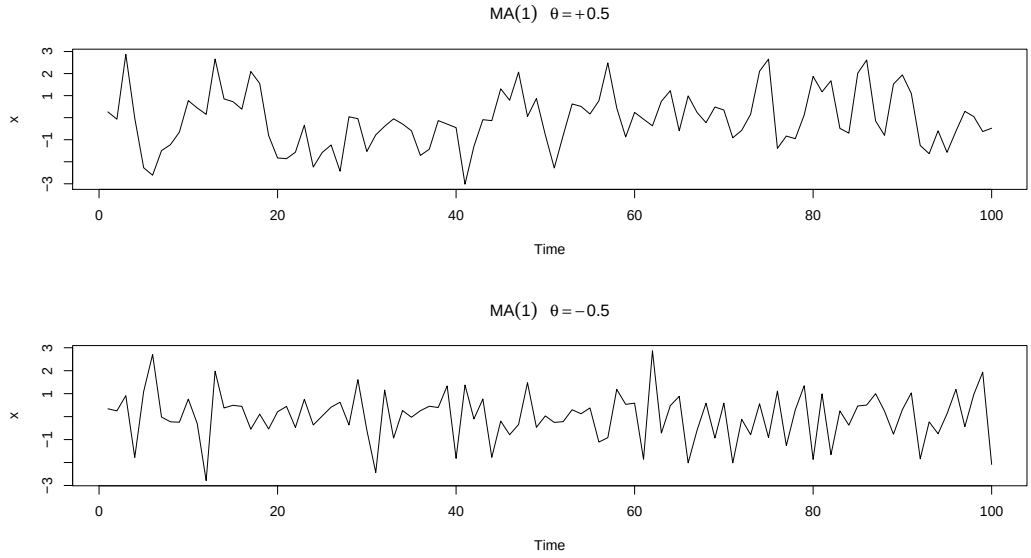


Figure 3.1: Simulated MA(1) models: $\theta = 0.5$ and $\sigma^2 = 1$ (top), $\theta = -0.5$ and $\sigma^2 = 1$ (bottom).

In the case of the MA(q) the existence of solution to (3.2) is obvious. We have to check if the process is stationary in order to verify that it is an ARMA process.

- $\mathrm{Var}(X_t) < \infty$. Trivial: all the Z_t have a finite variance, so the process has a finite variance as well.
- $$E[X_t] = \sum_{j=0}^{q} \theta_j E\left[Z_{t-j}\right] = 0 \quad \forall t;$$
- Defining $\theta_0 = 1$ a simple calculation shows (see Exercise 1.14)

$$\mathrm{Cov}\left(X_t, X_{t+h}\right) = \begin{cases} \sigma^2 \sum_{j=0}^{q-|h|} \theta_j \theta_{j+|h|} & \text{if } |h| \leq q, \\ 0 & \text{if } |h| > q, \end{cases}$$

  which is not depending on t.

Example 3.1.2 (AR(p) processes)**.** Here the moving average polynomial $\theta(z) = 1$ and $\phi(z) = 1 - \phi_1 z - \cdots - \phi_p z^p$. Thus

$$\phi(B)X_t = X_t - \phi_1 X_{t-1} - \cdots - \phi_p X_{t-p} = Z_t.$$

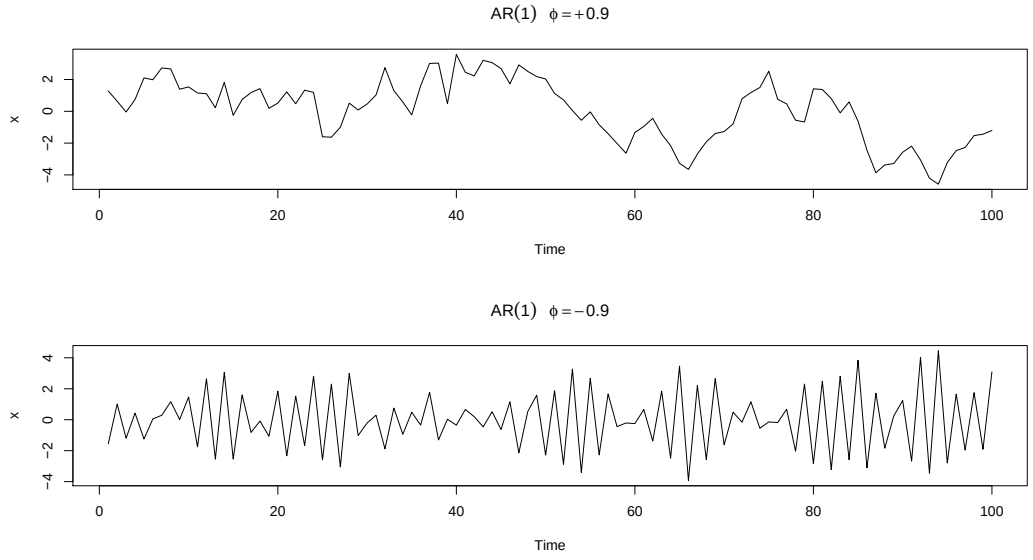


Figure 3.2: Simulated AR(1) models: $\phi = 0.9$ and $\sigma^2 = 1$ (top), $\phi = -0.9$ and $\sigma^2 = 1$ (bottom).

A bigger part of this chapter is devoted to the question whether such a process has a stationary solution. This is not evident! Let us for purpose of demonstration study the special case of an AR(1) process.

$$X_t = \phi X_{t-1} + Z_t. \tag{3.4}$$

Let us assume first that $|\phi| < 1$ *and that a stationary solution exists.* Then, by iteration

$$\begin{aligned}
X_t &= \phi X_{t-1} + Z_t \\
&= \phi(\phi X_{t-2} + Z_{t-1}) + Z_t \\
&= \phi^2 X_{t-2} + \phi Z_{t-1} + Z_t \\
&= \phi^2(\phi X_{t-3} + Z_{t-2}) + \phi Z_{t-1} + Z_t \\
&= \phi^3 X_{t-3} + \phi^2 Z_{t-2} + \phi Z_{t-1} + Z_t \\
&= \dots \\
&= Z_t + \phi Z_{t-1} + \phi^2 Z_{t-2} + \dots + \phi^{n-1} Z_{t-n+1} + \phi^n X_{t-n} \\
&= \underbrace{\sum_{k=0}^{n-1} \phi^k Z_{t-k}}_{X_t^{(n)}} + \phi^n X_{t-n}, \quad \forall n \geq 1.
\end{aligned}$$

It follows that

$$\begin{aligned}\|X_t - X_t^{(n)}\| &= \|\phi^n X_{t-n}\| \\ &= |\phi|^n \|X_{t-n}\| \\ &= |\phi|^n \sqrt{EX_0^2},\end{aligned}$$

because of stationarity and definition of the norm. This tends to 0 for $n \to \infty$ since $|\phi| < 1$. This shows that the stationary solution X_t is the L^2 limit of $X_t^{(n)}$, $n \to \infty$.

Define now as before $X_t^{(n)}$ *and assume* $|\phi| < 1$. Since the coefficients ϕ^k are square summable, we know from Section 2.2 that $\{X_t^{(n)}, n \geq 1\}$ defines a Cauchy sequence and thus it will have an L^2 limit. Let us denote this limit by X_t. We show X_t solves

$$X_t = \phi X_{t-1} + Z_t.$$

This is equivalent to

$$\lim_{n\to\infty} \left\| X_t^{(n)} - \phi X_{t-1}^{(n)} - Z_t \right\| = 0.$$

Now

$$\begin{aligned}&\lim_{n\to\infty} \left\| X_t^{(n)} - \phi X_{t-1}^{(n)} - Z_t \right\| \\ &\quad = \lim_{n\to\infty} \left\| \sum_{k=0}^{n-1} \phi^k Z_{t-k} - \phi \sum_{k=0}^{n-1} \phi^k Z_{t-k-1} - Z_t \right\| \\ &\quad = \lim_{n\to\infty} |\phi|^n \|Z_{t-n}\| = 0.\end{aligned}$$

We show finally stationarity. By continuity of the norm and inner product we have

$$\begin{aligned}E[X_t] &= \langle X_t | \mathbf{1} \rangle \\ &= \lim_{n\to\infty} \left\langle X_t^{(n)} | \mathbf{1} \right\rangle \\ &= \lim_{n\to\infty} \sum_{k=0}^{n-1} \phi^k E[Z_{t-k}] \\ &= 0,\end{aligned}$$

and

$$\begin{aligned}\gamma_X(h) &= \mathrm{Cov}(X_t, X_{t+h}) \\ &= \langle X_t, X_{t+h}\rangle \\ &= \lim_{n\to\infty} \left\langle X_t^{(n)}, X_{t+h}^{(n)} \right\rangle \text{(a moving average MA}(n)\text{)} \\ &= \lim_{n\to\infty} \sum_{k=0}^{n-|h|} \sigma^2 \phi^k \phi^{k+|h|} \\ &= \sigma^2 \phi^{|h|} \sum_{k=0}^{\infty} (\phi^2)^k \\ &= \sigma^2 \frac{\phi^{|h|}}{1-\phi^2}.\end{aligned}$$

We have shown that if $|\phi| < 1$ *then:*

- A stationary solution to $X_t = \phi X_{t-1} + Z_t$ must have the form $X_t = \lim_{n\to\infty} X_t^{(n)}$ (in L^2).
- The L^2 limit of $X_t^{(n)}$ exists and is a solution of $X_t = \phi X_{t-1} + Z_t$.
- This solution is stationary.
- That is: the L^2 limit of $X_t^{(n)}$ (which we shall now write as $\sum_{k\geq 0} \phi^k Z_{t-k}$) is the unique stationary solution of (3.4). We have $EX_t = 0$ and $\gamma_X(h) = \sigma^2 \left(\frac{\phi^{|h|}}{1-\phi^2}\right)$.

What happens if $|\phi| > 1$*?* Write

$$\frac{1}{\phi} X_t = X_{t-1} + \frac{1}{\phi} Z_t.$$

Then $X_{t-1} = \rho X_t + Z_t'$ where $\rho = \frac{1}{\phi}$ and $|\rho| < 1$ and $Z_t' = -\rho Z_t$. So we can write:

$$X_{t-1} = \rho(\rho X_{t+1} + Z_{t+1}') + Z_t' = Z_t' + \rho Z_{t+1}' + \rho^2 X_{t+1}.$$

Now we use the same argument:

$$\begin{aligned}X_{t-1} &= \lim_{n\to\infty} X_{t-1}^{(n)} \\ &= \lim_{n\to\infty} \sum_{k=0}^{n-1} \rho^k Z_{t+k}' \text{ (in } L^2\text{)}.\end{aligned}$$

This is the unique stationary solution but now $X_{t-1} = f(Z_t', Z_{t+1}', Z_{t+2}', \ldots)$ is not a *causal solution.*

Definition 3.1.3 (Causal processes)**.** Assume that $\{X_t\}$ is any time series that can be represented as

$$X_t = f(\ldots, Z_{t-1}, Z_t, Z_{t+1}, \ldots).$$

It is called *causal with respect to* $\{Z_t\}$ if

$$X_t = f(\ldots, Z_{t-1}, Z_t),$$

i.e. it depends only on the past.

Remark. Causality is a relationship between two processes $\{X_t\}$ and $\{Z_t\}$. When we say causal, we have to specify with respect to which process.

Proposition 3.1.1. *Consider the AR(1) model*

$$X_t = \phi X_{t-1} + Z_t \text{ with } Z_t \sim \mathrm{WN}(0, \sigma^2).$$

- *A stationary solution in* L^2 *exists* $\Leftrightarrow |\phi| \neq 1$*. This solution is unique.*
- *A stationary and causal solution in* L^2 *exists* $\Leftrightarrow |\phi| < 1$*. This solution is unique.*

Proof. The only thing that remains to show: there is no stationary solution if $|\phi| = 1$. (See Exercise 3.1 and Figure 3.3.) □

Remark. We can always find a non-stationary solution. Let $t > 0$.

$$X_t = \phi X_{t-1} + Z_t$$

set X_0 arbitrary.

$$X_t = \sum_{k=0}^{t-1} \phi^k Z_{t-k} + \phi^t X_0.$$

If $t < 0$ similar argument. This solution will usually not be stationary.

3.2 Stationary solutions for ARMA processes

The purpose of this section is to find conditions that guarantee the existence of a unique, stationary and causal solution of (3.2). We start with two preparatory propositions which allow us to

- give conditions for the convergence of $W_t = \sum_{k\in\mathbb{Z}} \psi_j X_{t-j}$ for some given process $\{X_t\}$;

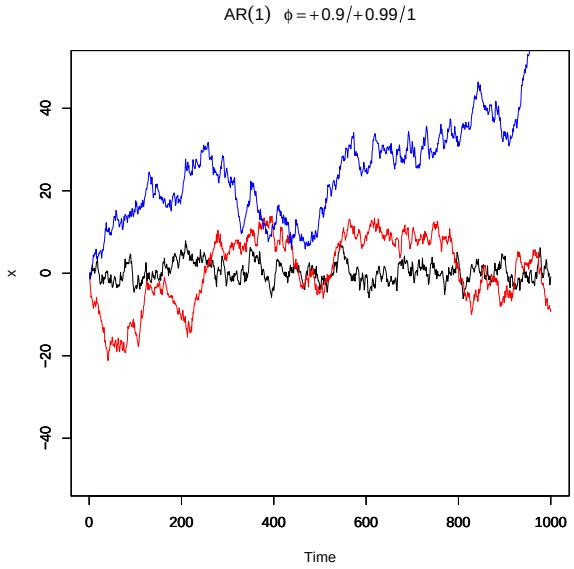


Figure 3.3: Simulated AR(1) models and random walk ($\phi = 1$): $\phi = 0.9$ (black), $\phi = 0.99$ (red) and $\phi = 1$ (blue).

- obtain a formula for the covariance structure of W_t as given above.

Proposition 3.2.1. *Let $\{X_t\}$ be a random process (not necessarily stationary). Further let $\{\psi_j\}$ be real coefficients such that $\sum_{j\in\mathbb{Z}} |\psi_j| < \infty$ and define $\psi(z) = \sum_{j\in\mathbb{Z}} \psi_j z^j$. (i) If the $\sup_{t\in\mathbb{Z}} E|X_t| < \infty$, then the series $\psi(B)X_t = \sum_{t\in\mathbb{Z}} \psi_j X_{t-j}$ is absolutely convergent with probability 1. (ii) If in addition $\sup_{t\in\mathbb{Z}} EX_t^2 < \infty$ is finite, then the series converges also in L^2. (iii) In the latter case the almost sure and the L^2 limit coincide.*

Proof. (i) We show that

$$W := \sum_{j\in\mathbb{Z}} |\psi_j||X_{t-j}| < \infty \quad \text{a.s.}$$

Since the above sum is positive, it will be finite if $EW < \infty$. By the monotone

convergence theorem

$$\begin{aligned} EW &= E \lim_{n\to\infty} \sum_{j=-n}^{n} |\psi_j||X_{t-j}| \\ &= \lim_{n\to\infty} E \sum_{j=-n}^{n} |\psi_j||X_{t-j}| \\ &\leq \sup_{t\in\mathbb{Z}} E|X_t| \sum_{j=-\infty}^{\infty} |\psi_j| \\ &< \infty. \end{aligned}$$

Then a.s. convergence follows.

(ii) We prove that $\{\sum_{j=-n}^{n} \psi_j X_{t-j} | n \geq 1\}$ is a Cauchy sequence. Let $m < n$. By the triangular inequality

$$\begin{aligned} \left\| \sum_{j=-n}^{n} \psi_j X_{t-j} - \sum_{j=-m}^{m} \psi_j X_{t-j} \right\| &= \left\| \sum_{|j|=m+1}^{n} \psi_j X_{t-j} \right\| \\ &\leq \sum_{|j|=m+1}^{n} |\psi_j| \|X_{t-j}\| \\ &\leq \sum_{|j|=m+1}^{\infty} |\psi_j| \sup_{t\in\mathbb{Z}} \left(EX_t^2\right)^{1/2}. \end{aligned}$$

Let $m \to \infty$ then $\sum_{|j|=m+1}^{\infty} |\psi_j| \to 0$. The completeness of L^2 guarantees the existence of the limit.

(iii) Let

$$\begin{aligned} S_t &= \lim_{L^2} S_t^{(n)} = \lim_{L^2} \sum_{j=-n}^{n} \psi_j X_{t-j}; \\ T_t &= \lim_{\text{a.s.}} S_t^{(n)}. \end{aligned}$$

We want to show that both limits are the same. By *Fatou's lemma*

$$\begin{aligned} E[S_t - T_t]^2 &= E \liminf_{n\to\infty} \left[S_t - S_t^{(n)}\right]^2 \\ &\leq \liminf_{n\to\infty} E \left[S_t - S_t^{(n)}\right]^2 = 0. \end{aligned}$$

□

Example 3.2.1. We found the AR(1) process $X_t = \phi X_{t-1} + Z_t$, $Z_t \sim \mathrm{WN}(0, \sigma^2)$ and $|\phi| < 1$ has the solution

$$X_t = \sum_{k \geq 0} \phi^k Z_{t-k}.$$

The series is absolutely convergent with probability 1 and it also converges in L^2 to the same limit.

More generally we have

Corollary 3.2.1. *Let $Z_t \sim \mathrm{WN}(0, \sigma^2)$. Then $\sum_{j \in \mathbb{Z}} \psi_j Z_{t-j}$ converges in $L^2 \Leftrightarrow \sum \psi_j^2 < \infty$. If $\sum |\psi_j| < \infty$ it converges a.s. and in L^2 to the same limit.*

Proposition 3.2.2. *Assume $\{Z_t\}$ is a stationary sequence with $EZ_t = \mu$ and ACF $\gamma_Z(h)$. Further let $\{\psi_j\}$ be absolutely summable $\left(\sum_{j \in \mathbb{Z}} |\psi_j| < \infty\right)$. Then $X_t = \psi(B) Z_t = \sum_{j \in \mathbb{Z}} \psi_j Z_{t-j}$ is again stationary with*

$$EX_t = \mu \sum_{j \in \mathbb{Z}} \psi_j,$$

$$\gamma_X(h) = \sum_{j \in \mathbb{Z}} \sum_{i \in \mathbb{Z}} \psi_j \psi_i \gamma_Z(h - i + j), \quad h \in \mathbb{Z}.$$

Proof. By Proposition 3.2.1, $\sum_{j \in \mathbb{Z}} \psi_j Z_{t-j}$ is absolutely convergent with probability 1 and it converges in L^2 to the same limit. Now

$$\begin{aligned} E[X_t] &= \langle X_t, \mathbf{1} \rangle \\ &= \lim_{n \to \infty} \left\langle \sum_{j=-n}^{n} \psi_j Z_{t-j}, \mathbf{1} \right\rangle \\ &= \lim_{n \to \infty} \sum_{j=-n}^{n} \psi_j \langle Z_{t-j}, \mathbf{1} \rangle \\ &= \mu \sum_{j \in \mathbb{Z}} \psi_j. \end{aligned}$$

Assume for simplicity that $\mu = 0$. Then by continuity of the inner product

$$\begin{aligned} \gamma_X(h) &= E[X_t X_{t+h}] \\ &= \langle X_t, X_{t+h} \rangle \\ &= \lim_{n \to \infty} \left\langle \sum_{j=-n}^{n} \psi_j Z_{t-j}, \sum_{i=-n}^{n} \psi_i Z_{t+h-i} \right\rangle \\ &= \lim_{n \to \infty} \sum_{j=-n}^{n} \sum_{i=-n}^{n} \psi_j \psi_i \underbrace{\langle Z_{t-j}, Z_{t+h-i} \rangle}_{\gamma_Z(h-i+j)}. \end{aligned}$$

□

Corollary 3.2.2. *If the assumptions in Proposition 3.2.2 hold with* $\{Z_t\} \sim \mathrm{WN}(0, \sigma^2)$, *then its statement holds with*

- $EX_t = 0$;
- $\gamma_X(h) = \sigma^2 \sum_{j\in\mathbb{Z}} \psi_j \psi_{j+|h|}$.

Example 3.2.2 (MA(q) processes)**.** Let

$$X_t = Z_t + \theta_1 Z_{t-1} + \cdots + \theta_q Z_{t-q}, \quad Z_t \sim \mathrm{WN}(0, \sigma^2).$$

Then defining $\theta_0 = 1$ and $\theta_j = 0$ for $j > q$ we get again

$$\gamma_X(h) = \begin{cases} \sigma^2 \sum_{j=0}^{q-|h|} \theta_j \theta_{j+|h|}, & \text{if } |h| \le q, \\ 0, & \text{else.} \end{cases}$$

Example 3.2.3 (AR(1) processes)**.**

$$\begin{aligned} X_t &= \phi X_{t-1} + Z_t \\ &= X_t = \sum_{j\ge 0} \phi^j Z_{t-j} \text{ (if } |\phi| < 1). \end{aligned}$$

Here

$$\begin{aligned} \gamma_X(h) &= \sigma^2 \sum_{j\ge 0} \phi^j \phi^{j+|h|} \\ &= \frac{\sigma^2 \phi^{|h|}}{1-\phi^2}. \end{aligned}$$

Corollary 3.2.3. *Assume* $\{Z_t\}$ *is stationary and* $\{\alpha_j\}$, $\{\beta_j\}$ *absolutely summable. Then the power series* $[\alpha\beta](z)$ *has again absolutely summable coefficients and defining* $\alpha(B)$ *and* $\beta(B)$ *the corresponding operators we get*

$$X_t := \alpha(B)[\beta(B)Z_t]$$

is stationary (converges a.s. and in L^2 *to same limit) and*

$$\alpha(B)[\beta(B)Z_t] = [\alpha\beta](B)Z_t = \beta(B)[\alpha(B)Z_t].$$

[Here $[\alpha\beta](z) = \alpha(z)\beta(z)$*.]*

Theorem 3.2.1 (Causal solution of ARMA(p,q))**.** *Assume* $\{Z_t\} \sim \mathrm{WN}(0,\sigma^2)$ *and let*

$$\phi(z) = 1 - \phi_1 z - \cdots - \phi_p z^p$$
$$\theta(z) = 1 + \theta_1 z + \cdots + \theta_q z^q$$

be two complex polynomials which have no common roots.

(i) If $\phi(z) \neq 0\ \forall z \in \bar{U}_1 := \{z \in \mathbb{C},\ |z| \leq 1\}$ *the ARMA equation*

$$\phi(B)X_t = \theta(B)Z_t \tag{3.5}$$

has a unique stationary and causal solution. It is given by $X_t = \psi(B)Z_t$ *where* $\psi(z) = \frac{\theta(z)}{\phi(z)} = \sum_{k\geq 0}\psi_k z^k,\ \ \forall z \in \bar{U}_1$ *with* $\sum_{k\in\mathbb{Z}}|\psi_k| < \infty$.

(ii) If a solution $X_t = \psi(B)Z_t$ *with* $\sum_{k\geq 1}|\psi_k| < \infty$ *exists, then*

$$\phi(z) \neq 0,\ \ \forall z \in \bar{U}_1. \tag{3.6}$$

Example 3.2.4 (AR(1) processes)**.** Assume

$$X_t - \phi X_{t-1} = Z_t.$$

Then $\phi(z) = 1 - \phi z$ and $\theta(z) = 1$.

- $1 - \phi z = 0 \ \Leftrightarrow z = \frac{1}{\phi}$ and $\frac{1}{|\phi|} > 1 \Leftrightarrow |\phi| < 1$.
- We have

$$\psi(z) = \frac{\theta(z)}{\phi(z)} = \frac{1}{1-\phi z} = \sum_{k\geq 0}\phi^k z^k;$$
$$\Rightarrow\ \psi(B) = \sum_{k\geq 0}\phi^k B^k;$$
$$\Rightarrow\ \psi(B)Z_t = \sum_{k\geq 0}\phi^k Z_{t-k}.$$

Proof of Theorem 3.2.1. We first show (i). Some standard results from analysis show that

$$\begin{aligned}
&\phi(z) \neq 0,\ \ \forall z \in \bar{U}_1 \\
\Rightarrow\ &\phi(z) \neq 0,\ \ \forall z \in U_{1+\epsilon} = \{z \in \mathbb{C}, |z| < 1+\epsilon\} \text{ for some small enough } \epsilon > 0. \\
\Rightarrow\ &\xi(z) := \frac{1}{\phi(z)} = \sum_{k\geq 0}\xi_k z^k,\ \ \forall z \in U_{1+\epsilon}.
\end{aligned}$$

Specifically, the convergence of $\xi(z)$ on $U_{1+\epsilon}$ implies that

$$
\begin{aligned}
& |\xi_k(1+\epsilon/2)^k| \xrightarrow{k\to\infty} 0; \\
\Rightarrow\ & |\xi_k| \left(1+\frac{\epsilon}{2}\right)^k \leq C \text{ for all } k \geq 0 \text{ and for some large enough constant } C; \\
\Rightarrow\ & |\xi_k| \leq C(1+\epsilon/2)^{-k} \\
\Rightarrow\ & \text{decays exponentially fast!} \\
\Rightarrow\ & \sum_{k\geq 0} |\xi_k| < \infty.
\end{aligned}
$$

Assume now first that $\{X_t\}$ *is a stationary solution of* (3.2). Using Corollaries 3.2.2 and 3.2.3 allows to apply $\xi(B)$ on both sides of the equation (3.5):

$$\xi(B)\left[\phi(B)X_t\right] = \xi(B)[\theta(B)Z_t].$$

The left hand side equals

$$[\xi\phi](B)X_t = X_t,$$

the right hand side equals

$$[\xi\theta](B)Z_t =: \psi(B)Z_t,$$

where $\psi(z) = \theta(z)/\phi(z)$, for $|z| \leq 1$ and $\sum_{j\geq 0}|\psi_j| < \infty$. This shows that any stationary solution must have the required form and is thus unique.
Now define $X_t = \psi(B)Z_t$. Again by Corollary 3.2.2 this process is stationary. Applying $\phi(B)$ on both sides of

$$X_t = \psi(B)Z_t,$$

shows by similar arguments as above that this is really a solution to (3.5) having the demanded form.

Now we show (ii). Assume X_t is a causal solution of the form

$$X_t = \psi(B)Z_t$$

with

$$\psi(z) = \sum_{k\geq 0} \psi_k z^k$$

and

$$\sum_{k\geq 0} |\psi_k| < \infty.$$

Then

$$\theta(B)Z_t = \phi(B)X_t = [\phi\psi](B)Z_t =: \eta(B)Z_t.$$

We apply for $j \geq 0$ $\langle \cdot, Z_{t-j} \rangle$ on both sides, and get by orthogonality of the Z_t that $\theta_j = \eta_j$ for all $j \geq 0$. Hence $\theta(z) = \phi(z)\psi(z)$ for all $z \in \bar{U}_1$.

Assume that for some $z_0 \in \bar{U}_1$ $\phi(z_0) = 0$. Then $\theta(z_0) = 0$, unless ψ has a pole in z_0. However, since the coefficients of ψ are absolutely summable, ψ cannot have a pole in $\bar{U}_1$. This means that ϕ and θ have common roots, which contradicts to our assumptions.

□

Remarks.

(i) If $\phi(z)$ and $\theta(z)$ have common roots NOT being on the unit circle, we can just cancel the common factors in both polynomials. If, however, the common roots are on the unit circle, then several stationary solutions might exist. The case of common roots is thus usually not considered.

(ii) If $\phi(z) = 0$ for some $|z| = 1$, then no stationary solution exists.

(iii) If $\phi(z)$ has zeros in U_1 and U_1^c, but no zeros on the unit circle, then a unique stationary solution of the form

$$X_t = \sum_{k \in \mathbb{Z}} \psi_k Z_{t-k}.$$

exists. Again $\psi(z) = \sum_{k \in \mathbb{Z}} \psi_k z^k = \theta(z)/\phi(z)$.

Example 3.2.5 (AR(2) processes)**.** Let

$$\phi(B)X_t = Z_t, \qquad \text{with } \phi(z) = 1 - \phi_1 z - \phi_2 z^2.$$

We look for

$$\{(\phi_1, \phi_2) \in \mathbb{R}^2 \,|\, 1 - \phi_1 z - \phi_2 z^2 \text{ has roots outside } \bar{U}_1\}.$$

We can assume that $\phi_2 \neq 0$, otherwise we have an AR(1) $\Longrightarrow$ already done. Then we can write

$$\phi(z) = \left(1 - \frac{1}{z_1}z\right)\left(1 - \frac{1}{z_2}z\right) \text{ with } z_1 \text{ and } z_2 \text{ are roots of } \phi.$$

Thus $\frac{1}{z_1} + \frac{1}{z_2} = \phi_1$ and $-\frac{1}{z_1}\frac{1}{z_2} = \phi_2$. We define

$$\hat{\phi}(z) = (1 - z_1 z)(1 - z_2 z) = 1 + \frac{\phi_1}{\phi_2}z - \frac{1}{\phi_2}z^2,$$

which has roots $\xi_1 = \frac{1}{z_1}$ and $\xi_2 = \frac{1}{z_2}$. So alternatively we can look for all $(\phi_1, \phi_2) \in \mathbb{R}^2$ such that $\hat{\phi}(z)$ has only roots inside unit circle. Then $\hat{\phi}(z) = 0 \Leftrightarrow z^2 - \phi_1 z - \phi_2 = 0$. The solutions to this equation are:

$$\xi_{1,2} = \frac{1}{2}\left(\phi_1 \pm \sqrt{\phi_1^2 + 4\phi_2}\right),$$

and we want ϕ_1, ϕ_2 such that

$$\left|\frac{1}{2}\left(\phi_1 \pm \sqrt{\phi_1^2 + 4\phi_2}\right)\right| < 1.$$

We consider 3 cases, see Figure 3.4.

I Case. $\phi_1^2 + 4\phi_2 = 0$ (red): Then the roots are $\xi_1 = \xi_2 = \frac{1}{2}\phi_1$ and we need

$$\begin{aligned} &\left|\frac{1}{2}\phi_1\right| < 1 \\ \Leftrightarrow\ &\frac{1}{4}\phi_1^2 < 1 \\ \Leftrightarrow\ &\phi_1^2 < 4 \\ \Leftrightarrow\ &-4\phi_2 < 4 \\ \Leftrightarrow\ &\phi_2 > -1. \end{aligned}$$

II Case. $\phi_1^2 + 4\phi_2 > 0$ (blue): Here we need

$$\begin{aligned} &-1 < \frac{1}{2}\left(\phi_1 \pm \sqrt{\phi_1^2 + 4\phi_2}\right) < 1 \\ \Leftrightarrow\ &-2 - \phi_1 < \pm\sqrt{\phi_1^2 + 4\phi_2} < 2 - \phi_1 \\ \Leftrightarrow\ &-2 - \phi_1 < -\sqrt{\phi_1^2 + 4\phi_2} \quad \text{and} \quad \sqrt{\phi_1^2 + 4\phi_2} < 2 - \phi_1 \\ \Leftrightarrow\ &-2 < \phi_1 < 2, \quad 4 + 4\phi_1 + \phi_1^2 > \phi_1^2 + 4\phi_2 \quad \text{and} \quad \phi_1^2 + 4\phi_2 < 4 - 4\phi_1 + \phi_1^2 \\ \Leftrightarrow\ &-2 < \phi_1 < 2, \quad \phi_2 - \phi_1 < 1 \quad \text{and} \quad \phi_2 + \phi_1 < 1 \\ \Leftrightarrow\ &\phi_2 > -1, \quad \phi_2 - \phi_1 < 1 \quad \text{and} \quad \phi_2 + \phi_1 < 1. \end{aligned}$$

III Case. $\phi_1^2 + 4\phi_2 < 0$ (yellow): Both roots are complex and $\xi_2 = \bar{\xi}_1$.

$$\begin{aligned} \xi_1 &= \frac{1}{2}\left(\phi_1 + \sqrt{\phi_1^2 + 4\phi_2}\right) \\ &= \frac{1}{2}\left(\phi_1 + i\sqrt{-\phi_1^2 - 4\phi_2}\right) \end{aligned}$$

Then $|\xi_1|^2 = \frac{1}{4}\phi_1^2 + \frac{1}{4}(-\phi_1^2 - 4\phi_2) = -\phi_2$, and we need $\phi_2 > -1$.

Summarizing this gives the so-called *Stralkowski region* (see Figure 3.4) for stationary causal solutions of AR(2) processes:

$$\text{Stationary causal solution exists if and only if } \begin{cases} \phi_2 - \phi_1 < 1; \\ \phi_2 + \phi_1 < 1; \\ |\phi_2| < 1. \end{cases} \tag{3.7}$$

Remark. For any ARMA$(2, q)$ process where the AR and MA polynomials are coprime, we get a unique, stationary and causal solution if (3.7) holds.

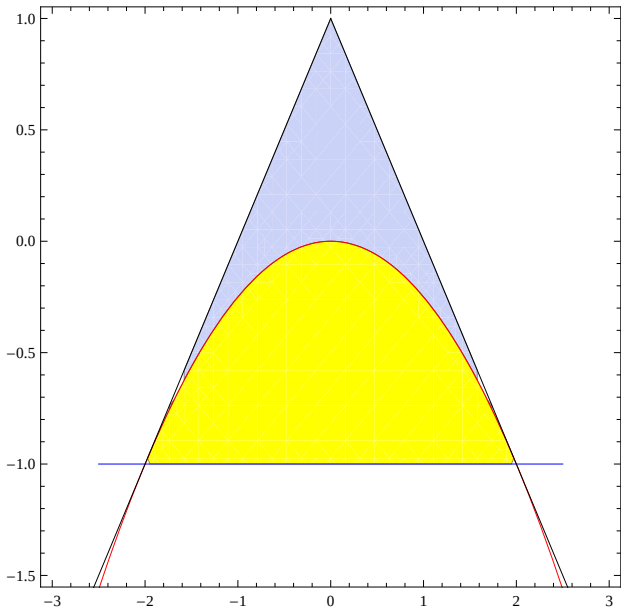


Figure 3.4: The Stralkowski region. Here the x-axis shows values of ϕ_1 and the y-axis values of ϕ_2. The inner points of the triangular give stationary solutions.

3.3 Computing the ACF of an ARMA process

Assume that we have the ARMA model

$$\phi(B)X_t = \theta(B)Z_t,$$

with $\phi(z)$ and $\theta(z)$ given (satisfying stationarity conditions). A central object for our investigations is the ACF. Then the important question is how do we practically compute the ACF?

Method 1. We first compute the coefficients of $\psi(z) = \frac{\theta(z)}{\phi(z)}$ and then use Corollary 3.2.2:

$$\gamma_X(h) = \sigma_Z^2 \sum_{k\geq 0} \psi_k \psi_{k+|h|} \quad \text{where} \quad \sigma_Z^2 = \text{Var}(Z_t).$$

In order to compute the coefficients we can proceed as follows:

Compare coefficients. We have $\psi(z)\phi(z) = \theta(z)$, or more explicitly

$$(\psi_0 + \psi_1 z + \psi_2 z^2 + \ldots)(1 - \phi_1 z - \phi_2 z^2 - \ldots - \phi_p z^p) = 1 + \theta_1 z + \theta_2 z^2 + \ldots + \theta_q z^q.$$

We get

1. $\psi_0 = 1$;
2. $(\psi_1 - \psi_0\phi_1)z = \theta_1 z \;\Rightarrow\; \psi_1 = \phi_1 + \theta_1$;
3. $(\psi_2 - \psi_1\phi_1 - \psi_0\phi_2)z^2 = \theta_2 z^2 \;\Rightarrow\; \psi_2 = (\phi_1 + \theta_1)\phi_1 + \phi_2 + \theta_2$;
4. ...

Example 3.3.1 (ARMA(1,1))**.** We have

$$X_t - \phi X_{t-1} = Z_t + \theta Z_{t-1} \quad \text{with} \quad Z_t \sim \text{WN}(0, \sigma^2).$$

I.e.

$$\psi(z) = \frac{1 + \theta z}{1 - \phi z} = (1 + \theta z)(1 + \phi z + \phi^2 z^2 + \cdots),$$

giving

$$\begin{aligned} \psi_0 &= 1 \\ \psi_k &= \phi^{k-1}(\phi + \theta), \quad k \geq 1, \end{aligned}$$

and hence

$$\begin{aligned} \gamma_X(h) &= \sigma^2 \sum_{k\geq 0} \psi_k \psi_{k+|h|} \\ &= \sigma^2 \left(\psi_0 \psi_{|h|} + \sum_{k\geq 1} \left(\phi^2\right)^{k-1} (\phi + \theta)^2 \phi^{|h|} \right) \\ &= \sigma^2 \left(\psi_{|h|} + \frac{(\phi + \theta)^2}{1 - \phi^2} \phi^{|h|} \right). \end{aligned}$$

Problem. Unlike in the previous example, getting explicit expressions for the ψ_k of higher order ARMA(p,q) processes is tricky.

Method 2. We assume to have a stationary ARMA(p,q) process $\{X_t\}$ of the form

$$X_t - \phi_1 X_{t-1} - \cdots - \phi_p X_{t-p} = Z_t + \theta_1 Z_{t-1} + \cdots + \theta_q Z_{t-q}.$$

We multiply both sides of this equation with X_{t-k}, $k \geq 0$, and take expectations. This gives with the convention that $\theta_0 = 1$

$$\gamma_X(k) - \phi_1\gamma_X(k-1) - \cdots - \phi_p\gamma_X(k-p) = \begin{cases} 0 \quad \text{if } k > q; \\ \theta_k EZ_{t-k}X_{t-k} + \theta_{k+1}EZ_{t-k-1}X_{t-k} + \ldots \theta_q EZ_{t-q}X_{t-k} \quad \text{if } k \in \{0,\ldots,q\}. \end{cases}$$

Here we used that $X_{t-k} = \sum_{j\geq 0}\psi_j Z_{t-k-j}$ is uncorrelated with any Z_s, when $s > t-k$. Note also that causal representation implies that

$$EZ_{t-k-j}X_{t-k} = \sigma^2\psi_j, \quad j \geq 1.$$

And hence

$$\gamma_X(k) - \phi_1\gamma_X(k-1) - \cdots - \phi_p\gamma_X(k-p) = \begin{cases} 0 \quad \text{if } k > q; \\ \sigma^2(\theta_k\psi_0 + \theta_{k+1}\psi_1 + \ldots \theta_q\psi_{q-k}) \quad \text{if } k \in \{0,\ldots,q\}. \end{cases}$$

We can thus use **Method 1** to explicitly compute $\psi_0(=1)$, $\psi_1,\ldots,\psi_q$ and then $\gamma_X(h)$ can be obtained solving simple linear equations and recursions.

Example 3.3.2 (ARMA(2,2))**.** Assume we have

$$X_t - \phi_1 X_{t-1} - \phi_2 X_{t-2} = Z_t + \theta_1 Z_{t-1} + \theta_2 Z_{t-2}, \quad Z_t \sim \text{WN}(0,\sigma^2).$$

We assume that the AR and MA polynomials are coprime and that ϕ_1 and ϕ_2 satisfy (3.7) (hence stationary and causal solution exists). Multiplying the ARMA(2,2) equation with with X_t, X_{t-1} and X_{t-2} and taking expectations gives the following equation system

$$\begin{aligned} \gamma_X(0) - \phi_1\gamma_X(1) - \phi_2\gamma_X(2) &= \sigma^2(1 + \theta_1\psi_1 + \theta_2\psi_2) \\ \gamma_X(1) - \phi_1\gamma_X(0) - \phi_2\gamma_X(1) &= \sigma^2(\theta_1 + \theta_2\psi_1) \\ \gamma_X(2) - \phi_1\gamma_X(1) - \phi_2\gamma_X(0) &= \sigma^2\theta_2. \end{aligned}$$

We saw before that $\psi_1 = \phi_1 + \theta_1$ and $\psi_2 = (\phi_1 + \theta_1)\phi_1 + \phi_2 + \theta_2$. So we can solve the equation system for $\gamma_X(0)$,$\gamma_X(1)$ and $\gamma_X(2)$. Since for $k \geq 3$ we have

$$\gamma_X(k) - \phi_1\gamma_X(k-1) - \phi_2\gamma_X(k-2) = 0,$$

or equivalently

$$\gamma_X(k) = \phi_1\gamma_X(k-1) + \phi_2\gamma_X(k-2),$$

we can recursively calculate the ACF at any lag.

Example 3.3.3 (AR(2))**.** Using the previous approach and multiplying with $X_{t-1}, X_{t-2}, \ldots$ we obtain in case of an AR(2) model

$$\gamma_X(h) - \phi_1\gamma_X(h-1) - \phi_2\gamma_X(h-2) = 0, \quad h \geq 1,$$

and by dividing by $\gamma_X(0)$

$$\rho_X(h) - \phi_1\rho_X(h-1) - \phi_2\rho_X(h-2) = 0, \quad h \geq 1.$$

The initial condition $\rho_X(0) = 1$ gives $\rho_X(1) = \frac{\phi_1}{1-\phi_2}$. To get an explicit form for $\rho_X(h)$ we have to solve a *homogeneous difference equation* of the form

$$u_n - \phi_1 u_{n-1} - \phi_2 u_{n-2} = 0, \quad n \geq 2, \tag{3.8}$$

with the initial conditions $u_0 = 1$ and $u_1 = \frac{\phi_1}{1-\phi_2}$. One can show (Exercise 3.10) that the solution of (3.8) has the form

$$u_n = c_1 z_1^{-n} + c_2 z_2^{-n}(1 + (n-1)I\{z_1 = z_2\}), \quad n \geq 0,$$

where z_1 and z_2 are the two roots of the AR polynomial $1 - \phi_1 z - \phi_2 z^2$. The constants c_1 and c_2 are determined by the initial conditions via

$$\begin{aligned} u_0 &= c_1 + c_2 I\{z_1 \neq z_2\} \\ u_1 &= c_1 z_1^{-1} + c_2 z_2^{-1}. \end{aligned}$$

We end up with

(i) If z_1 and z_2 are real and distinct, then

$$\rho_X(h) = c_1 z_1^{-h} + c_2 z_2^{-h}.$$

(ii) If $z_1 = z_2$ (then the roots are real) we have $c_1 = u_0 = 1$ and

$$\rho_X(h) = z_1^{-h}(1 + c_2 h).$$

(iii) When $z_1 = \bar{z}_2$ are complex, then $c_2 = \bar{c}_1$ (because $\rho_X(h)$ is real), and

$$\rho_X(h) = c_1 z_1^{-h} + \bar{c}_1 \bar{z}_1^{-h}.$$

Note that we require $|z_1| > 1$ and $|z_2| > 1$ in order to obtain a stationary and causal solution. So in all cases we have that $\rho_X(h) \to 0$ exponentially fast. The closer z_1 or z_2 is to a unit root, the slower the speed of decay.

In the complex case we can write $z_1 = |z_1|e^{i\theta}$ and $\bar{z}_1 = |z_1|e^{-i\theta}$. Hence for $c_1 = a + ib$ we have

$$\begin{aligned}\rho_X(h) &= |z_1|^{-h}(c_1 e^{-ih\theta} + \bar{c}_1 e^{ih\theta}) \\ &= a|z_1|^{-h}(e^{-ih\theta} + e^{ih\theta}) + ib|z_1|^{-h}(e^{-ih\theta} - e^{ih\theta}) \\ &= 2|z_1|^{-h}\left(a\cos(h\theta) + b\sin(h\theta)\right) \\ &= 2\alpha|z_1|^{-h}\cos(h\theta + \beta),\end{aligned}$$

where α and β are given by the relation $\alpha\cos(\beta) = a$ and $-\alpha\sin(\beta) = b$. This shows that $\rho_X(h)$ dampens again to zero exponentially fast if $h \to \infty$ but it does so in a sinusoidal fashion (see Figure 3.5).

3.4 The partial autocorrelation function

We will develop in Chapter 5 methods to estimate the unknown parameters of an ARMA(p, q) process from a data set. However, in statistics it is often not advisable to do estimation without some preliminary empirical data analysis. E.g. it would be good to get some idea which order p and q should be choosen, or if we do some numerical maximization using a likelihood method, a good starting value can be very helpful. A plot of the data and it's ACF gives often very good indiction of the order (p, q) and the parameters.

Example 3.4.1 (Sunspot numbers and AR(2))**.** In Figure 3.6 we see the sunspot numbers along with its empirical ACF, compared to a simulated AR(2) process. The fact that the empirical ACF shows a sinusoidal behavior and the data a periodic pattern could indicate (see Example 3.3.3) that an AR(2) process with complex roots might be an appropriate model. For the simulated AR(2) process in Figure 3.6 we chose parameters $\phi_1 = 1.88$ and $\phi_2 = 0.95$ (without using any estimation!) and indeed, the ACF and also the sample paths look quite similar (with exception of the mean which should be removed from the sunspot numbers). If we wish to fit a model using the likelihood approach developed in Chapter 5, we could use $\phi_1 = 1.88$ and $\phi_2 = 0.95$ as starting values for a numerical maximization.

In the last example, we can of course not be sure that some AR(3), AR(4) more general ARMA(p, q) wouldn't do a much better job. A useful method to determine the order of an AR process is to use the *partial autocorrelation function*, short PACF.

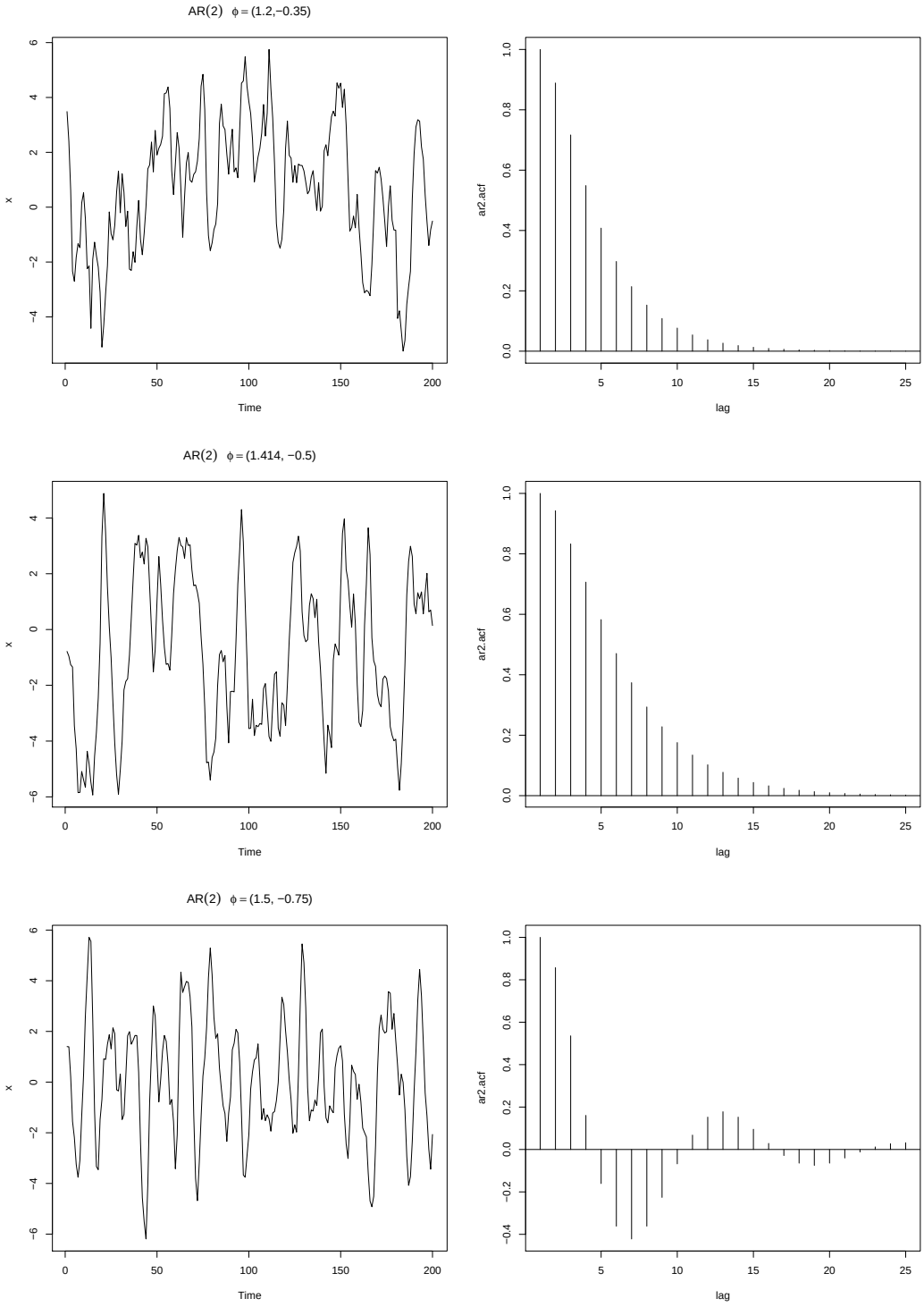


Figure 3.5: Three different AR(2) processes with 2 different real roots (top), two coinciding roots (middle) and two complex roots (bottom). [R-code].

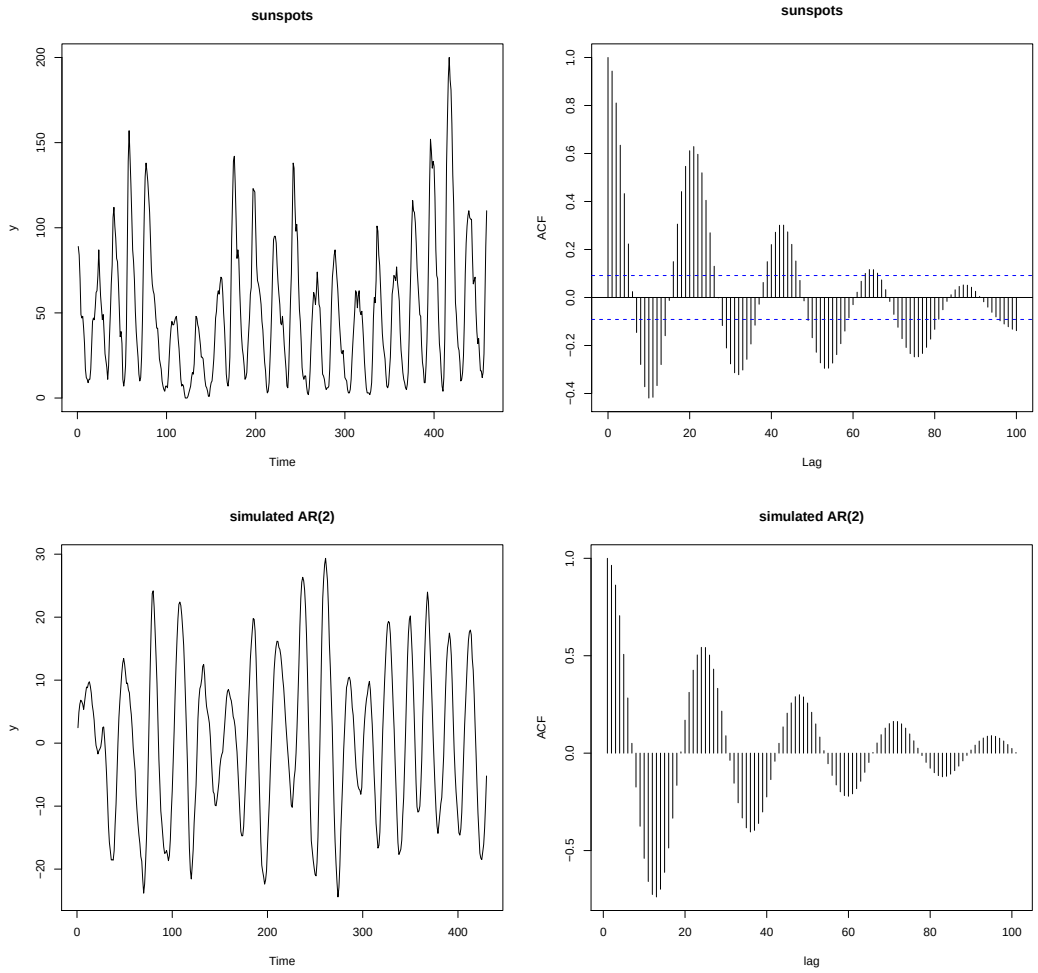


Figure 3.6: Sunspot numbers and empirical ACF (top) and simulated AR(2) model with ACF (bottom). For the simulated process we used $\phi_1 = 1.88$ and $\phi_2 = 0.95$.

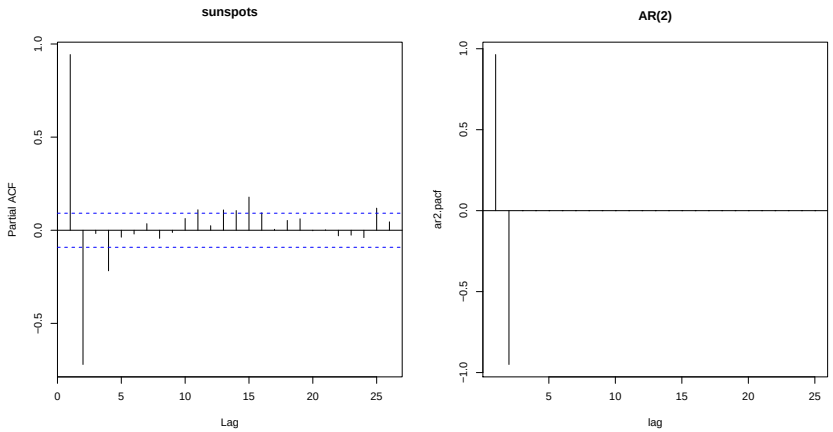


Figure 3.7: Empirical PACF of sunspot numbers (left) and PACF of simulated AR(2) model (right). [R-code]

Definition 3.4.1 (Partial autocorrelation function, PACF)**.** Let $\{X_t\}$ be a stationary process. Then the PACF α_X is defined as

$$\begin{aligned}\alpha_X(1) &= \mathrm{Corr}(X_2, X_1) = \rho_X(1);\\ \alpha_X(k) &= \mathrm{Corr}\left(X_{k+1} - P_{\overline{\mathrm{sp}}\{X_2,\ldots,X_k\}}(X_{k+1}), X_1 - P_{\overline{\mathrm{sp}}\{X_2,\ldots,X_k\}}(X_1)\right), \quad k \geq 2.\end{aligned}$$

Example 3.4.2 (PACF for AR(p) processes)**.** Let $\{X_t\}$ be a causal AR(p) process (zero mean). If $k > p$, then by using the projection theorem, one can easily show that

$$P_{\overline{\mathrm{sp}}\{X_2,\ldots,X_k\}}(X_{k+1}) = \phi_1 X_k + \cdots + \phi_p X_{k-p+1},$$

and thus

$$X_{k+1} - P_{\overline{\mathrm{sp}}\{X_2,\ldots,X_k\}}(X_{k+1}) = X_{k+1} - (\phi_1 X_k + \cdots + \phi_p X_{k-p+1}) = Z_{k+1}.$$

By causality $Z_{k+1} \perp \overline{\mathrm{sp}}\{X_2, \ldots, X_k\}$ and $Z_{k+1} \perp X_1$, and thus $\alpha_X(h) = 0$ for $h > p$.

The example shows that we can determine the order of an AR(p) process using $\hat{\alpha}_X$ in a similar way as we can determine the order of an MA(q) process using $\hat{\rho}_X$. Figure 3.7 indicates that we should maybe use an AR(3) process to model the sunspot numbers, though the AR(2) seems reasonable.

In Section 4.4 we will give a practical method to compute and estimate $\alpha_X(h)$.

3.5 Identification of ARMA models

In Section 3.3 we learned how to obtain the ACF γ_Y of a certain ARMA(p,q) process $\{Y_t\}$. Assume now that for some other process $\{X_t\}$ we know that $\gamma_X(h) = \gamma_Y(h)$ for all h. What can we say about $\{X_t\}$?

Proposition 3.5.1. *Assume that $\{Y_t\}$ is an is an ARMA(p,q) process with ACF γ_Y. Let $\{X_t\}$ be stationary with $EX_t = 0$ and with ACF γ_X. If $\gamma_Y = \gamma_X$, then it follows that $\{X_t\}$ is also an ARMA(p,q) process.*

The proof is based on

Proposition 3.5.2. *If $\{X_t\}$ is stationary with $EX_t = 0$ and if $\gamma_X(h) = 0 \ \ \forall |h| > q$, then $\{X_t\}$ is an MA(q) process.*

Proof of Proposition 3.5.1. The proof uses Proposition 3.5.2 which we show below. By our assumption $W_t := \phi(B)Y_t = \theta(B)Z_t$ with $Z_t \sim \mathrm{WN}(0,\sigma^2)$. By definition $\{W_t\}$ is an MA(q) process. Define $V_t := \phi(B)X_t$. By Proposition 3.2.2 $\{V_t\}$ has the same ACF as $\{W_t\}$. By assumption $EX_t = 0$, it follows that $EV_t = 0$. Since $\{W_t\}$ is an MA(q) process, it follows via Proposition 3.5.2 that $\{V_t\}$ is an MA(q) process. Hence

$$V_t = \phi(B)X_t = Z'_t + \theta'_1 Z'_{t-1} + \cdots + \theta'_q Z'_{t-q},$$

where θ'_i are real coefficients and $\{Z'_t\}$ is some white noise. □

Proof of Proposition 3.5.2. We need to show that there exists $\{Z_t\} \sim \mathrm{WN}(0,\sigma^2)$ and coefficients $\theta_1, \ldots, \theta_q$ such that

$$X_t = Z_t + \theta_1 Z_{t-1} + \ldots + \theta_q Z_{t-q}.$$

Define $S_t = \overline{\mathrm{sp}}\{X_t, X_{t-1}, \ldots\}$ and $S_{t,n} = \overline{\mathrm{sp}}\{X_t, X_{t-1}, \ldots, X_{t-n}\}$. We use below that $P_{S_t}(X) = \lim_{n\to\infty} P_{S_{t,n}}(X)$.

Now define

$$Z_t = X_t - P_{S_{t-1}}(X_t) = \lim_{n\to\infty} Z_{t,n} = \lim_{n\to\infty} (X_t - P_{S_{t-1,n}}(X_t)).$$

First we note that $EX_t = 0$ implies $EZ_{t,n} = 0$. Also it is clear that $\{Z_{t,n},\, t \in \mathbb{Z}\}$ is stationary. From this we deduce:

1: $EZ_t = \lim_n EZ_{t,n} = 0$.

2: $EZ_t^2 = \lim EZ_{t,n}^2 = \lim EZ_{0,n}^2 = EZ_0^2$.

3: $\mathrm{Cov}(Z_t, Z_s) = \langle Z_t, Z_s\rangle = \langle X_t - P_{S_{t-1}}(X_t), X_s - P_{S_{s-1}}(X_s)\rangle = 0$ for $t \neq s$.

4: $X_t = P_{S_t}(X_t) = P_{\overline{\text{sp}}\{Z_t,\ldots,Z_{t-q},X_{t-q-1},X_{t-q-2},\ldots\}}(X_t) = \lim_n P_{\overline{\text{sp}}\{Z_t,\ldots,Z_{t-q},X_{t-q-1},\ldots,X_{t-q-n}\}}(X_t)$.

Now note that for $n > q$ and $\{X_t, Z_t, \ldots, Z_{t-q}\} \perp \{X_{t-q-1}, X_{t-q-2}, \ldots\}$

$$\begin{aligned} P_{\overline{\text{sp}}\{Z_t,\ldots,Z_{t-q},X_{t-q-1},\ldots,X_{t-q-n}\}}(X_t) &= \sum_{k=0}^{q} \frac{\langle Z_{t-k}, X_t\rangle}{\|Z_{t-k}\|^2} Z_{t-k} + P_{\overline{\text{sp}}\{X_{t-q-1},\ldots,X_{t-q-n}\}}(X_t) \\ &= \sum_{k=0}^{q} \frac{\langle Z_{0-k}, X_0\rangle}{\|Z_0\|^2} Z_{t-k}. \end{aligned}$$

In the last step we used

$$\begin{aligned} \langle Z_{t-k}, X_t\rangle &= \lim_m \langle Z_{t-k,m}, X_t\rangle \\ &= \lim_m \langle X_{t-k} - P_{S_{t-k-1,m}}(X_{t-k}), X_t\rangle, \end{aligned}$$

which by stationarity of the process $\{X_t\}$ is equal to $\lim_m \langle X_{0-k} - P_{S_{0-k-1,m}}(X_{0-k}), X_0\rangle$. □

We still need the following lemma

Lemma 3.5.1. *Let $X \in L^2$ and S_t as before. Then $P_{S_{t,n}}(X) \overset{n\to\infty}{\longrightarrow} P_{S_t}(X)$.*

Proof. When $Y \in S_t$ it means that there are coefficients $a_k^{(n)}$, such that

$$Y = \lim_n \sum_{k=0}^{n} a_k^{(n)} X_{t-k}.$$

Obviously $Y = P_{S_t}(X) \in S_t$. And thus by the projection theorem and Exercise 2.8 we get

$$\|P_{S_t}(X) - P_{S_{t,n}}(X)\| = \|P_{S_t}(X) - P_{S_{t,n}}(P_{S_t}(X))\| \leq \|P_{S_t}(X) - \sum_{k=0}^{n} a_k^{(n)} X_{t-k}\| \to 0.$$

□

3.6 Invertibility

Consider the MA(1) processes $X_t = Z_t + \theta Z_{t-1}$ with $\{Z_t\}$ are i.i.d. $N(0, \sigma^2)$ and $Y_t = U_t + \frac{1}{\theta} U_{t-1}$ where $\{U_t\}$ are i.i.d. $N(0, \theta^2\sigma^2)$. Both are apparently zero-mean Gaussian processes and a simple calculation shows that $\gamma_X(h) = \gamma_Y(h)$. Hence both processes have exactly the same marginal distributions! Since we don't observe the innovations of the process, how can we decide from which process data have been sampled? How should we estimate parameters if they are not unique? In such a situation we will go for the solution that is *invertible*.

Definition 3.6.1 (Invertibility for ARMA(p,q))**.** An ARMA(p,q) process is called invertible if there are coefficients $\{\pi_k,\, k \geq 0\}$ such that

$$Z_t = \sum_{k\geq 0} \pi_k X_{t-k} \quad \text{with} \quad \sum_{k\geq 0} |\pi_k| < \infty.$$

In other words: We can always recover the innovations from the observations.

Theorem 3.6.1. *Let $\{X_t\}$ be an ARMA(p,q) given by $\phi(B)X_t = \theta(B)Z_t$ where ϕ and θ are coprime complex polynomials. Then $\{X_t\}$ is invertible, iff $\theta(z) \neq 0\ \forall z \in \bar{U}_1$. The coefficients π_j are determined by:*

$$\pi(z) = \sum_{j\geq 0} \pi_j z^j = \frac{\phi(z)}{\theta(z)}, \quad \forall z \in \bar{U}_1.$$

Proof. The proof is quite similar to the Proof of Theorem 3.2.1. □

Example 3.6.1 (AR(p) processes)**.** Assume

$$X_t = \phi_1 X_{t-1} + \ldots + \phi_p X_{t-p} + Z_t.$$

Such a process is trivially invertible.

Example 3.6.2 (MA(1) processes)**.** Assume

$$X_t = Z_t + \theta Z_{t-1}$$

Then

$$\theta(z) = 1 + \theta z;$$
$$\phi(z) = 1.$$

Thus the process is invertible iff $|\theta| < 1$ and we have

$$\pi(z) = \frac{1}{1+\theta z} = \sum_{k\geq 0} (-\theta)^k z^k.$$

Hence

$$Z_t = \sum_{k\geq 0} (-\theta)^k X_{t-k}.$$

In the introductory example we would thus choose the process $\{X_t\}$ if $|\theta| < 1$ and $\{Y_t\}$ if $\theta > 1$.

Remark. If we extend the definition of invertibility to

$$Z_t \in \overline{\text{sp}}\,\{X_t, X_{t-1}, \ldots\},$$

Then

$$\phi(z) \neq 0 \quad \text{for} \quad |z| = 1$$

is no longer necessary. See Exercise 3.5.

3.7 Exercises

Ex 3.1 Let $\{Z_t | t \in \mathbb{Z}\} \sim \mathrm{WN}(0, \sigma^2)$ and define the AR(1) process

$$X_t = \phi X_{t-1} + Z_t.$$

Show:

- If $|\phi| < 1$ a unique stationary causal solution exists.
- If $|\phi| > 1$ a unique stationary solution exists, but it is not causal.
- If $|\phi| = 1$ no stationary solution exists.

Ex 3.2 Assume that $\{Z_t | t \in \mathbb{Z}\}$ is iid and $|\phi| < 1$. Try to find an as good as possible condition on the moments of Z_t [e.g. only the existence of logarithmic moments $E \log^m(1 + |Z|)$ for some large enough m] which guarantees that the AR(1) process

$$\sum_{i \geq 0} \phi^i Z_{t-i}$$

converges absolutely with probability 1. Try at least to improve the condition $E|Z_0| < \infty$.

Ex 3.3 Let $\{X_t\}$ be an ARMA(p, q) process with ACF $\gamma_X(\cdot)$. Show that there are constants $C = C(\{X_t\}) > 0$ and $0 < \rho = \rho(\{X_t\}) < 1$ such that

$$|\gamma_X(h)| \leq C\rho^{|h|} \quad \forall h \in \mathbb{Z}.$$

Consequently $\sum_{h \in \mathbb{Z}} |\gamma_X(h)| < \infty$.

Ex 3.4 Which of the following processes are causal/invertible?
(a) $X_t + 0.2X_{t-1} - 0.48X_{t-2} = Z_t$,
(b) $X_t + 1.9X_{t-1} + 0.88X_{t-2} = Z_t + 0.2Z_{t-1} + 0.7Z_{t-2}$,
(c) $X_t + 1.6X_{t-1} = Z_t - 0.4Z_{t-1} + 0.04Z_{t-2}$.

Ex 3.5 Show that if the definition of invertibility is extended to the following: "The process $\{X_t\}$ is invertible if $Z_t \in \overline{\mathrm{sp}}\{X_i,\, i \leq t\}$" then the process

$$X_t = Z_t - Z_{t-1}$$

is invertible.
Hint: show that $\sum_{j=0}^{n}(1 - j/n)X_{t-j}$ *converges in mean square to* Z_t.

Ex 3.6 In view of Ex. 3.5 do you think that causality of an AR(1) process given by $X_t = X_{t-1} + Z_t$ could be established, if we only require that $X_t \in \overline{\text{sp}}\{Z_i,\, i \le t\}$?

Ex 3.7 Find the coefficients ψ_k, $k \ge 0$, in the representation

$$X_t = \sum_{k \ge 0} \psi_k Z_{t-k}$$

of the ARMA(2,1) process

$$X_t - 0.5X_{t-1} + 0.04X_{t-2} = Z_t + 0.25Z_{t-1} \quad \{Z_t\} \sim \text{WN}(0, \sigma^2).$$

Ex 3.8 Let

$$Z_t = (1 - 0.5B)(1 - 0.2B)(1 - 0.8B)X_t \quad \{Z_t\} \sim \text{WN}(0, 1).$$

Find the ACF of $\{X_t\}$.

Ex 3.9 Let $\{X_t\}$ be the ARMA(1,1) process,

$$X_t - \phi X_{t-1} = Z_t + \theta Z_{t-1}, \quad \{Z_t\} \sim \text{WN}(0, 1),$$

where $|\phi| < 1$ and $|\theta| < 1$. Determine the coefficients in the representation

$$X_t = \sum_{k \ge 0} \psi_k Z_{t-k}$$

and obtain an explicit formula expression for $\gamma_X(h)$, $h \in \mathbb{Z}$. For some values of ϕ, θ, h and n compare (using a computer) with the rough approximation for $\gamma(h)$

$$\kappa_n(h) = \sum_{k=0}^{n} \psi_k \psi_{k+|h|}.$$

Ex 3.10 In Section 3.3 we gave the solution of a homogeneous difference equation of the form $u_n - \phi u_{n-1} - \phi_2 u_{n-2} = 0$, with some given starting conditions u_0 and u_1. Verify correctness of the solution.

Chapter 4

Forecasting

4.1 Prediction equations

Assume that $\{X_t,\ t \in \mathbb{Z}\}$ is a stationary time series with $EX_t = \mu$ and ACF γ_X. We would like to forecast X_{n+h}, $h \geq 1$, given the sample $X_1, \ldots, X_n$:

$$\hat{X}_{n+h} = f_h(X_1, \ldots, X_n).$$

The question is, which f_h is "optimal"? If we measure the quality of the forecast $\hat{X}_{n+h}$ in terms of the means squared prediction error,

$$E(\hat{X}_{n+h} - X_{n+h})^2 = \|\hat{X}_{n+h} - X_{n+h}\|^2,$$

then by the projection theorem

$$\hat{X}_{n+h} = P_S(X_{n+h}),$$

where $S = \{Z \in L^2 | Z$ is a measurable function of $X_1, \ldots, X_n\}$. One can show that S defines a closed subspace of $L^2 = L^2(\Omega, \mathcal{A}, P)$ (see Exercises 4.1, 4.2, 4.3). $P_S(X_{n+h})$ is the so-called *conditional expectation of* X_{n+h} *given* $X_1, \ldots, X_n$.

The problem is that in general the explicit computation of $P_S(X_{n+h})$ *is hopelessly difficult.*

An alternative approach is to project onto a smaller subspace, which is easier to handle. More specifically we define

$$\hat{X}_{n+h} := P_{\overline{\text{sp}}\{1, X_1, \ldots, X_n\}}(X_{n+h}).$$

Then $\hat{X}_{n+h}$ minimizes

$$E\left(X_{n+h} - [\alpha_0 + \alpha_1 X_1 + \cdots + \alpha_n X_n]\right)^2,$$

over all $\alpha_0, \ldots, \alpha_n \in \mathbb{R}$. For this $\hat{X}_{n+h}$ we can derive explicit expressions.

We start by defining

$$Y_t := X_t - \mu.$$

Then $\{Y_t\}$ is stationary with $EY_t = 0$ and $\gamma_Y = \gamma_X$. It can be easily shown that (Exercise 4.4)

$$\hat{X}_{n+h} = P_{\overline{\text{sp}}\{1, X_1, \ldots, X_n\}}(X_{n+h}) = \mu + \hat{Y}_{n+h} = \mu + P_{\overline{\text{sp}}\{Y_1, \ldots, Y_n\}}(Y_{n+h}).$$

Hence we can assume in the sequel that the stationary sequence $\{X_t\}$ has a zero mean.

We focus now on $h = 1$, the derivation for general h are completely analogue. (See Exercise 4.8). We write $\gamma_X = \gamma$ and

$$\hat{X}_{n+1} = \phi_{n1} X_n + \phi_{n2} X_{n-1} + \cdots + \phi_{nn} X_1 = \sum_{j=1}^{n} \phi_{nj} X_{n+1-j}.$$

The goal is to find ϕ_{nj}, $1 \le j \le n$. By the projection theorem $\hat{X}_{n+1}$ is characterized by

$$\langle X_{n+1} - \hat{X}_{n+1}, Y \rangle = 0, \quad \forall \quad Y \in \overline{\text{sp}}\{X_1, \ldots, X_n\}.$$

It is clear that the latter is equivalent to

$$\begin{aligned} &\langle X_{n+1} - \hat{X}_{n+1}, X_{n+1-k} \rangle = 0, \quad \forall\, k \in \{1, \ldots, n\}; \\ \Leftrightarrow \quad &\langle X_{n+1}, X_{n+1-k} \rangle = \langle \hat{X}_{n+1}, X_{n+1-k} \rangle \\ &\qquad = \sum_{j=1}^{n} \phi_{nj} \langle X_{n+1-j}, X_{n+1-k} \rangle, \quad \forall\, k \in \{1, \ldots, n\}. \end{aligned}$$

Using $\langle X_n, X_{n+k} \rangle = \gamma(k)$ this is

$$\gamma(k) = \sum_{j=1}^{n} \phi_{nj} \gamma(k-j), \quad \forall\, k \in \{1, \ldots, n\},$$

and in matrix form

$$\underbrace{\begin{pmatrix} \gamma(1) \\ \gamma(2) \\ \vdots \\ \gamma(n) \end{pmatrix}}_{\boldsymbol{\gamma}_n} = \underbrace{\begin{bmatrix} \gamma(0) & \gamma(1) & \ldots & \gamma(n-1) \\ \gamma(1) & \gamma(0) & \ldots & \gamma(n-2) \\ \vdots & \vdots & \ddots & \vdots \\ \gamma(n-1) & \gamma(n) & \ldots & \gamma(0) \end{bmatrix}}_{\Gamma_n} \underbrace{\begin{pmatrix} \phi_{n1} \\ \phi_{n2} \\ \vdots \\ \phi_{nn} \end{pmatrix}}_{\boldsymbol{\phi}_n}.$$

Note that

$$\Gamma_n = \mathrm{Cov}(\mathbf{X}_n), \quad \mathbf{X}_n = (X_1, \ldots, X_n)^T.$$

Now if $|\Gamma_n| > 0$ then

$$\boldsymbol{\phi}_n = \Gamma_n^{-1}\boldsymbol{\gamma}_n.$$

If $|\Gamma_n| = 0$ the equation system will have no unique solution, BUT all solutions will result in the same $\hat{X}_{n+1}$! This follows by uniqueness of the projection. (See Theorem 2.3.1.)
If Γ_n is non-singular then the *mean squared prediction error*

$$\begin{aligned}
\mathrm{Var}(X_{n+1} - \hat{X}_{n+1}) &= \|X_{n+1} - \hat{X}_{n+1}\|^2 \\
&= \langle X_{n+1} - \hat{X}_{n+1}, X_{n+1}\rangle - \underbrace{\langle X_{n+1} - \hat{X}_{n+1}, \hat{X}_{n+1}\rangle}_{=0} \\
&= \|X_{n+1}\|^2 - \langle \hat{X}_{n+1}, X_{n+1}\rangle \\
&= \gamma(0) - \sum_{i=1}^{n} \phi_{ni} \underbrace{\langle X_{n+1-i}, X_{n+1}\rangle}_{\gamma(i)} \\
&= \gamma(0) - \boldsymbol{\phi}_n^T \boldsymbol{\gamma}_n \\
&= \gamma(0) - \boldsymbol{\gamma}_n^T \Gamma_n^{-1} \boldsymbol{\gamma}_n =: \nu_n.
\end{aligned}$$

Here is a simple criterion for Γ_n non-singular:

Lemma 4.1.1. *Let $\{X_t\}$ be stationary and let γ_X be its ACF. If $\gamma_X(0) > 0$ and $\gamma_X(h) \to 0$ for $h \to \infty$ [holds true for all stationary and causal ARMA(p,q)] then Γ_n is non singular (i.e. $|\Gamma_n| > 0$) for all $n \geq 1$.*

Proof. By assumption

$$|\Gamma_1| = \gamma(0) > 0.$$

We proceed by contradiction: assume there is an $1 \leq r < \infty$ being the largest number such that Γ_r is non-singular. That means $|\Gamma_k| > 0$ if $k \leq r$ and $|\Gamma_{r+1}| = 0$. This implies (see Exercise 4.6) that $X_1, \ldots, X_{r+1}$ are linearly dependent and hence

$$X_{r+1} \in \overline{\mathrm{sp}}\{X_1, \ldots X_r\}. \tag{4.1}$$

By stationarity the same argument applies for $X_2, \ldots, X_{r+2}$ and thus

$$X_{r+2} \in \overline{\mathrm{sp}}\{X_2, \ldots X_{r+1}\} \overset{\text{by (4.1)}}{\Rightarrow} X_{r+2} \in \overline{\mathrm{sp}}\{X_1, \ldots X_r\}.$$

Thus by induction

$$X_n \in \overline{\mathrm{sp}}\{X_1, \ldots X_r\}, \ \forall n \geq 1,$$

or equivalently

$$X_n = a_1^{(n)} X_1 + \ldots + a_r^{(n)} X_r =: \left(\mathbf{a}^{(n)}\right)^T \mathbf{X}_r, \quad n \geq 1.$$

Therefore we get

$$\begin{aligned}\gamma(0) &= \mathrm{Var}(X_n) \\ &= \mathrm{Var}\left(\left(\mathbf{a}^{(n)}\right)^T \mathbf{X}_r\right) \\ &= \left(\mathbf{a}^{(n)}\right)^T \Gamma_r \, \mathbf{a}^{(n)}.\end{aligned}$$

Letting λ_r be the smallest eigenvalue of Γ_r (note that $\lambda_r > 0$ as we assume $|\Gamma_r| > 0$!), then using the spectral theorem, one can show that (Exercise)

$$\left(\mathbf{a}^{(n)}\right)^T \Gamma_r \, \mathbf{a}^{(n)} \geq \lambda_r \left(\mathbf{a}^{(n)}\right)^T \mathbf{a}^{(n)} \geq \lambda_r \left(a_j^{(n)}\right)^2, \ \forall j \in \{1, \ldots, r\}.$$

This implies that

$$\left|a_j^{(n)}\right| \leq \sqrt{\frac{\gamma(0)}{\lambda_r}}, \ \forall j \in \{1, \ldots, r\}, \ \forall n \geq 1. \tag{4.2}$$

Now using (4.2) we get that

$$\begin{aligned}\gamma(0) &= \|X_n\|^2 \\ &= \langle X_n, X_n \rangle \\ &= \left|\left\langle X_n, a_1^{(n)} X_1 + \cdots + a_r^{(n)} X_r \right\rangle\right| \\ &\leq \left|a_1^{(n)}\right| \cdot |\langle X_n, X_1 \rangle| + \cdots + \left|a_r^{(n)}\right| \cdot |\langle X_n, X_r \rangle| \\ &\leq \sqrt{\frac{\gamma(0)}{\lambda_r}} \left(|\gamma(n-1)| + \cdots + |\gamma(n-r)|\right), \ \forall n \geq 1.\end{aligned}$$

Letting $n \to \infty$ shows that $\gamma(0) = 0$ which contradicts our assumption! □

Let us summarize the previous results in the following:

Proposition 4.1.1. *Assume $\{X_t\}$ is stationary and zero mean with ACF γ_X such that $\gamma_X(0) > 0$ and $\gamma_X(h) \to 0$ as $h \to \infty$. Define Γ_n and $\boldsymbol{\gamma}_n$ as before. Then*

$$\hat{X}_{n+1} = P_{\overline{\mathrm{sp}}\{X_1, \ldots, X_n\}}(X_{n+1}) = \sum_{i=1}^{n} \phi_{ni} X_{n+1-i},$$

where

$$(\phi_{n1}, \ldots, \phi_{nn})^T = \boldsymbol{\phi}_n = \Gamma_n^{-1} \boldsymbol{\gamma}_n,$$

and

$$\nu_n := \mathrm{Var}(X_{n+1} - \hat{X}_{n+1}) = \gamma(0) - \boldsymbol{\gamma}_n' \Gamma_n^{-1} \boldsymbol{\gamma}_n.$$

Example 4.1.1 (AR(p) processes with $n \geq p$ observations)**.** Assume that

$$X_t = \phi_1 X_{t-1} + \cdots + \phi_p X_{t-p} + Z_t, \quad Z_t \sim \mathrm{WN}(0, \sigma^2).$$

Assume we want to predict X_{n+1} given $X_1, \ldots, X_n$. If $n \geq p$, the most natural forecast seems to be

$$\hat{X}_{n+1} = \phi_1 X_n + \cdots + \phi_p X_{n-p+1}.$$

We can directly check this by showing that

$$\langle X_{n+1} - \hat{X}_{n+1}, X_i \rangle = 0 \quad \forall\, i \in \{1, \ldots, n\}.$$

But clearly, by definition

$$X_{n+1} - \hat{X}_{n+1} = Z_{n+1},$$

and if the process X_t has a causal solution

$$X_t = \psi_0 Z_t + \psi_1 Z_{t-1} + \cdots,$$

then

$$\langle Z_{n+1}, X_i \rangle = 0 \quad \forall\, i \in \{1, \ldots, n\}.$$

Besides we have $\nu_n = E(X_{n+1} - \hat{X}_{n+1})^2 = EZ_{n+1}^2 = \sigma^2$.

Example 4.1.2 (AR(p) processes with $< p$ observations)**.** Assume again that

$$X_t = \phi_1 X_{t-1} + \cdots + \phi_p X_{t-p} + Z_t, \quad Z_t \sim \mathrm{WN}(0, \sigma^2),$$

and that we want to predict X_{n+1} only given $\{X_n, \ldots, X_{n-m+1}\}$, $m < p$. The previous approach doesn't work anymore. (By stationarity we can assume that we have observed $\mathbf{X}_m = (X_1, \ldots, X_m)^T$ and that we want to predict X_{m+1}.) The coefficient vector $(\phi_{m1}, \ldots, \phi_{mm})^T$ can be obtained using Proposition 4.1.1. Let us consider the case $m = 1$. Then the prediction equation becomes

$$\gamma_X(1) = \Gamma_1 \phi_{11} = \gamma_X(0)\phi_{11},$$

i.e. $\phi_{11} = \frac{\gamma_X(1)}{\gamma_X(0)} = \rho_X(1)$ and $\hat{X}_{n+1} = \rho_X(1) X_n$.

4.2 Innovations algorithm

The prediction equations of the previous section have general validity for stationary processes and can be in principle always used. However,

1. $\boldsymbol{\phi}_n = \Gamma_n^{-1} \boldsymbol{\gamma}_n$ is numerically costly (we might need to invert huge matrix);

2. $\boldsymbol{\phi}_n$ has to be calculated from scratch whenever we add one observation (i.e. change the n).

The *innovations algorithm* will give us a recursive scheme that allows us to efficiently update the forecasts. Besides it can be conveniently applied to forecast ARMA(p, q) and for *Gaussian likelihood estimation* (see Chapter 5). The algorithm doesn't require stationarity!

Setup. We assume that $\{X_t\}$ is zero mean, but not necessarily stationary, with covariance function

$$K(i, j) = \mathrm{Cov}(X_i, X_j) = E[X_i X_j] = \langle X_i, X_j \rangle.$$

Idea. Use again

$$\hat{X}_{n+1} = P_{\overline{\mathrm{sp}}\{X_1,\ldots,X_n\}}(X_{n+1})$$

but change (*orthogonalize*) the basis $X_1, \ldots, X_n$. We set $\hat{X}_1 = 0$ and note that then

$$\begin{aligned}\mathcal{H}_n &= \overline{\mathrm{sp}}\{X_1, \ldots, X_n\} \\ &= \overline{\mathrm{sp}}\{X_1 - \hat{X}_1, X_2 - \hat{X}_2, \ldots, X_n - \hat{X}_n\}.\end{aligned}$$

(This is the so-called *Gram-Schmidt orthogonalization.*) In fact, it is easy to see by using the projection theorem that the variables

$$X_i - \hat{X}_i \perp X_j - \hat{X}_j, \quad i \neq j.$$

We then have

$$\begin{aligned}\hat{X}_{n+1} &= P_{\overline{\mathrm{sp}}\{X_1 - \hat{X}_1, \ldots, X_n - \hat{X}_n\}}\left(X_{n+1}\right) \\ &= \theta_{n1}\left(X_n - \hat{X}_n\right) + \theta_{n2}\left(X_{n-1} - \hat{X}_{n-1}\right) + \cdots + \theta_{nn}\left(X_1 - \hat{X}_1\right) \\ &= \sum_{j=1}^{n} \theta_{nj}\left(X_{n+1-j} - \hat{X}_{n+1-j}\right), \quad (\star)\end{aligned}$$

and our goal is to determine the coefficients θ_{nj}, $1 \leq j \leq n$.

Proposition 4.2.1 (Innovations algorithm)**.** *Assume that $\{X_t\}$ is zero mean with covariance function $\mathrm{Cov}(X_i, X_j) = K(i, j)$. Assume that $((K(i, j)))_{i,j}^n$ is non singular $\forall n \geq 1$. Then the coefficients for $\hat{X}_{n+1}$ given in $(\star)$ can be obtained by the*

following recursion:

$$\begin{aligned}
\nu_0 &= K(1,1) \qquad \textit{(Recall: } \nu_k = E[X_{k+1} - \hat{X}_{k+1}]^2\textit{)};\\
\theta_{n,n-k} &= \frac{1}{\nu_k}\left\{K(n+1,k+1) - \sum_{j=0}^{k-1}\theta_{k,k-j}\theta_{n,n-j}\nu_j\right\},\ \textit{for } k = 0,\dots,n-1;\\
\nu_n &= K(n+1,n+1) - \sum_{j=0}^{n-1}\theta_{n,n-j}^2\nu_j.
\end{aligned}$$

Here $\sum_{j=0}^{-1}(\quad) := 0$.

The array blow depicts in which order to evaluate the coefficients:

$$\begin{array}{llll}
\text{Level } 0 & \nu_0 & & \\
\text{Level } 1 & \theta_{11} \to & \nu_1 & \\
\text{Level } 2 & \theta_{22} \to & \theta_{21} \to & \nu_2\\
\text{Level } 3 & \theta_{33} \to & \theta_{32} \to & \dots\\
\dots & \dots & \dots & \dots
\end{array}$$

Proof of Proposition 4.2.1. That $\nu_0 = K(1,1)$ is trivial by the definition $\hat{X}_1 = 0$. We use that $X_k - \hat{X}_k$ are orthogonal. We get immediately (use Exercise 2.9 (a)) that

$$\begin{aligned}
\theta_{n,n-k} &= \frac{\left\langle X_{k+1} - \hat{X}_{k+1}, X_{n+1}\right\rangle}{\|X_{k+1} - \hat{X}_{k+1}\|^2} = \frac{\left\langle X_{k+1} - \hat{X}_{k+1}, X_{n+1}\right\rangle}{\nu_k}\\
&= \frac{1}{\nu_k}\left\{K(n+1,k+1) - \left\langle \hat{X}_{k+1}, X_{n+1}\right\rangle\right\}\\
&= \frac{1}{\nu_k}\left\{K(n+1,k+1) - \left\langle \sum_{j=1}^{k}\theta_{kj}\left(X_{k+1-j} - \hat{X}_{k+1-j}\right), X_{n+1}\right\rangle\right\}\\
&= \frac{1}{\nu_k}\Bigg\{K(n+1,k+1) - \underbrace{\sum_{j=1}^{k}\theta_{kj}\left\langle X_{k+1-j} - \hat{X}_{k+1-j}, X_{n+1}\right\rangle}_{=\sum_{j=0}^{k-1}\theta_{k,k-j}\langle X_{j+1} - \hat{X}_{j+1}, X_{n+1}\rangle}\Bigg\},
\end{aligned}$$

giving the formula for $\theta_{n,n-k}$.

The formula for ν_n is obtained by observing

$$\begin{aligned}
K(n+1,n+1) &= \|X_{n+1}\|^2 \\
&= \left\|X_{n+1} - \hat{X}_{n+1} + \hat{X}_{n+1}\right\|^2 \\
&= \left\|X_{n+1} - \hat{X}_{n+1}\right\|^2 + \left\|\hat{X}_{n+1}\right\|^2 \quad \text{since } X_{n+1} - \hat{X}_{n+1} \perp \hat{X}_{n+1}; \\
&= \nu_n + \left\|\hat{X}_{n+1}\right\|^2 \\
&= \nu_n + \left\|\sum_{j=1}^{n} \theta_{nj}\left(X_{n+1-j} - \hat{X}_{n+1-j}\right)\right\|^2 \\
&= \nu_n + \sum_{j=1}^{n} \theta_{nj}^2 \nu_{n-j} \\
&= \nu_n + \sum_{j=0}^{n-1} \theta_{n,n-j}^2 \nu_j.
\end{aligned}$$

Here we repeatedly used the Pythagorean theorem (Exercise 2.4). □

4.3 Prediction of ARMA processes

We have already shown in Examples (4.1.1) and (4.1.2) how to forecast AR(p) processes. Let us apply the innovations algorithm to forecast an MA(1) process:

$$X_t = Z_t + \theta Z_{t-1}, \quad Z_t \sim \mathrm{WN}(0, \sigma^2).$$

Here

$$K(i,j) = \begin{cases} \sigma^2(1+\theta^2) & \text{if } i = j; \\ \sigma^2\theta & \text{if } |i-j| = 1; \\ 0 & \text{otherwise.} \end{cases}$$

So the innovation algorithm here gives:

$$\begin{aligned}
\nu_0 &= \sigma^2(1+\theta^2)\\
\theta_{11} &= \frac{1}{\nu_0}\sigma^2\theta\\
\nu_1 &= \sigma^2(1+\theta^2) - \theta_{11}^2\nu_0 = \sigma^2(1+\theta^2) - \frac{1}{\nu_0}\sigma^4\theta^2 = \sigma^2\left(1+\theta^2-\frac{1}{\nu_0}\theta^2\sigma^2\right)\\
\theta_{22} &= 0\\
\theta_{21} &= \frac{1}{\nu_1}\sigma^2\theta\\
\nu_2 &= \sigma^2\left(1+\theta^2-\frac{1}{\nu_1}\theta^2\sigma^2\right)\\
&\ldots
\end{aligned}$$

Continuing this way one easily finds that $\theta_{nj} = 0$ if $2 \le j \le n$ and

$$\begin{aligned}
\theta_{n1} &= \frac{1}{\nu_{n-1}}\sigma^2\theta;\\
\nu_n &= \sigma^2\left(1+\theta^2-\frac{1}{\nu_{n-1}}\theta^2\sigma^2\right).
\end{aligned}$$

Consequently

$$\hat{X}_{n+1} = \theta_{n1}\left(X_n - \hat{X}_n\right) = \frac{1}{\nu_{n-1}}\theta\sigma^2\left(X_n - \hat{X}_n\right).$$

Now if $|\theta| < 1$ (i.e. if the process is invertible) then one can show that $\nu_n \overset{n\to\infty}{\longrightarrow} \sigma^2$, and consequently $\theta_{n1} \overset{n\to\infty}{\longrightarrow} \theta$ and we have $\hat{X}_{n+1} \approx \theta(X_n - \hat{X}_n)$. (Exercise 4.11)

Why is this result plausible?
For an ARMA process we expect that the optimal forecast satisfies

$$X_{n+1} - \hat{X}_{n+1} = Z_{n+1} \quad \text{[like in the AR}(p)\text{ case]}$$

or at least

$$X_{n+1} - \hat{X}_{n+1} \approx Z_{n+1}.$$

Then for the model $X_{n+1} = Z_{n+1} + \theta Z_n$ we get by replacing the Z's

$$X_{n+1} \approx X_{n+1} - \hat{X}_{n+1} + \theta(X_n - \hat{X}_n) \Leftrightarrow \hat{X}_{n+1} \approx \theta(X_n - \hat{X}_n).$$

Why do we need here an invertible process?
Since $\hat{X}_{n+1}$ should be in $\overline{\text{sp}}\{X_n, X_{n-1}, \ldots\}$, we can have $X_{n+1} - \hat{X}_{n+1} = Z_{n+1}$ only

if $Z_{n+1} \in \overline{\text{sp}}\{X_{n+1}, X_n, \ldots\}$, i.e. if the process is invertible! The reason why we have only $\approx$ instead of $=$ is that we need the "infinite past" to recover Z_n from $X_n, X_{n-1}, \ldots$. However, in real life we have only $X_1, \ldots, X_n$ in our sample!

Forecasting a general ARMA(p, q) model. Assume now that

$$\phi(B)X_t = \theta(B)Z_t, \quad Z_t \sim \text{WN}(0, \sigma^2),$$

where as usual $\phi(z) = 1 - \phi_1 z - \cdots - \phi_p z^p$ and $\theta(z) = 1 + \theta_1 z + \cdots + \theta_q z^q$. We define

$$W_n = \begin{cases} \frac{1}{\sigma} X_n & \text{if } 1 \le n \le \max(p,q); \\ \frac{1}{\sigma}\phi(B)X_n & \text{if } n > \max(p,q). \end{cases}$$

Note that $\{W_t,\, t > \max(p,q)\}$ defines an MA(q) process. The process $\{W_t,\, t \ge 1\}$ is not stationary but the innovations algorithm applies!

We remark first that

$$\overline{\text{sp}}\{X_1, \ldots, X_n\} = \mathcal{H}_n = \overline{\text{sp}}\{W_1, \ldots, W_n\}.$$

As before we set $\hat{X}_1 = \hat{W}_1 = 0$ and then

$$\hat{W}_{n+1} = P_{\mathcal{H}_n}(W_{n+1}) = \sum_{j=1}^{n} \theta_{nj} \left(W_{n+1-j} - \hat{W}_{n+1-j} \right), \tag{4.3}$$

where we note that $\theta_{nj} = 0$ if $n > \max(p,q)$ and $j > q$. (See Exercise 4.10). Coefficients can be obtained from Proposition 4.2.1. By linearity of the projection we get

$$\hat{W}_{n+1} = \begin{cases} \frac{1}{\sigma}\hat{X}_{n+1} \text{ if } 0 \le n < \max(p,q); \\ \frac{1}{\sigma}\left(\hat{X}_{n+1} - \phi_1 X_n - \phi_2 X_{n-1} - \ldots - \phi_p X_{n-p+1} \right) \text{ if } n \ge \max(p,q), \end{cases} \tag{4.4}$$

since

$$\phi(B)X_{n+1} = X_{n+1} - \phi_1 X_{n-1} - \phi_2 X_{n-2} - \ldots - \phi_p X_{n-p},$$

if $n \ge \max\{p,q\}$. Thus, combining (4.3) and (4.4) we obtain

$$\hat{X}_{n+1} = \begin{cases} \sigma \sum_{j=1}^{n} \theta_{nj}(W_{n+1-j} - \hat{W}_{n+1-j}) & \text{if } 0 \le n < \max(p,q); \\ \phi_1 X_n + \ldots + \phi_p X_{n-p+1} + \sigma \sum_{j=1}^{q} \theta_{nj}(W_{n+1-j} - \hat{W}_{n+1-j}) & \text{else.} \end{cases}$$

Also we have that $\forall t \ge 1$

$$W_t - \hat{W}_t = \frac{1}{\sigma}(X_t - \hat{X}_t),$$

and therefore

$$\hat{X}_{n+1} = \begin{cases} \sum_{j=1}^{n} \theta_{nj}(X_{n+1-j} - \hat{X}_{n+1-j}) & \text{if } 0 \leq n < \max(p,q) \\ \phi_1 X_n + \ldots + \phi_p X_{n-p+1} + \sum_{j=1}^{q} \theta_{nj}(X_{n+1-j} - \hat{X}_{n+1-j}) & \text{else.} \end{cases}$$

Remark. One can show that if $\{X_t\}$ is invertible, then the θ_{nj}'s converge exponentially fast to θ_j. Hence for larger n we may safely replace $\hat{X}_{n+1}$ by

$$\tilde{X}_{n+1} = \phi_1 X_n + \ldots + \phi_p X_{n-p+1} + \sum_{j=1}^{q} \theta_j (X_{n+1-j} - \hat{X}_{n+1-j}).$$

Remark. It is not difficult to see that he covariances of the $\{W_t\}$ process don't depend on σ^2 and can be solely obtained from the ϕ_i's and the θ_j's. Also, $r_k = E(W_{k+1} - \hat{W}_{k+1})^2 = \frac{1}{\sigma^2} v_k$ doesn't depend on σ^2. This is an important fact we will use later for estimation.

4.4 An alternative method to compute the PACF

While we have already provided an estimator $\hat{\rho}_X$ in Chapter 1, it is not so clear, how we should define $\hat{\alpha}_X$. The following proposition gives a more convenient alternative definition.

Proposition 4.4.1 (Alternative definition of the PACF)**.** *Let $\{X_t\}$ be a stationary mean zero process such that its ACF satisfies $\gamma_X(h) \to 0$, as $h \to \infty$ and $\gamma_X(0) > 0$. Then*

$$\alpha_X(h) = \phi_{hh},$$

where ϕ_{hh} comes from the prediction equation

$$(\phi_{h1}, \ldots, \phi_{hh})^T = \Gamma_h^{-1} \boldsymbol{\gamma}_h.$$

Under the setup of Proposition 4.4.1, we can use

$$\hat{\alpha}_X(h) = \hat{\phi}_{hh},$$

with

$$(\hat{\phi}_{h1}, \ldots, \hat{\phi}_{hh})^T = \hat{\Gamma}_h^{-1} \hat{\boldsymbol{\gamma}}_h,$$

provided $|\hat{\Gamma}_h| \neq 0$. This latter equation only involves the estimated autocovariances $\hat{\gamma}_X(j)$, $1 \leq j \leq h$.

Proof of Proposition 4.4.1. We notice first that by Lemma 4.1.1 the parameters ϕ_{ki} are unique. (Otherwise $\alpha_X(k)$ would not be well defined!) Next we note that

$$\overline{\text{sp}}\{X_1, \ldots, X_k\} = \overline{\text{sp}}\{X_2, \ldots, X_k\} \oplus \overline{\text{sp}}\{X_1 - P_{\overline{\text{sp}}\{X_2,\ldots,X_k\}}(X_1)\}.$$

Observing that $P_{\overline{\text{sp}}\{X_2,\ldots,X_k\}}(X_1)$ and $P_{\overline{\text{sp}}\{X_2,\ldots,X_k\}}(X_{k+1})$ are both linear combination depending only on $X_2, \ldots, X_k$, it follows that

$$\begin{aligned}
\hat{X}_{k+1} &= P_{\overline{\text{sp}}\{X_1,\ldots,X_k\}}(X_{k+1}) \\
&= P_{\overline{\text{sp}}\{X_2,\ldots,X_k\}}(X_{k+1}) + \frac{\langle X_{k+1}, X_1 - P_{\overline{\text{sp}}\{X_2,\ldots,X_k\}}(X_1)\rangle}{\|X_1 - P_{\overline{\text{sp}}\{X_2,\ldots,X_k\}}(X_1)\|^2}\left[X_1 - P_{\overline{\text{sp}}\{X_2,\ldots,X_k\}}(X_1)\right] \\
&= \phi_{k1}X_k + \cdots + \phi_{k,k-1}X_2 + \underbrace{\frac{\langle X_{k+1}, X_1 - P_{\overline{\text{sp}}\{X_2,\ldots,X_k\}}(X_1)\rangle}{\|X_1 - P_{\overline{\text{sp}}\{X_2,\ldots,X_k\}}(X_1)\|^2}}_{\phi_{kk}} X_1.
\end{aligned}$$

Since $P_{\overline{\text{sp}}\{X_2,\ldots,X_k\}}(X_{k+1}) \perp X_1 - P_{\overline{\text{sp}}\{X_2,\ldots,X_k\}}(X_1)$ and since it can be shown (Exercise) that

$$\|X_1 - P_{\overline{\text{sp}}\{X_2,\ldots,X_k\}}(X_1)\| = \|X_{k+1} - P_{\overline{\text{sp}}\{X_2,\ldots,X_k\}}(X_{k+1})\|,$$

it follows that $\alpha_X(k) = \phi_{kk}$. □

4.5 Exercises

Ex 4.1 Consider the space $L^2 = L^2(\Omega, \mathcal{A}, P)$. Let $Z \in L^2$ and let $S(Z)$ be the set of all $X \in L^2$ that can be written in the form

$$X = \phi(Z)$$

for some Borel measurable function ϕ. In other words: $S(Z)$ is the set of square integrable random variables that are measurable with respect to $\mathcal{A}_0 := Z^{-1}(\mathcal{B}(\mathbb{R}))$. Show that $S(Z)$ is a closed linear subspace.

Ex 4.2 Continuation of 4.1. Show that for all $A \in \mathcal{A}_0$ we have

$$\int_A E[X|S(Z)]dP = \int_A X dP,$$

where $E[X|M] = P_M(X)$ is the orthogonal projection of X onto M. Hence $E[X|S(Z)]$ is version of the usual conditional expectation $E[X|S]$.

Ex 4.3 Continuation of Ex 4.1. Let $Z_1, \ldots, Z_n$ in L^2. Let $L = \overline{\text{sp}}(1, Z_1, \ldots, Z_n)$ and $S = \{Y \in L^2 : Y = \phi(Z_1, \ldots, Z_n)\}$ for some Borel measurable ϕ. Prove the following "sum of squares partition"

$$\|X - E[X|L]\|^2 = \|X - E[X|S]\|^2 + \|E[X|S] - E[X|L]\|^2.$$

Hence: if $E[X|S]$ is our prediction of X given the information $Z_1, \ldots, Z_n$ then it is usually giving a better prediction than $E[X|L]$ (but much harder to compute).

Ex 4.4 Let $\{X_t\}$ be stationary with mean μ and let $Y_t = X_t - \mu$. Show that

$$P_{\overline{\text{sp}}\{1, X_1, \ldots, X_n\}}(X_{n+h}) = \mu + P_{\overline{\text{sp}}\{Y_1, \ldots, Y_n\}}(Y_{n+h}).$$

Ex 4.5 Let $\mathbf{X}_n = (X_1, \ldots, X_n)^T$ be the first n observations from the stationary sequence $\{X_t\}$. Construct an example where $\gamma(h)$ doesn't converge to 0 if $h \to \infty$ and for which $\Gamma_n = \text{Cov}(\mathbf{X}_\text{n})$ is non-singular for all $n \geq 1$.

Ex 4.6 Refer to Ex 4.5. Assume that $EX_t = 0$. Show that Γ_n is singular if and only if there are constants $b = (b_1, \ldots, b_{n-1})^T$ such that $b^T \mathbf{X}_{n-1} = X_n$.

Ex 4.7 Let $\{X_t\}$ be an AR(3) process. Compute the linear forecast for X_{n+1} based on $\{X_n\}$, $\{X_n, X_{n-1}\}$, $\{X_n, X_{n-1}, X_{n-2}\}$ and finally on $\{X_n, \ldots, X_1\}$, when $n > 3$. Compute in each case the mean-squared prediction error.

Ex 4.8 Let $\{X_t\}$ be a stationary mean zero process. Obtain the formula for the h-step prediction equations by imitating the 1-step prediction approach. Also obtain a formula for mean-squared prediction error $\nu_n^{(h)} = E[X_{n+h} - \hat{X}_{n+h}]^2$.

Ex 4.9 Apply the innovations algorithm to an AR(1) process

$$X_t = \phi X_{t-1} + Z_t, \quad |\phi| < 1, \quad \{Z_t\} \sim \text{WN}(0, \sigma^2).$$

Ex 4.10 Apply the innovations algorithm to an MA(q) process and show that

$$\hat{X}_{n+1} = \sum_{j=1}^{q} \theta_{nj}(X_{n+1-j} - \hat{X}_{n+1-j}),$$

if $n \geq q$.

Ex 4.11 For an MA(1) process $X_t = Z_t + \theta Z_{t-1}$ we obtained by applying the innovations algorithm that

$$\hat{X}_{n+1} = \frac{\sigma^2\theta}{\nu_n}(X_n - \hat{X}_n),$$

with

$$\nu_n = \sigma^2\left(1 + \theta^2 - \frac{1}{\nu_{n-1}}\theta^2\sigma^2\right),$$

and $\nu_0 = \sigma^2(1+\theta^2)$. Show that if $|\theta| < 1$ then $|\nu_n - \sigma^2| \leq C\rho^n$ for some $0 < \rho < 1$ and some large enough constant C (exponentially fast convergence). Also the convergence $\theta_{n1} \to \theta$ is exponentially fast.

What happens to ν_n if $|\theta| > 1$ or if $|\theta| = 1$?

Chapter 5

Parameter estimation for ARMA processes

Up to now we have developed the entire theory under the (very unrealistic) assumption that we know the parameters $(\phi_1, \dots, \phi_p, \theta_1, \dots, \theta_q, \sigma^2)$ of our model:

$$X_t - \phi_1 X_{t-1} - \cdots - \phi_p X_{t-p} = Z_t + \theta_1 Z_{t-1} + \cdots + \theta_q Z_{t-q}, \quad \text{WN}(0, \sigma^2).$$

In this chapter will provide an elementary insight to the important estimation theory.

The first step in estimating the ARMA model is to determine the order of the model. Since any ARMA(p, q) is also an ARMA(p', q') process when $p' \geq p$ and $q' \geq q$, it might be tempting to fit a "large" enough model i.e. choose big values p and q. However, as usual in statistics, *overfitting* can have serious impact on the quality of the model. Although it might perfectly mimic the process for a given sample, it can be extremely bad when it comes to predict new values. While theoretical methods to assess the order p, q exist (e.g. Akaike's AIC criterion, see Section 5.3) an efficient applied method, is based on the inspection of the ACF and the *partial ACF*. Once the order is determined we can employ next to empirical methods, preliminary estimation methods (see Section 5.1) to get estimators that may later be improved by likelihood methods (Section 5.2).

5.1 Preliminary estimation method for AR processes

Assume we have a zero mean and causal AR(p) process, p known. In Example 4.1.1 we have seen that

$$\hat{X}_{p+1} = \phi_1 X_p + \cdots + \phi_p X_1. \tag{5.1}$$

Combining with Proposition 4.1.1 we conclude that

$$(\phi_1, \ldots, \phi_p) = \Gamma_p^{-1} \boldsymbol{\gamma}_p.$$

Similarly we obtain from (5.1) that

$$E(X_{p+1} - \hat{X}_{p+1})^2 = EZ_{p+1}^2 = \sigma^2,$$

and using again Proposition 4.1.1 we conclude that

$$\sigma^2 = \gamma(0) - \boldsymbol{\gamma}_p' \Gamma_p^{-1} \boldsymbol{\gamma}_p.$$

Idea. Replace population $\gamma(h)$ with its sample version

$$\hat{\gamma}(h) = \frac{1}{n} \sum_{k=1}^{n-|h|} (X_k - \bar{X})(X_{k+h} - \bar{X}).$$

The resulting estimators

$$\hat{\boldsymbol{\phi}} = \hat{\Gamma}_p^{-1} \hat{\boldsymbol{\gamma}}_p \quad \text{and} \quad \hat{\sigma}^2 = \hat{\gamma}(0) - \hat{\boldsymbol{\gamma}}_p^T \hat{\Gamma}_p^{-1} \hat{\boldsymbol{\gamma}}_p$$

are called *Yule-Walker* estimators.

Example 5.1.1 (Yule-Walker estimator for AR(1) processes)**.** Assume

$$X_t = \phi X_{t-1} + Z_t, \quad Z_t \sim \mathrm{WN}(0, \sigma^2), \quad |\phi| < 1.$$

Then we get immediately

$$\hat{\phi} = \frac{\hat{\gamma}_X(1)}{\hat{\gamma}_X(0)} = \hat{\rho}_X(1),$$

and

$$\hat{\sigma}^2 = \hat{\gamma}_X(0) - \frac{\hat{\gamma}_X^2(1)}{\hat{\gamma}_X(0)} = \hat{\gamma}_X(0) \left(1 - \hat{\rho}_X^2(1)\right).$$

Note that we have used $|\hat{\Gamma}_p| \neq 0$! The following proposition shows among others, when this is true:

Proposition 5.1.1. *Assume that $\{X_t\}$ is a stationary and causal AR(p) process. Let $\hat{\gamma}(0) > 0$. Then*

(a) $\hat{\Gamma}_p$ is not singular and thus the Yule-Walker equations have a unique solution.

(b) The fitted model $\{X_t^F\}$, given by the equations

$$X_t^F = \hat{\phi}_1 X_{t-1}^F + \ldots + \hat{\phi}_p X_{t-p}^F + Z_t^F \text{ with } \left\{Z_t^F\right\} \sim \mathrm{WN}(0, \hat{\sigma}^2)$$

has again a stationary and causal solution.

(c) The ACF of $\{X_t^F\}$ *coincides with the sample ACF at lags* $h = 0, \ldots, p$*:*

$$\gamma_{X^F}(h) = \hat{\gamma}(h), \quad h = 0, \ldots, p.$$

Proof. Part (a). For the entire proof we assume that n is fixed, and that $\hat{\gamma}(h)$ are the estimates obtained from a realization of $X_1, \ldots, X_n$. Usually $\hat{\gamma}(h) = \frac{1}{n}\sum_{k=1}^{n-|h|}(X_k - \bar{X})(X_{k+h} - \bar{X})$ is only defined for $|h| < n$, but we extend its definition by setting $\hat{\gamma}(h) = 0$ for $|h| \geq n$. We show first that $\{\hat{\gamma}(h),\, h \in \mathbb{Z}\}$ is n.n.d., which is equivalent of saying that $\hat{\Gamma}_k = ((\hat{\gamma}(i-j)))_{i,j=1}^k$ is n.n.d. for all $k \geq 1$, where $k \geq 1$. But $\hat{\Gamma}_k = \frac{1}{n}T_k'T_k$, with

$$T_k' = \begin{pmatrix} Y_1 & Y_2 & \cdots & Y_{n-1} & Y_n & 0 & 0 & \cdots & 0 \\ 0 & Y_1 & Y_2 & \cdots & Y_{n-1} & Y_n & 0 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & Y_1 & Y_2 & \cdots & \cdots & Y_{n-1} & Y_n \end{pmatrix} \in \mathbb{R}^{k\times(n+k-1)},$$

and $Y_k = X_k - \bar{X}$. Hence for $\mathbf{a} \in \mathbb{R}^k$ we have

$$\mathbf{a}'\hat{\Gamma}_k\mathbf{a} = \frac{1}{n}\mathbf{a}'T_k'T_k\mathbf{a},$$

which is easily seen to be non-negative. Consequently, by Theorem 1.3.1, we know that $\hat{\gamma}(h)$ is the ACF of a stationary process. We also know that $\hat{\gamma}(h) = 0$ if $|h| \geq n$, which implies (Proposition 3.5.2) that it is an MA$(n-1)$ process and (Lemma 4.1.1) that $|\hat{\Gamma}_n| \neq 0$ for all n. This shows part (a).

Part (b): Define $\hat{\phi}(z) = 1 - \hat{\phi}_1 z - \cdots - \hat{\phi}_p z^p$, where the $\hat{\phi}_i$ are the Yule-Walker estimates obtained from the realization of $X_1, \ldots, X_n$. We assume now that there exists $|z_0| \leq 1$ with $\hat{\phi}(z_0) = 0$, i.e. that the estimated parameters don't define a stationary and causal process. We assume here for simplicity that z_0 is real. We then factorize $\hat{\phi}(z) = (1 - z_0^{-1}z)\xi(z)$ and remark that $\xi(z)$ must be of the form $\xi(z) = 1 + \xi_1 z + \cdots + \xi_{p-1}z^{p-1}$. Now we define $Y_j = \xi(B)W_j$ where $\{W_t\}$ is a stationary process that has as its ACF the function $\hat{\gamma}(h)$. Above we proved that such a process exists and is an MA$(n-1)$ process. Consequently Y_j is again stationary. In fact, it is easy to see that it is an MA(q) process (of some order q which is not important). We define $\rho = \mathrm{Corr}(Y_t, Y_{t+1})$. Clearly $\rho \in [-1, 1]$ and in fact for an MA(q) one can show that $\rho \in (-1, 1)$ (Exercise 5.6). Now we define

$\tilde{\phi}(z) = (1-\rho z)\xi(z) = 1 - \tilde{\phi}z - \cdots - \tilde{\phi}z^p$. It then follows that

$$\begin{aligned} E|W_{p+1} - \tilde{\phi}_1 W_p - \cdots - \tilde{\phi}_p W_1|^2 &= E|\tilde{\phi}(B)W_{p+1}|^2 \\ &= E|Y_{p+1} - \rho Y_p|^2 \\ &\leq E|Y_{p+1} - z_0^{-1} Y_p|^2 \qquad (5.2) \\ &= E|\hat{\phi}(B)W_{p+1}|^2 \\ &= E|W_{p+1} - \hat{\phi}_1 W_p - \cdots - \hat{\phi}_p W_1|^2. \end{aligned}$$

Here (5.2) follows from the fact that ρ is the coefficient in the best 1-step forecast, and that this forecast is unique by the projection theorem, so we can have equality if and only if $z_0^{-1} = \rho$. This cannot happen, since $|z_0^{-1}| \geq 1$ by our assumptions. On the other hand, by the very definition of the process $\{W_t\}$ we have that $\hat{\phi}_1 W_p + \cdots + \hat{\phi}_p W_1 = P_{\overline{\text{sp}}\{W_1,\ldots,W_p\}}(W_{p+1})$ which gives the optimal linear approximation of W_{p+1} in terms of $W_1, \ldots, W_p$. This leads to a contradiction.

Part (c): Let $\gamma_F(h)$ be the ACF of $\{X_t^F\}$. Using Method 2 of Section 3.3 to compute the ACF, we obtain that

$$\gamma_F(h) - \hat{\phi}_1\gamma_F(h-1) - \cdots - \hat{\phi}_p\gamma_F(h-p) = \begin{cases} 0, & h = 1, \ldots, p \\ \hat{\sigma}^2, & h = 0. \end{cases} \qquad (5.3)$$

At the same time we have by the definition of the $\hat{\phi}_i$'s that

$$\hat{\gamma}(h) - \hat{\phi}_1\hat{\gamma}(h-1) - \cdots - \hat{\phi}_p\hat{\gamma}(h-p) = \begin{cases} 0, & h = 1, \ldots, p \\ \hat{\sigma}^2, & h = 0. \end{cases} \qquad (5.4)$$

Hence: $x = (\gamma_F(0), \ldots, \gamma_F(p))'$ and $y = (\hat{\gamma}(0), \ldots, \hat{\gamma}(p))'$ are both solutions of a linear equation system

$$Ax = b \quad \text{and} \quad Ay = b,$$

for some $(p+1)$-square-matrix A and some $(p+1)$-vector b. Here A and b are determined by $\hat{\phi}_i$ and $\hat{\sigma}^2$. In order to be sure that x and y are identical, i.e. that the equation system $Ax = b$ has a unique solution, we need to sure that A has full rank. It seems that this step has been ignored in the book of Brockwell and Davis. We propose to solve this issue as follows.

It is not hard to show (Exercise 5.7) that one can equivalently rewrite the equation systems (5.3) and (5.4) as

$$\Gamma_{p+1} = \Theta\Gamma_{p+1}\Theta + \Omega, \qquad (5.5)$$

where

$$\Theta = \begin{pmatrix} \boldsymbol{\theta}' & 0 \\ I_p & 0_{p\times 1} \end{pmatrix} \quad \text{and} \quad \Omega = \begin{pmatrix} \sigma^2 & 0_{1\times p} \\ 0_{p\times p} & 0_{p\times 1} \end{pmatrix} \qquad (5.6)$$

and $\boldsymbol{\theta}' = (\hat{\phi}_1, \ldots, \hat{\phi}_p)$. Here Γ_{p+1} can either be Γ^F_{p+1} or $\hat{\Gamma}_{p+1}$. To show uniqueness of the solution, we iterate the recursion (5.5). This yields for all $n \geq 1$

$$\Gamma_{p+1} = \Omega + \Theta\Omega\Theta' + \cdots + \Theta^n\Omega(\Theta')^n + \Theta^{n+1}\Gamma_{p+1}(\Theta')^{n+1}.$$

Using the fact that we have a stationary and causal AR process, one can show that the eigenvalues of Θ are inside the complex unit-circle and hence $\Theta^{n+1} \to 0_{(p+1)\times(p+1)}$. This yields uniqueness.

□

Example 5.1.2. Figure 5.1 below shows the "Recruits series" (number of new fish) given in Example 1.5 of Shumway and Stoffer. The plots of acf and pacf suggest that an AR(2) model might fit. We use the Yule-Walker estimators and for an AR(2) and obtained

	$\hat{\phi}_1$	$\hat{\phi}_2$	$\hat{\mu}$	$\hat{\sigma}^2$
estimates	1.33	-0.44	62.26	95.8
sd	0.042	0.042		

[R-code]

Although moment estimators are usually not very efficient, it turns out that the Yule-Walker estimator is asymptotically efficient (has same asymptotic variance as ML-estimator). We formulate now the general theorem and will give the proof in the special case of an AR(1) process in Section 5.4.

Theorem 5.1.1. *Let $\{X_t\}$ be a stationary and causal AR(1) process with innovations $\{Z_t\} \sim \text{IID}(0, \sigma^2)$. Then the Yule-Walker estimator is consistent:*

$$\hat{\boldsymbol{\phi}} \xrightarrow{P} \boldsymbol{\phi} \text{ and } \hat{\sigma}^2 \xrightarrow{P} \sigma^2.$$

Moreover

$$\sqrt{n}(\hat{\boldsymbol{\phi}} - \boldsymbol{\phi}) \xrightarrow{d} \mathbf{N}_p(\mathbf{0}, \sigma^2\Gamma_p^{-1}).$$

5.2 Recursive calculation of Gaussian likelihood

Likelihood estimation is one of the most powerful estimation methods in statistics. However, in the dependence setup it is usually very complicated to compute the likelihood (i.e. the join density of $X_1, \ldots, X_n$). Besides that, numerical maximization procedures can be quite intensive. We show how the innovations algorithm can be conveniently used to recursively compute the likelihood function for Gaussian processes.

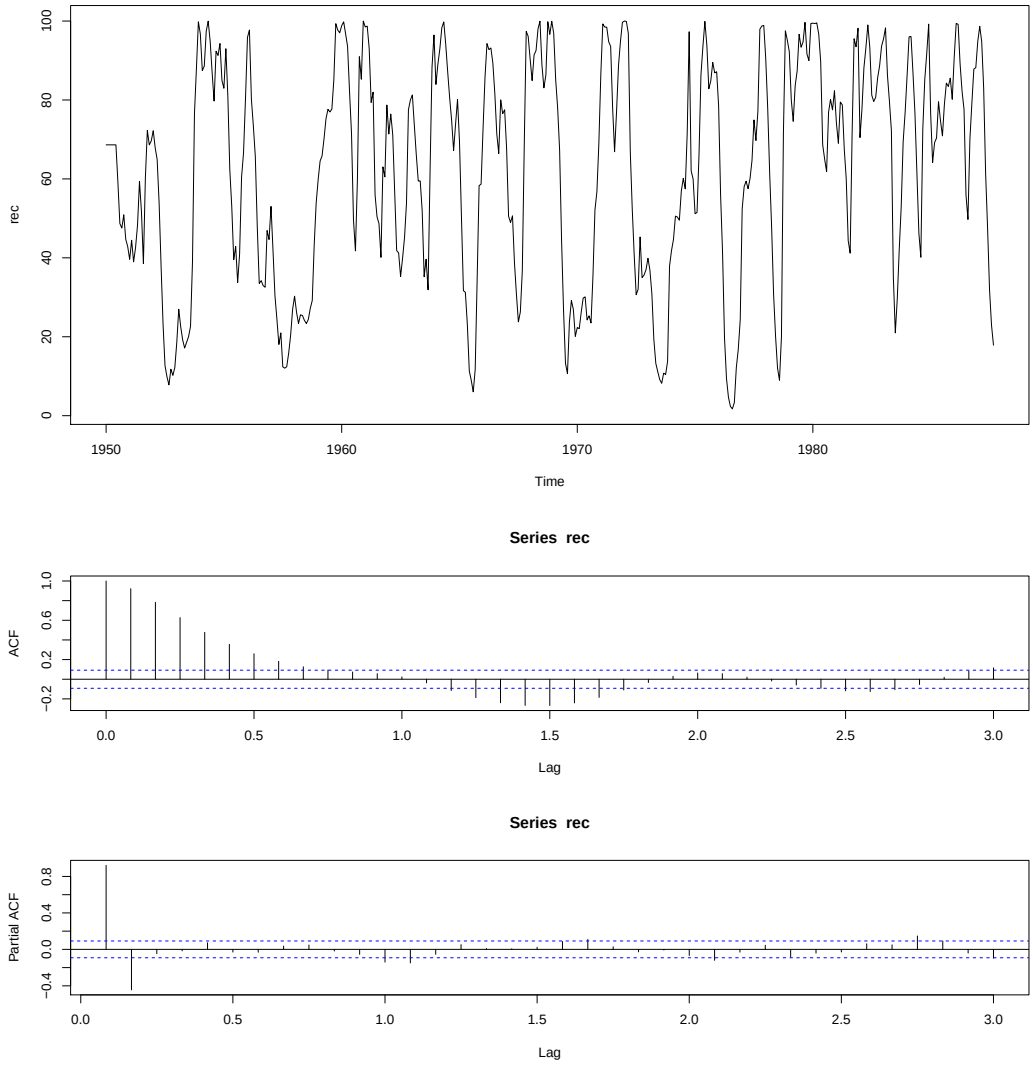


Figure 5.1: The recruitment time series (top) and its acf (middle) and pacf (bottom). Lag is expressed in years.

Let $\{X_t\}$ be a zero mean *Gaussian* process and denote $\gamma(i,j) = \mathrm{Cov}(X_i, X_j) = \langle X_i, X_j \rangle$. (We don't require stationarity).

The likelihood of $\mathbf{X}_n = (X_1, \ldots, X_n)^T$ is given as:

$$L(\Gamma_n | \mathbf{X}_n) = \left(\frac{1}{\sqrt{2\pi}}\right)^n \frac{1}{\sqrt{|\Gamma_n|}} \exp\left(-\frac{1}{2}\mathbf{X}_n^T \Gamma_n^{-1} \mathbf{X}_n\right),$$

with $\Gamma_n = \mathrm{Cov}(\mathbf{X}_n)$.

Idea. If $\{X_t\}$ is an ARMA(p,q) with Gaussian innovations $\{Z_t\}$, then $\{X_t\}$ is Gaussian and $\Gamma_n = \Gamma_n(\boldsymbol{\phi}, \boldsymbol{\theta}, \sigma^2)$. The ML-estimator is the maximizer of $L(\Gamma_n(\boldsymbol{\phi}, \boldsymbol{\theta}, \sigma^2)|\mathbf{X}_n)$ with respect to $\boldsymbol{\phi}, \boldsymbol{\theta}, \sigma^2$. Still, this is numerically intense, as we will usually not get a closed expression for $L(\Gamma_n(\boldsymbol{\phi}, \boldsymbol{\theta}, \sigma^2)|\mathbf{X}_n)$. (Remember that Γ_n depends usually in a quite complicated way on $\boldsymbol{\phi}, \boldsymbol{\theta}, \sigma^2$!)

Now set again $\hat{X}_1 = 0$ and recall that

$$\begin{aligned}
\hat{X}_{k+1} &= P_{\overline{\mathrm{sp}}\{X_1,\ldots,X_k\}}(X_{k+1}) \\
&= P_{\overline{\mathrm{sp}}\{X_1-\hat{X}_1,\ldots,X_k-\hat{X}_k\}}(X_{k+1}) \\
&= \sum_{j=1}^{k} \theta_{kj}(X_{t+1-j} - \hat{X}_{k+1-j}),
\end{aligned}$$

where θ_{kj} can be obtained from the innovations algorithm. Define

$$C_n = \begin{bmatrix}
1 & 0 & 0 & 0 & 0 & \ldots & 0 \\
\theta_{11} & 1 & 0 & 0 & 0 & \ldots & 0 \\
\theta_{22} & \theta_{21} & 1 & 0 & 0 & \ldots & 0 \\
\ldots & \ldots & \ldots & \ldots & \ldots & \ldots & \ldots \\
\theta_{n-1,n-1} & \theta_{n-1,n-2} & \ldots & \ldots & \ldots & \ldots & 1
\end{bmatrix}$$

and

$$I_n = \mathrm{diag}(1, \ldots, 1) \in \mathbb{R}^{n \times n} \quad \text{and} \quad D_n = \mathrm{diag}(\nu_0, \ldots, \nu_{n-1}).$$

Letting $\hat{\mathbf{X}}_n = \left(\hat{X}_1, \ldots, \hat{X}_n\right)^T$ we get

$$\hat{\mathbf{X}}_n = (C_n - I_n)(\mathbf{X}_n - \hat{\mathbf{X}}_n),$$

and thus

$$\begin{aligned}
\mathbf{X}_n &= \hat{\mathbf{X}}_n + (\mathbf{X}_n - \hat{\mathbf{X}}_n) \\
&= (C_n - I_n)(\mathbf{X}_n - \hat{\mathbf{X}}_n) + I_n(\mathbf{X}_n - \hat{\mathbf{X}}_n) \\
&= C_n(\mathbf{X}_n - \hat{\mathbf{X}}_n).
\end{aligned}$$

Hence

$$\begin{aligned}\Gamma_n &= \mathrm{Cov}(\mathbf{X}_n) = \mathrm{Cov}(C_n(\mathbf{X}_n - \hat{\mathbf{X}}_n)) \\ &= C_n \mathrm{Cov}(\mathbf{X}_n - \hat{\mathbf{X}}_n) C_n^T \\ &= C_n D_n C_n^T\end{aligned}$$

Consequently

$$\Gamma_n^{-1} = (C_n^T)^{-1} D_n^{-1} C_n^{-1}$$

and

$$\mathbf{X}_n^T \Gamma_n^{-1} \mathbf{X}_n = (\mathbf{X}_n - \hat{\mathbf{X}}_n)^T C_n^T (C_n^T)^{-1} D_n^{-1} C_n^{-1} C_n (\mathbf{X}_n - \hat{\mathbf{X}}_n) = \sum_{k=1}^{n} \frac{(X_k - \hat{X}_k)^2}{\nu_{k-1}}.$$

Furthermore

$$|\Gamma_n| = \underbrace{|C_n|^2}_{=1} |D_n| = \nu_0 \nu_1 \dots \nu_{n-1}.$$

Summing up we have

$$L(\Gamma_n | \mathbf{X}_n) = \left(\frac{1}{\sqrt{2\pi}}\right)^n \frac{1}{\sqrt{\nu_0 \dots \nu_{n-1}}} \exp\left(-\frac{1}{2} \sum_{k=1}^{n} \frac{(X_k - \hat{X}_k)^2}{\nu_{k-1}}\right). \tag{5.7}$$

Application to causal ARMA(p, q). In this case $L(\Gamma_n|\mathbf{X}_n) = L(\Gamma_n(\boldsymbol{\phi}, \boldsymbol{\theta}, \sigma^2)|\mathbf{X}_n)$. The dependence of the likelihood on the parameters is implicit in $\nu_k = \nu_k(\boldsymbol{\phi}, \boldsymbol{\theta}, \sigma^2)$ and $\hat{X}_k = \hat{X}_k(\boldsymbol{\phi}, \boldsymbol{\theta}, \sigma^2)$. We remarked at the end of Section 4.3 that

$$\hat{X}_k = \phi_1 X_{k-1} + \dots + \phi_p X_{k-p} + \theta_{n1}(X_{k-1} - \hat{X}_{k-1}) + \dots + \theta_{nq}(X_{k-q} - \hat{X}_{k-q}),$$

where the θ_{ni} are functions of $\boldsymbol{\phi}$ and $\boldsymbol{\theta}$, but not of σ^2. Also $\nu_k = \sigma^2 r_k$, where $r_k = r_k(\boldsymbol{\phi}, \boldsymbol{\theta})$. Hence we can then write (5.7) as

$$L(\Gamma_n(\boldsymbol{\phi}, \boldsymbol{\theta}, \sigma^2)|\mathbf{X}_n) = \left(\frac{1}{\sqrt{2\pi\sigma^2}}\right)^n \frac{1}{\sqrt{r_0 \dots r_{n-1}}} \exp\left(-\frac{1}{2\sigma^2} \sum_{k=1}^{n} \frac{(X_k - \hat{X}_k)^2}{r_{k-1}}\right),$$

where $\hat{X}_k = \hat{X}_k(\boldsymbol{\phi}, \boldsymbol{\theta})$ and $r_k = r_k(\boldsymbol{\phi}, \boldsymbol{\theta})$. The maximum likelihood estimator (MLE) are the parameters that maximize $L(\Gamma_n(\boldsymbol{\phi}, \boldsymbol{\theta}, \sigma^2)|\mathbf{X}_n)$ or equivalently that minimize

$$\ell(\boldsymbol{\phi}, \boldsymbol{\theta}, \sigma^2) := -\frac{2}{n} \log L(\Gamma_n(\boldsymbol{\phi}, \boldsymbol{\theta}, \sigma^2)|\mathbf{X}_n) - 2\log(\sqrt{2\pi}).$$

Since we were able to factorize out the σ^2, we get by denoting

$$I(\boldsymbol{\phi}, \boldsymbol{\theta}) = \sum_{k=1}^{n} \frac{(X_k - \hat{X}_k)^2}{r_{k-1}},$$

that

$$\ell(\boldsymbol{\phi}, \boldsymbol{\theta}, \sigma^2) = \log \sigma^2 + \frac{1}{n} \sum_{k=1}^{n} \log r_{k-1}(\boldsymbol{\phi}, \boldsymbol{\theta}) + \frac{1}{n\sigma^2} I(\boldsymbol{\phi}, \boldsymbol{\theta}).$$

If we fix $\boldsymbol{\phi}, \boldsymbol{\theta}$ for the moment and take partial derivative with respect to σ^2 we get

$$\frac{\partial}{\partial \sigma^2} \ell(\boldsymbol{\phi}, \boldsymbol{\theta}, \sigma^2) = \frac{1}{\sigma^2} - \frac{1}{n\sigma^4} I(\boldsymbol{\phi}, \boldsymbol{\theta}).$$

The minimum is obtained at $\sigma^2_{\min}(\boldsymbol{\phi}, \boldsymbol{\theta})$ which satisfies

$$\frac{1}{\sigma^2_{\min}(\boldsymbol{\phi}, \boldsymbol{\theta})} - \frac{1}{n\sigma^4_{\min}(\boldsymbol{\phi}, \boldsymbol{\theta})} I(\boldsymbol{\phi}, \boldsymbol{\theta}) = 0.$$

Thus

$$\sigma^2_{\min}(\boldsymbol{\phi}, \boldsymbol{\theta}) = \operatorname{argmin}_{\sigma^2 > 0} \{\ell(\boldsymbol{\phi}, \boldsymbol{\theta}, \sigma^2)\} = \frac{1}{n} I(\boldsymbol{\phi}, \boldsymbol{\theta}).$$

The MLE $(\hat{\boldsymbol{\phi}}, \hat{\boldsymbol{\theta}})$ for $(\boldsymbol{\phi}, \boldsymbol{\theta})$ is obtained by minimizing the reduced likelihood

$$\ell(\boldsymbol{\phi}, \boldsymbol{\theta}, \sigma^2_{\min}(\boldsymbol{\phi}, \boldsymbol{\theta})) - 1 = \ell_{\text{red}}(\boldsymbol{\phi}, \boldsymbol{\theta}) = \log \left(\frac{1}{n} I(\boldsymbol{\phi}, \boldsymbol{\theta}) \right) + \frac{1}{n} \sum_{k=1}^{n} \log r_{k-1}(\boldsymbol{\phi}, \boldsymbol{\theta}).$$

From this we go back to obtain the MLE $\hat{\sigma}^2 = \sigma^2_{\min}(\hat{\boldsymbol{\phi}}, \hat{\boldsymbol{\theta}}) = \frac{1}{n} I(\hat{\boldsymbol{\phi}}, \hat{\boldsymbol{\theta}})$.

Remarks.

- Even if innovations are non-Gaussian the estimation via Gaussian likelihood makes sense and produces under quite general conditions consistent and asymptotically normal estimators.
- The maximization procedure can be speed up with good starting values. We can use, for example, the Yule-Walker or other preliminary estimators.
- Typically one should put as a starting value a causal solution (the software may only allow causal solutions).
- The intuition behind (5.7) is this: roughly speaking the likelihood is large, if $(X_k - \hat{X}_k)^2$ is small in average. This is the case, when the mean squared prediction error is small. I.e. the parametrization that gives in average (here we have a weighted average!) the smallest prediction error, is the optimal one.
- An attractive alternative method that is closely related is the *weighted least squares* estimation. It is based on minimizing

$$I(\boldsymbol{\phi}, \boldsymbol{\theta}) = \sum_{k=1}^{n} \frac{(X_k - \hat{X}_k)^2}{r_{k-1}}. \tag{5.8}$$

  If $\tilde{\boldsymbol{\phi}}, \tilde{\boldsymbol{\theta}}$ are the minimizers in (5.8) then

$$\tilde{\sigma}^2 := \frac{1}{n} I(\tilde{\boldsymbol{\phi}}, \tilde{\boldsymbol{\theta}}).$$

  For this approach one should require $(\boldsymbol{\phi}, \boldsymbol{\theta})$ such that a causal and invertible solution exists. Without requiring invertibility there will be in general no minimizer. (See Exercise 5.1).
- For invertible processes $r_n \to 1$. Then weighted least squares and Gaussian MLE are asymptotically equivalent.

5.3 Order selection

The auto-correlation and the partial-autocorrelation can be used as an empirical tool to get some idea of the order of an ARMA model. There exist also some theoretical tools which we shall discuss now briefly. We start by assuming to have an AR(p) process with some oder p:

$$X_{t+1} - \phi_1 X_t - \cdots - \phi_p X_{t-p+1} = Z_{t+1}.$$

Assume now that we have estimated the parameters $\phi_1, \dots, \phi_p$ from our sample $X_1, \dots, X_n$, e.g. by using the Yule-Walker approach. To check how good the model is, we assume that we would apply it to another realization $\{Y_n\}$ which is independent of the original process, i.e.

$$Y_{t+1} - \phi_1 Y_t - \cdots - \phi_p Y_{t-p+1} = Z'_{t+1},$$

where $\{Z'_t\} \stackrel{d}{=} \{Z_t\}$ and $\{Z'_t\}$ is independent of $\{Z_t\}$. We do this to see if the model is able to give also good forecasts for new data, that haven't been used in the estimation.

The expected prediction error when using the parameters that have been estimated from our sample $X_1, \dots, X_n$ is

$$\begin{aligned}
&E(Y_{n+1} - \hat{\phi}_1 Y_n - \cdots - \hat{\phi}_p Y_{n-p+1})^2 \\
&\quad = E(Z'_{n+1} + (\phi_1 - \hat{\phi}_1) Y_n + \cdots + (\phi_p - \hat{\phi}_p) Y_{n-p+1})^2 \\
&\quad = E\left((Z'_{n+1})^2 + ((\boldsymbol{\phi} - \hat{\boldsymbol{\phi}})' \boldsymbol{Y})^2 + 2 Z'_{n+1} (\boldsymbol{\phi} - \hat{\boldsymbol{\phi}})' \boldsymbol{Y}) \right) \\
&\quad = \sigma^2 + E(\boldsymbol{\phi} - \hat{\boldsymbol{\phi}})' \boldsymbol{Y} \boldsymbol{Y}' (\boldsymbol{\phi} - \hat{\boldsymbol{\phi}}) \\
&\quad = \sigma^2 + E(\boldsymbol{\phi} - \hat{\boldsymbol{\phi}})' \Gamma_p (\boldsymbol{\phi} - \hat{\boldsymbol{\phi}}).
\end{aligned}$$

Here $\boldsymbol{Y} = (Y_n, \dots, Y_{n-p+1})'$. We have used in the last step, that $E\mathbf{A}'B\mathbf{A} = E\mathbf{A}'(EB)\mathbf{A}$, for a random vector $\mathbf{A} \in \mathbb{R}^k$ which is independent of the random matrix $B \in \mathbb{R}^{k \times k}$. (Exercise 5.5.) Now if n is big, we know by Theorem 5.1.1 that

$$\sqrt{n} \Gamma_p^{1/2} (\boldsymbol{\phi} - \hat{\boldsymbol{\phi}}) \sim \mathbf{N}_p(0, \sigma^2 I_p),$$

indicating that

$$(\boldsymbol{\phi} - \hat{\boldsymbol{\phi}})' \Gamma_p (\boldsymbol{\phi} - \hat{\boldsymbol{\phi}}) \quad \text{and} \quad \frac{1}{n} \sigma^2 \chi_p^2,$$

should have approximately the same law. The expected value of the right hand side above is $\sigma^2 \frac{p}{n}$ and so this suggests

$$E(Y_{n+1} - \hat{\phi}_1 Y_n - \cdots - \hat{\phi}_p Y_{n-p+1})^2 \approx \sigma^2 \left(1 + \frac{p}{n}\right), \tag{5.9}$$

indicating that larger p will result in a larger mean squared prediction error. This is not surprising, since large p requires estimation of more parameters.

So far we were assuming that we know the true value of p. To get a rule for choosing p that compromises between a good fit and small prediction error we replace σ^2 in (5.9) by its estimate $\hat{\sigma}^2$. More precisely we use $n\hat{\sigma}^2/(n-p)$ (here $n/(n-p)$ corrects for the bias) and will chose p which minimizes

$$\texttt{FPE} = \hat{\sigma}^2 \frac{n+p}{n-p}.$$

The FPE stands for *final prediction error*. It was introduced by *Akaike* (1969) who later (1973) introduced the famous, more general *Akaike information criterion* (AIC). As its name indicates, this criterion is based on information theoretical arguments. It is given by

$$\mathtt{AIC}_{p,q}(\boldsymbol{\phi},\boldsymbol{\theta}) := -2\log L(\Gamma_n(\boldsymbol{\phi},\boldsymbol{\theta},\frac{1}{n}I(\boldsymbol{\phi},\boldsymbol{\theta}))|X_1,\ldots,X_n) + 2(p+q+1).$$

Instead of maximizing the (Gaussian) likelihood in $\boldsymbol{\phi}$ and $\boldsymbol{\theta}$ we minimize $\mathtt{AIC}_{p,q}(\boldsymbol{\phi},\boldsymbol{\theta})$ over $\boldsymbol{\phi}$, $\boldsymbol{\theta}$ and p and q. Roughly speaking we do maximum likelihood estimation adding a penalty term for too big p and q. Alternatively, if n is small, its bias corrected version

$$\mathtt{AICC}_{p,q}(\boldsymbol{\phi},\boldsymbol{\theta}) := -2\log L(\Gamma_n(\boldsymbol{\phi},\boldsymbol{\theta},\frac{1}{n}I(\boldsymbol{\phi},\boldsymbol{\theta}))|X_1,\ldots,X_n)+2n(p+q+1)/(n-p-q-2)$$

should be used.

Another famous criterion is the BIC which we shall not discuss here.

5.4 Asymptotics for AR(1) estimators

Setup. We assume that

$$X_t = \phi X_{t-1} + Z_t, \quad Z_t \sim \mathrm{IID}(0,\sigma^2).$$

The Yule-Walker estimator. As we saw in Example 5.1.1, the Yule-Walker estimator is given by $\hat{\phi} = \frac{\hat{\gamma}(1)}{\hat{\gamma}(0)}$. As we assume that the process $\{X_t\}$ is zero mean we will use a simplified version of $\hat{\gamma}(h)$:

$$\begin{aligned}
\hat{\phi} = \frac{\hat{\gamma}(1)}{\hat{\gamma}(0)} &= \frac{\frac{1}{n}\sum_{k=1}^{n-1} X_k X_{k+1}}{\frac{1}{n}\sum_{k=1}^{n} X_k^2} \\
&= \frac{\frac{1}{n}\phi\sum_{k=1}^{n} X_k^2 + \frac{1}{n}\sum_{k=1}^{n} X_k Z_{k+1}}{\frac{1}{n}\sum_{k=1}^{n} X_k^2} - \frac{\frac{1}{n}(\phi X_n^2 + X_n Z_{n+1})}{\frac{1}{n}\sum_{k=1}^{n} X_k^2} \\
&= \phi + \frac{\frac{1}{n}\sum_{k=1}^{n} X_k Z_{k+1}}{\frac{1}{n}\sum_{k=1}^{n} X_k^2} - \frac{\frac{1}{n}(\phi X_n^2 + X_n Z_{n+1})}{\frac{1}{n}\sum_{k=1}^{n} X_k^2} \\
&=: \phi + V_n - R_n.
\end{aligned}$$

Hence

$$\sqrt{n}(\hat{\phi}-\phi) = \sqrt{n}V_n - \sqrt{n}R_n.$$

We would like to prove Theorem 5.1.1 for the case $p=1$.

Proof. We start observing that by the ergodic theorem (Proposition 1.4.2) we have that

$$\frac{1}{n}\sum_{k=1}^{n} X_k^2 \xrightarrow{\text{a.s.}} EX_1^2 = \gamma(0). \tag{5.10}$$

In the following we make use of *Slutzky's theorem.*

Recall. If X_n and Y_n are two random sequences, where $X_n \xrightarrow{d} X$ and $Y_n \xrightarrow{P} c$ for a random variable X and a constant c, then $X_n + Y_n \xrightarrow{d} X + c$ and $X_n Y_n \xrightarrow{d} Xc$. Note also that $Y_n \xrightarrow{d} c$ is equivalent to $Y_n \xrightarrow{P} c$.

By the Markov inequality first and then by the Cauchy-Schwarz inequality we have for any $\varepsilon > 0$ that

$$\begin{aligned} P\left(\frac{1}{\sqrt{n}}|\phi X_n^2 + X_n Z_{n+1}| > \epsilon\right) &\le \frac{1}{\epsilon\sqrt{n}} E|\phi X_n^2 + X_n Z_{n+1}| \\ &\le \frac{1}{\epsilon\sqrt{n}}\left(E|X_n^2| + E|X_n Z_{n+1}|\right) \\ &\le \frac{1}{\epsilon\sqrt{n}}\left(\mathrm{Var}(X_1) + \sqrt{\mathrm{Var}(X_1)\mathrm{Var}(Z_1)}\right) \overset{n\to\infty}{\longrightarrow} 0. \end{aligned}$$

[We used that $|\phi| < 1$.] Combining this with (5.10) and Slutzky's theorem this implies that

$$\sqrt{n} R_n \xrightarrow{P} 0, \tag{5.11}$$

and it remains to study the weak convergence of $\sqrt{n}V_n$. Again in view of Slutzky's theorem and (5.10) it remains to prove that

$$\frac{1}{\sqrt{n}}\sum_{k=1}^{n} X_k Z_{k+1} \xrightarrow{d} N(0, \sigma^2\gamma(0)).$$

(Note that weak consistency of $\hat{\phi}$ follows then immediately.) To show this, we use the following method:

Martingale method. Let $\mathscr{F}_k = \sigma(Z_k, Z_{k-1}, \ldots)$ be the σ-algebra generated by $Z_k, Z_{k-1}, \ldots$. Then we have

$$E[X_k Z_{k+1} | \mathscr{F}_k] = E[Z_{k+1}] X_k = 0,$$

and it follows that $\sum_{k=1}^{n} X_k Z_{k+1}$ is a *martingale* with respect to the filtration $\mathscr{F}_k$. Now we can use a version of the central limit theorem for martingales (see e.g. p. 476 in Billingsley (1995) *Probability and Measure.* Wiley) to obtain

$$\frac{1}{\sqrt{n}}\sum_{k=1}^{n} X_k Z_{k+1} \xrightarrow{d} N(0, \tau^2),$$

where $\tau^2 = E[Z_{k+1}^2 X_k^2] = E[Z_{k+1}^2]E[X_k^2] = \sigma^2\gamma(0)$. □

The likelihood estimator. We have to first maximize

$$\ell_{\text{red}}(\phi) = \log\left(\frac{1}{n}I(\phi)\right) + \frac{1}{n}\sum_{k=1}^{n}\log r_{k-1}(\phi).$$

Here:

- $r_0(\phi) = \nu_0/\sigma^2 = \gamma(0)/\sigma^2 = \frac{1}{1-\phi^2}$; $\hat{X}_1 = 0$;
- $\hat{X}_k = \phi X_{k-1},\ \forall k \geq 2$;
- $r_k(\phi) = 1, \forall k \geq 1$.

Hence

$$\ell_{\text{red}}(\phi) = \log\left(\frac{1}{n}\sum_{k=2}^{n}(X_k - \phi X_{k-1})^2 + \frac{1}{n}(1-\phi^2)X_1\right) - \frac{1}{n}\log(1-\phi^2).$$

If ϕ is assumed to be in $[-1+\delta, 1-\delta]$, $\delta > 0$ we may argue that for large n

$$\ell_{\text{red}}(\phi) \approx \log\left(\frac{1}{n}\sum_{k=2}^{n}(X_k - \phi X_{k-1})^2\right),$$

which has the same minimum as $\frac{1}{n}\sum_{k=2}^{n}(X_k - \phi X_{k-1})^2$, namely

$$\hat{\phi} = \frac{\sum_{k=1}^{n-1} X_k X_{k+1}}{\sum_{k=1}^{n-1} X_k^2}.$$

The latter is asymptotically equivalent to the Yule-Walker estimator.

5.5 Exercises

Ex 5.1 Referring to estimation via the Gaussian likelihood approach: Suppose that the covariance matrix Γ_n depends on the parameter $\boldsymbol{\beta}$. Further assume that the n values $\nu_0, \ldots, \nu_{n-1}$ are unbounded functions of $\boldsymbol{\beta}$. Show that the function $\mathbf{X}_n^T \Gamma_n^{-1} \mathbf{X}_n$ can be made arbitrarily close to zero for a suitable choice of $\boldsymbol{\beta}$.

Ex 5.2 Show that if $n > p$ the likelihood of the observations $X_1, \ldots, X_n$ of the causal AR(p) process

$$X_t - \phi_1 X_{t-1} - \cdots - \phi_p X_{t-p} = Z_t, \quad \{Z_t\} \sim \text{WN}(0, \sigma^2),$$

is

$$L(\boldsymbol{\phi}, \sigma^2) = (2\pi\sigma^2)^{-n/2}|G_p|^{-1/2} \times \exp\left(-\frac{1}{2\sigma^2}\left[\mathbf{X}_p^T G_p^{-1}\mathbf{X}_p + \sum_{t=p+1}^{n}(X_t - \phi_1 X_{t-1} - \cdots - \phi_p X_{t-p})^2\right]\right),$$

where $\mathbf{X}_p = (X_1, \ldots, X_p)^T$ and $G_p = \sigma^{-2}E(\mathbf{X}_p\mathbf{X}_p^T)$.

Ex 5.3 Consider the time series $X_t = Z_t + \theta Z_{t-1}$ where $\{Z_t\} \sim \text{WN}(0, \sigma^2)$. Try to derive moment estimators for θ and σ^2.

Ex 5.4 Estimate the parameters of the recruitment series (Example 5.1.2) by the ML approach.

Ex 5.5 Consider a random vector $\mathbf{A} \in \mathbb{R}^k$ which is independent of the random matrix $B \in \mathbb{R}^{k\times k}$. Suppose that $E\|\mathbf{A}\|^2 < \infty$ and $E\|B\| < \infty$. Show by elementary arguments (not using conditional expectations) that $E\mathbf{A}'B\mathbf{A} = E\mathbf{A}'(EB)\mathbf{A}$.

Ex 5.6 Consider the MA(q) process $X_t = Z_t + \theta_1 Z_{t-1} + \cdots + \theta_q Z_{t-q}$, with $\{Z_t\} \sim \text{WN}(0, \sigma^2)$. Show that $\rho_X(1) \in (-1, 1)$.

Ex 5.7 Show that (5.4) is equivalent to $\hat{\Gamma}_{p+1} = \Theta\hat{\Gamma}_{p+1}\Theta' + \Omega$, where Θ and Ω have been defined in (5.6).

Chapter 6

Wold decomposition

Let $\{X_t\}$ be a stationary mean zero process with ACF γ_X. We saw in Chapter 3.5 that if $\gamma_X(h) = 0$ for $|h| > q$, then $\{X_t\}$ is necessarily an MA(q) process. It seems plausible, that if we let q tend to ∞, so that we just have $\gamma_X(h) \to 0$, we should be able to represent stationary processes as MA(∞) processes:

$$X_t = \sum_{k \geq 0} \psi_k Z_{t-k}, \tag{6.1}$$

where $\{Z_t\} \sim \text{WN}(0, \sigma^2)$. It is relatively clear, that $\gamma_X(h) \to 0$ is a necessary condition for this. This is because X_t with representation (6.1) can only be in L^2 (i.e. have a finite variance) if $\sum_{k \geq 0} \psi_k^2 < \infty$ (see Chapter 2). Since

$$\text{Cov}(X_t, X_{t+h}) = \sigma^2 \sum_{j \geq 0} \psi_j \psi_{j+|h|}, \tag{6.2}$$

it follows from Cauchy-Schwarz that

$$|\text{Cov}(X_t, X_{t+h})| \leq \sigma^2 \left(\sum_{j \geq 0} \psi_j^2 \right)^{1/2} \left(\sum_{j \geq 0} \psi_{j+|h|}^2 \right)^{1/2} \to 0 \quad (h \to \infty).$$

Here is an example where $\gamma_X(h) \to 0$ doesn't hold.

Example 6.0.1. *Let $Y_t = \rho Y_{t-1} + Z_t$ with $|\rho| < 1$, $\{Z_t\} \sim \text{WN}(0, 1)$ and $V_t = W$ for all $t \in \mathbb{Z}$, where $W \sim WN(0, \tau^2)$, $\tau^2 > 0$, and $W \perp \{Z_t\}$. Now assume*

$$X_t = Y_t + V_t = \underbrace{\sum_{j \geq 0} \rho^j Z_{t-j}}_{\text{MA}(\infty)} + V_t.$$

Clearly $\{X_t\}$ is stationary, but in addition to the MA(∞) part, we have the component V_t. Due to $Y_t \perp V_t$ we have

$$\begin{aligned}\mathrm{Cov}(X_t, X_{t+h}) &= \mathrm{Cov}(Y_t + V_t, Y_{t+h} + V_{t+h}) \\ &= \mathrm{Cov}(Y_t, Y_{t+h}) + \mathrm{Cov}(V_t, V_{t+h}) \\ &= \frac{\rho^{|h|}}{1-\rho^2} + \tau^2 \to \tau^2, \quad (|h| \to \infty).\end{aligned}$$

The problem in the previous example was caused by the sequence V_t, which is in some sense *perfectly predictable* in terms of $\{X_t\}$. Indeed,

$$V_{n+j} = \lim_{k\to\infty} \frac{1}{k} \sum_{\ell=0}^{k-1} X_{n-\ell} \in \mathcal{M}_n,$$

where $\mathcal{M}_n = \overline{\mathrm{sp}}\{X_t,\, t \le n\}$. This holds true for all $j \in \mathbb{Z}$. Thus if we observe the entire past of the process $\{X_t\}$ up to some time n, we can predict any V_{n+j} without error. Notice that the above argument holds for any n, thus

$$V_t \in \mathcal{M}_{-\infty} = \bigcap_{n\ge 1} \mathcal{M}_{-n} \quad \text{[this is a closed subspace!]} \tag{6.3}$$

We remark, that trivially in this example V_t is also perfectly predictable in terms of $\widetilde{\mathcal{M}}_n = \overline{\mathrm{sp}}\{V_t,\, t \le n\}$:

$$V_t \in \widetilde{\mathcal{M}}_{-\infty} = \bigcap_{n\ge 1} \widetilde{\mathcal{M}}_{-n}. \tag{6.4}$$

Definition 6.0.1 (Deterministic process)**.** A stationary process $\{V_t\}$, which satisfies (6.4), is called *deterministic*.

Lemma 6.0.1. *A stationary process $\{V_t\}$ is deterministic, if and only if for all n*

$$E[V_{n+1} - P_{\widetilde{\mathcal{M}}_n}(V_{n+1})]^2 = 0.$$

Proof. First, if $\{V_t\}$ is deterministic, then $V_{n+1} \in \widetilde{\mathcal{M}}_{-\infty}$, and thus $V_{n+1} \in \widetilde{\mathcal{M}}_n$. Consequently $P_{\widetilde{\mathcal{M}}_n}(V_{n+1}) = V_{n+1}$.

Second, $E[V_{n+1} - P_{\widetilde{\mathcal{M}}_n}(V_{n+1})]^2 = 0$ implies $V_{n+1} = P_{\widetilde{\mathcal{M}}_n}(V_{n+1})$ a.s. Since this holds for all n, we can replace the $V_n \in \widetilde{\mathcal{M}}_n$, by an element in $\widetilde{\mathcal{M}}_{n-1}$ and thus we have $V_{n+1} \in \widetilde{\mathcal{M}}_{n-1}$. By the same argument we see that $V_{n+1} \in \widetilde{\mathcal{M}}_{n-k}$, for all $k \ge 0$. □

The following famous theorem tells us, that exactly like in our Example 6.0.1, we can decompose any stationary process in the sum of an MA(∞) process $\{Y_t\}$ and a deterministic process $\{V_t\}$, such that $\{Y_t\} \perp \{V_t\}$. We use the notation introduced above.

Theorem 6.0.1 (Wold decomposition). *Let $\{X_t\}$ be a zero mean stationary process. Set $\sigma^2 = E|X_{n+1} - P_{\mathcal{M}_n}(X_{n+1})|^2$. If $\sigma^2 > 0$, then X_t can be written as*

$$X_t = \sum_{j\geq 0} \psi_j Z_{t-j} + V_t, \tag{6.5}$$

where

(i) $\psi_0 = 1$ *and* $\sum_{j\geq 0} \psi_j^2 < \infty$;

(ii) $\{Z_t\} \sim \mathrm{WN}(0, \sigma^2)$;

(iii) $Z_t \in \mathcal{M}_t \quad \forall t \in \mathbb{Z}$;

(iv) $EZ_tV_s = 0 \quad \forall s, t \in \mathbb{Z}$;

(v) $\{V_t\} \in \mathcal{M}_{-\infty} \cap \widetilde{\mathcal{M}}_{-\infty}$;

The sequences $\{\psi_j\}$, $\{Z_j\}$ and $\{V_j\}$ are uniquely determined by (6.5) and (i)–(v).

It is easy to see that our Example 6.0.1 satisfies all the claims of Theorem 6.0.1. For example, condition (v) follows from (6.3) and (6.4).

Before we give the proof, let us discuss the result in more detail. First consider the case of a stationary and causal ARMA(p, q) process. It has representation (6.5) with $V_t = 0$ and conditions (iv) and (v) are then trivial. We have also verified (i). If the process is assumed to be invertible, it can be easily seen that

$$X_{n+1} - P_{\mathcal{M}_n}(X_{n+1}) = Z_{n+1},$$

which shows (ii). An interesting condition is (iii): it says the the process is in fact invertible! That seems surprising. For example we know that the process

$$X_t = Z_t^* + 2Z_{t-1}^*, \quad Z_t^* \sim \mathrm{WN}(0, 1), \tag{6.6}$$

is not invertible, so how does this match? The answer is, that the white noise $\{Z_t^*\}$ in (6.6), does not have to coincide with the white noise $\{Z_t\}$ of the theorem. We will have that

$$Z_t^* + 2Z_{t-1}^* = \sum_{k\geq 0} \psi_k Z_{t-k},$$

when Z_t is defined like in the proof of Theorem 6.0.1. It is thus important to stress, that *invertibility is a relation between a process $\{X_t\}$ and a given white noise sequence $\{Z_t\}$.*

Proof. We claim that

$$Z_t = X_t - P_{\mathcal{M}_{t-1}}(X_t), \quad \psi_j = \frac{\langle X_t, Z_{t-j}\rangle}{\sigma^2} \quad \text{and} \quad V_t = X_t - \sum_{j\geq 0} \psi_j Z_{t-j}$$

satisfy the conditions of the theorem and show uniqueness of the three sequences. Let us collect the facts:

- It is not difficult to see that by the stationarity assumption the ψ_j don't depend on t.

- We have already defined a Z_t of this form in Chapter 3. Hence we can skip the proof of (ii).

- Furthermore, by the definition of Z_t it is clear that (iii) holds.

- Since $Z_t \sim \mathrm{WN}(0, \sigma^2)$, it follows that Z_t/σ is an orthonormal subset of L^2. Thus by Bessel's inequality we have

  $$\gamma_X(0) = \|X_t\|^2 \geq \sum_{j=0}^{m} \left| \left\langle X_t, \frac{Z_{t-j}}{\sigma} \right\rangle \right|^2 \quad \forall m \geq 0.$$

  Consequently $\sum_{j\geq 0}(\sigma\psi_j)^2 < \infty$. Furthermore

  $$\psi_0 = \frac{\langle X_t, X_t - P_{\mathcal{M}_{t-1}}(X_t)\rangle}{\|X_t - P_{\mathcal{M}_{t-1}}(X_t)\|^2} = \frac{\langle X_t - P_{\mathcal{M}_{t-1}}(X_t), X_t - P_{\mathcal{M}_{t-1}}(X_t)\rangle}{\|X_t - P_{\mathcal{M}_{t-1}}(X_t)\|^2} = 1.$$

  This shows (i).

- Next we show (iv). To this end we write

  $$\begin{aligned} V_t &= X_t - \sum_{j\geq 0} \psi_j Z_{t-j} \\ &= X_t - \sum_{j\geq 0} \frac{\langle X_t, Z_{t-j}\rangle}{\|Z_{t-j}\|^2} Z_{t-j} = X_t - P_{\overline{\mathrm{sp}}\{Z_j, j\leq t\}}(X_t). \end{aligned}$$

  By the Projection Theorem, we have that $Z_s \perp V_t$ for all $s \leq t$. On the other hand, when $s > t$, then $Z_s \perp \sum_{j\geq 0} \psi_j Z_{t-j}$, since $\{Z_t\}$ is white noise. The only thing that remains is to prove is that $Z_s = X_s - P_{\mathcal{M}_{s-1}}(X_s) \perp X_t$, $s > t$. This follows again from the projection theorem, since in this case $X_t \in \mathcal{M}_{s-1}$.

- We have seen in (iii) that Z_{t-j}, $j \geq 0$, is in $\mathcal{M}_t$. Thus $V_t = X_t - \sum_{j\geq 0} \psi_j Z_{t-j} \in \mathcal{M}_t$. Furthermore, $Z_t \perp \mathcal{M}_{t-1}$ and $\mathcal{M}_t = \mathcal{M}_{t-1} \oplus \overline{\text{sp}}\{Z_t\}$. Thus we have

$$V_t = P_{\mathcal{M}_t}(V_t) = P_{\mathcal{M}_{t-1}}(V_t) + \underbrace{P_{\overline{\text{sp}}\{Z_t\}}(V_t)}_{=0 \quad \text{by (iv)}}.$$

  Hence $V_t \in \mathcal{M}_{t-1}$. Continuing this argument shows that $V_t \in \mathcal{M}_n$ for all n, and thus $V_t \in \mathcal{M}_{-\infty}$.

  It remains to prove that $V_t \in \widetilde{\mathcal{M}}_{-\infty}$. We will show that $\widetilde{\mathcal{M}}_{-\infty} = \mathcal{M}_{-\infty}$. First, as we have just shown, $V_t \in \mathcal{M}_{-\infty}$. This implies that

$$\widetilde{\mathcal{M}}_t = \overline{\text{sp}}\{V_j,\, j \leq t\} \subset \mathcal{M}_{-\infty} \quad \forall t.$$

  And hence $\widetilde{\mathcal{M}}_{-\infty} \subset \mathcal{M}_{-\infty}$.

  On the other hand, assume that $Y \in \mathcal{M}_{-\infty}$. Then $Y \in \mathcal{M}_s$ for all s. Now by $X_t = V_t + \sum_{j\geq 1} \psi_j Z_{t-j}$ it follows that

$$\mathcal{M}_t = \overline{\text{sp}}\{V_j,\, j \leq t\} \oplus \overline{\text{sp}}\{Z_j,\, j \leq t\}. \tag{6.7}$$

  Hence

$$Y = P_{\overline{\text{sp}}\{V_j, j\leq t\}}(Y) + \underbrace{P_{\overline{\text{sp}}\{Z_j, j\leq t\}}(Y)}_{=0 \,(\text{if } Y \in \mathcal{M}_s \text{ then } \langle Y, Z_j\rangle = 0 \,\forall j > s)},$$

  and thus

$$Y \in \overline{\text{sp}}\{V_j,\, j \leq t\} = \widetilde{\mathcal{M}}_t \quad \forall t.$$

  Consequently $Y \in \widetilde{\mathcal{M}}_{-\infty}$, which implies that $\mathcal{M}_{-\infty} \subset \widetilde{\mathcal{M}}_{-\infty}$.

In summary, our collected facts show, that any stationary mean zero process, can be represented as in (6.5), such that (i)–(v) hold. To prove uniqueness, we assume that $\{Z_t\}$ and $\{V_t\}$ are some processes such that (6.5) and (i)–(v) hold. Due to (6.7) in combination with (ii) and (iv) we have that $Z_t \perp \mathcal{M}_{t-1}$.

$$\begin{aligned} X_t - P_{\mathcal{M}_{t-1}}(X_t) &= \\ &= \sum_{j\geq 0} \psi_j Z_{t-j} + V_t - \underbrace{P_{\mathcal{M}_{t-1}}(\psi_0 Z_t)}_{=0 \text{ by } Z_t \perp \mathcal{M}_{t-1}} - \underbrace{P_{\mathcal{M}_{t-1}}\left(\sum_{j\geq 1} \psi_j Z_{t-j}\right)}_{=\sum_{j\geq 1} \psi_j Z_{t-j} \text{ by (iii)}} - \underbrace{P_{\mathcal{M}_{t-1}}(V_t)}_{=V_t \text{ by (v)}} \\ &= \psi_0 Z_t = Z_t \text{ by (i)}. \end{aligned}$$

Again assuming (6.5) we have that

$$\langle X_t, Z_{t-i}\rangle = \psi_i \|Z_{t-i}\|^2,$$

showing that $\psi_i = \langle X_t, Z_{t-i}\rangle / \sigma^2$.

Finally it is obvious that under (6.5) we have $V_t = X_t - \sum_{j\geq 0} \psi_j Z_{t-j}$. □

The fact that any stationary process can be represented as the sum of a linear process and some deterministic component, shows in some way the strength of ARMA models, which form arguably the most important class of linear processes. This theorem doesn't say however, that fitting linear processes is the top of the time series world. The noise $\{Z_t\}$ is uncorrelated, which means that there is no linear dependence left. However, for many applications correlation is a way too rough measure of dependence, and in fact the $\{Z_t\}$ can have still a very complicated dependence structure, that one usually wishes to remove, and so it is worthwhile to study other time series models as well.

6.1 Exercises

Ex 6.1 Let $X_t = Z_t + \theta Z_{t-1}$ where $\{Z_t\} \sim \mathrm{WN}(0, \sigma^2)$. What can you say about the Wold decomposition with respect to the cases $|\theta| < 1$, $|\theta| = 1$ and $|\theta| > 1$?

Ex 6.2 Let $\{Y_t\}$ be an MA(1) process

$$Y_t = Z_t + 2Z_{t-1}, \quad \{Z_t\} \sim \mathrm{WN}(0, \sigma^2),$$

and define $X_t = A\cos(\omega t) + B\sin(\omega t) + Y_t$, where A and B are uncorrelated with mean zero and variance one which are uncorrelated with $\{Z_t\}$.

(a) Is this process deterministic?

(b) Determine the Wold decomposition of $\{X_t\}$.

Ex 6.3 Let $\{X_t\}$ be a stationary Markov chain with states ± 1 and transition probabilities $P(X_{t+1} = j | X_t = i) = p$ if $i = j$ and $1 - p$ if $i \neq j$. Find the explicit form of the Wold decomposition of this process.

Chapter 7

Statistical analysis in the frequency domain

Recall from harmonic analysis that any complex-valued function $x \in L^2 = L^2([-\pi, \pi])$ can be expanded in its Fourier series:

$$x = \sum_{k \in \mathbb{Z}} \langle x, e_k \rangle e_k,$$

where $\langle x, e_k \rangle = \int_{-\pi}^{\pi} x(t)\overline{e}_k(t)dt$ and $e_k(t) = \frac{1}{\sqrt{2\pi}} \exp(ikt) = \frac{1}{\sqrt{2\pi}} \left(\cos(kt) + i \sin(kt)\right)$. (As usual $\overline{z}$ denotes the comlpex conjugate of z.)

- The sinusoidal functions $e_k(t)$ form an orthonormal basis (ONB) of L^2.
- There is a 1:1 correspondence between x and $\{\langle x, e_k \rangle,\ k \in \mathbb{Z}\}$.
- If $|\langle x, e_k \rangle|$ (here and in the sequel $|z|$ is the modulus of a complex number z) is big, this indicates that sinusoid $e_k(t)$ is dominant in the expansion.

We wish to obtain a similar transformation for a time series sample $\mathbf{X} = (X_1, \ldots, X_n)'$.

7.1 The periodogram

Like in the above example, the theory will be most elegant when we assume complex data (of course the real data are contained as a special case). If $\mathbf{u}$ and $\mathbf{v}$ are vectors in $\mathbb{C}^n$ then we set $\langle \mathbf{u}, \mathbf{v} \rangle = \sum_{i=1}^{n} u_i \overline{v}_i$. To proceed similar as above we search an ONB of $\mathbb{C}^n$ that is based on sinusoids. We define

$$\mathbf{e}_k = \frac{1}{\sqrt{n}} (\exp(i\omega_k), \exp(i2\omega_k), \ldots, \exp(in\omega_k))', \tag{7.1}$$

where

$$\omega_k = \frac{2\pi k}{n}, \quad k \in \mathcal{F}_n := \left\{ -\left[\frac{n-1}{2}\right], \ldots, \left[\frac{n}{2}\right] \right\}.$$

We call the frequencies ω_k, $k \in \mathcal{F}_n$, the *fundamental frequencies* . Clearly $|\mathcal{F}_n| = n$ and

Lemma 7.1.1. *The vectors* $\{\mathbf{e}_k,\, k \in \mathcal{F}_n\}$ *form an ONB of* $\mathbb{C}^n$.

Proof. We have

$$\begin{aligned}
\langle \mathbf{e}_k, \mathbf{e}_j \rangle &= \frac{1}{n} \sum_{t=1}^{n} \exp(it(\omega_k - \omega_j)) \\
&= \frac{1}{n} \sum_{t=1}^{n} \exp\left(i \frac{2\pi t}{n} (k-j) \right) \\
&= \begin{cases} 1 & \text{if} \quad k = j; \\ 0 & \text{else,} \end{cases}
\end{aligned}$$

where we used $\sum_{t=1}^{n} \exp(im2\pi t/n) = 0$ for all integers $m \notin n\mathbb{Z}$. □

It follows that $\mathbf{X} = \sum_{k \in \mathcal{F}_n} \langle \mathbf{X}, \mathbf{e}_k \rangle \mathbf{e}_k$, hence $X_t = \frac{1}{\sqrt{n}} \sum_{k \in \mathcal{F}_n} \langle \mathbf{X}, \mathbf{e}_k \rangle \exp(it\omega_k)$, with

$$\langle \mathbf{X}, \mathbf{e}_k \rangle = \frac{1}{\sqrt{n}} \sum_{t=1}^{n} X_t \exp(-it\omega_k).$$

Definition 7.1.1. The sequence defined by the coefficients $d(\omega_k) := \langle \mathbf{X}, \mathbf{e}_k \rangle$, $k \in \mathcal{F}_n$, is called the *discrete Fourier transform (DFT)* of the sequence $X_1, \ldots, X_n$.

Figure 7.1 shows a realization $\mathbf{x}$ of a time series of length $n = 50$ and the approximations $\sum_{k=-m}^{m} \langle \mathbf{x}, \mathbf{e}_k \rangle \mathbf{e}_k$ for different orders m.

Remark 7.1.1. *The so-called* Fast Fourier Transformation (FFT) *is an algorithm which allows a very efficient computation of the DFT.*

Apparently, there is a 1-1-correspondence between $(d(\omega_k) \colon k \in \mathcal{F}_n)$ and $(X_k \colon 1 \leq k \leq n)$, so that any statistical analysis can in principle be performed on either sequence. When we are interested to see which underlying frequencies dominate the data generating process, it is an advantage to consider the DFT coefficients. To see this, notice that (exercise) for $k \notin n\mathbb{Z}$ we have

$$d(\omega_k) = \hat{\sigma} \sqrt{n} \hat{\rho}(\mathbf{X}, \mathbf{e}_k),$$

where $\hat{\rho}(\mathbf{X}, \mathbf{e}_k)$ denotes the empirical correlation between the sequences $\mathbf{X}$ and $\mathbf{e}_k$ and $\hat{\sigma}^2$ is the empirical variance of $\mathbf{X}$. In other words, $d(\omega_k)$ measures how much

the trajectory of our sample is correlated with sinusoidal functions at a given frequency.

Note that $d(\omega_k)$ is complex valued. To interpret a complex correlation is not so convenient and hence we look at the *squared modulus* of the $d(\omega_k)$. Recall that for a complex number z, the modulus is $|z| = \sqrt{z\bar{z}}$.

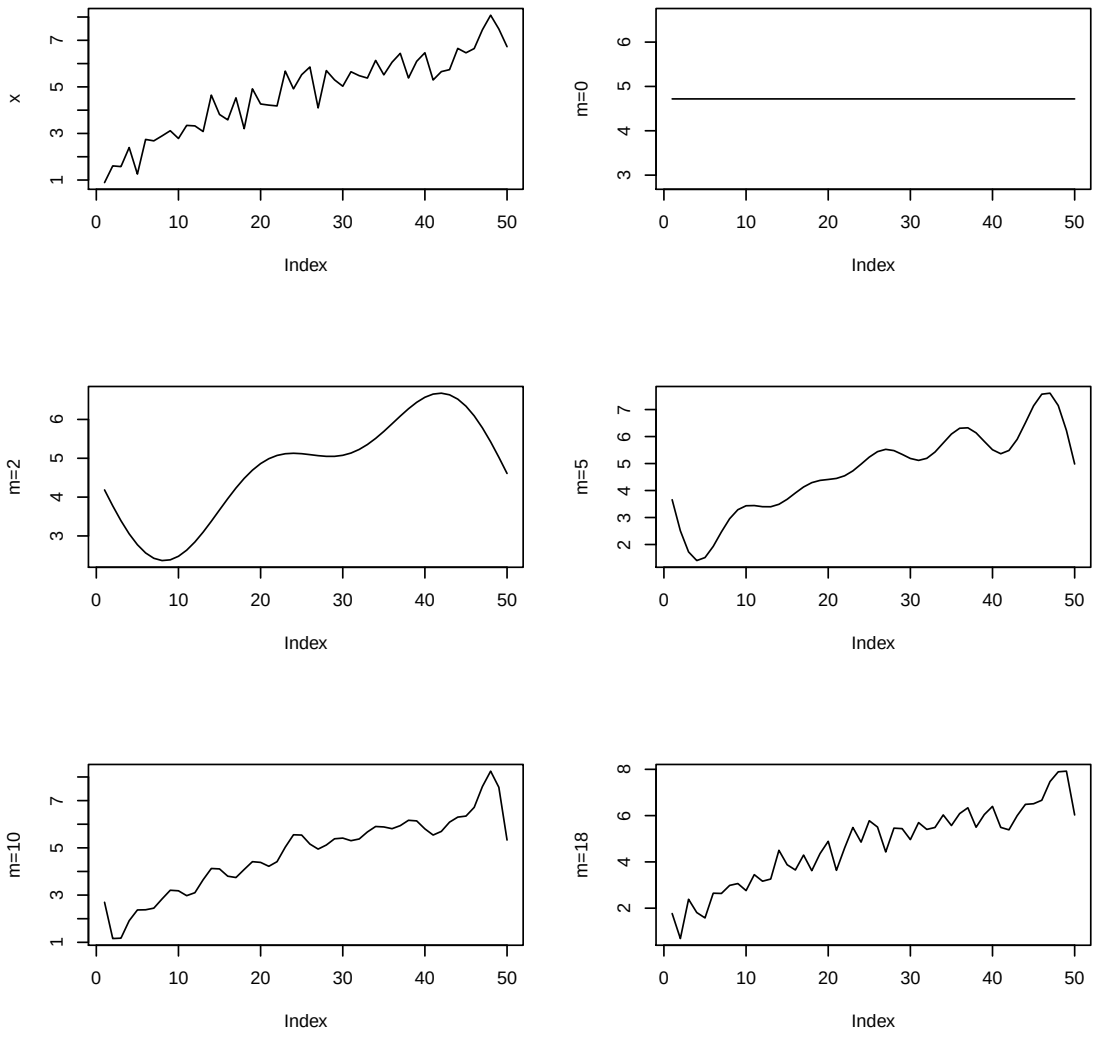


Figure 7.1: Vector $\mathbf{x}$ of length 50 (upper left) and Fourier approximations $\sum_{k=-m}^{m} \langle \mathbf{x}, \mathbf{e}_k \rangle \mathbf{e}_k$ for the orders $m = 0, 1, 5, 10, 18$. (Note that if $\mathbf{x}$ is real, then this is always real. Exercise.)

Definition 7.1.2. We call

$$I(\omega_k) := |\langle \mathbf{X}, \mathbf{e}_k \rangle|^2 = \frac{1}{n} \left| \sum_{t=1}^{n} X_t \exp(-it\omega_k) \right|^2$$

the *periodogram of* $\mathbf{X}$ *at frequency* $\omega_k = \frac{2\pi k}{n}$, $k \in \mathcal{F}_n$.

Example 7.1.1. Suppose that $X_t = A\cos(t\omega_k + \Theta)$ is a cosine wave, with period n/k, random amplitude A and random phase shift Θ. Then (exercise) $I(\omega_k) = I(\omega_{-k}) = A^2 n/4$, while $I(\omega_j) = 0$ for $j \neq \pm k$. In particular, a phase shift has no impact for the value of the periodogram.

Remarks.

- $\sum_{k \in \mathcal{F}_n} I(\omega_k) = \|\mathbf{X}\|^2$ (by Parseval's identity).
- $I(\omega_0) = \frac{1}{n} \left| \sum_{t=1}^{n} X_t \right|^2 = n|\bar{X}|^2$.
- If $\mathbf{X}$ is real, then

$$I(\omega_k) = \frac{1}{n} \left(\sum_{t=1}^{n} X_t \cos(t\omega_k) \right)^2 + \frac{1}{n} \left(\sum_{t=1}^{n} X_t \sin(t\omega_k) \right)^2.$$

  and hence $I(\omega_k) = I(-\omega_k)$. It follows that

$$\|\mathbf{X}\|^2 = I(0) + 2 \sum_{k=1}^{[(n-1)/2]} I(\omega_k) + \begin{cases} I(\omega_{[n/2]}) & \text{if } n \text{ is even,} \\ 0 & \text{if } n \text{ is odd.} \end{cases}$$

Example 7.1.2 (The Southern Oscillation Index (SOI))**.** Figure 7.2 shows the SOI, which is a monthly recorded environmental series for the period 1950–1987, 453 observations. The SOI measures changes in air pressure related to sea surface temperature in the Pacific, which warms due to the El Niño effect every 3-7 years. On top of the El Niño effect we expect an annual pattern. This expected behavior of the series is confirmed in Figure 7.3.

The following proposition relates the periodogram to the empirical ACF of $\mathbf{X}$.

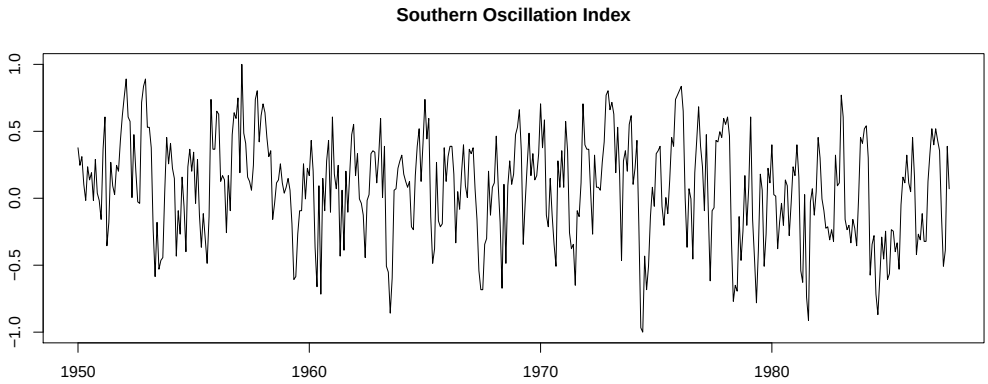


Figure 7.2: Monthly SOI series, 1950–1987.

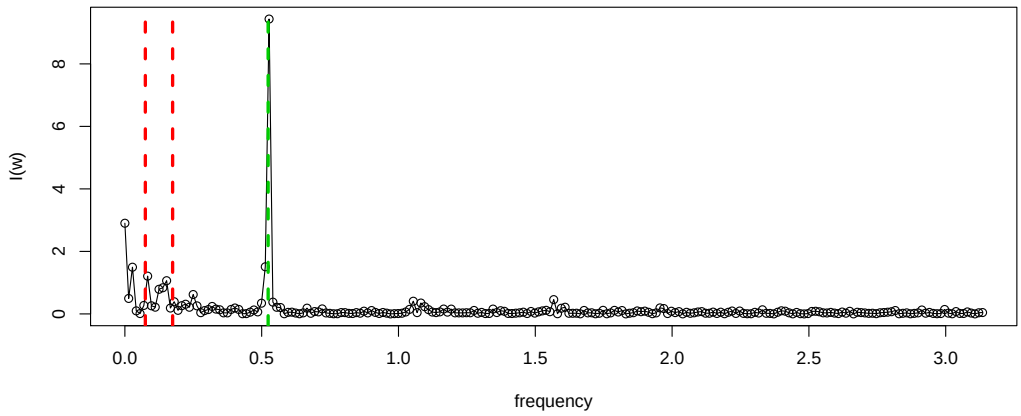


Figure 7.3: Periodogram of the SOI time series. The green vertical line was made at frequency $2\pi/12$ (annual periodicity) while red lines were drawn at $2\pi/36$ and $2\pi/84$ (3 and 7 years periodicity).

Proposition 7.1.1. *The periodogram of* $\mathbf{X}$ *can be alternatively calculated as*

$$I(\omega_k) = \sum_{|h|<n} \hat{\gamma}(h) \exp(-ih\omega_k) \quad \text{if } k \neq 0,$$
$$I(0) = n|\bar{X}|^2,$$

where for $h \geq 0$ *we set* $\hat{\gamma}(h) = \frac{1}{n}\sum_{t=1}^{n-h}(X_{t+h} - m)(\overline{X}_t - \overline{m})$, $m = \frac{1}{n}\sum_{t=1}^{n} X_t$ *and* $\hat{\gamma}(-h) = \overline{\hat{\gamma}(h)}$.

Proof. Noticing again that $\sum_{t=1}^{n}\exp(-it\omega_k) = 0$ if $\omega_k \neq 0$ we get

$$\begin{aligned} I(\omega_k) &= |\langle \mathbf{X}, \mathbf{e}_k\rangle|^2 = \langle \mathbf{X}, \mathbf{e}_k\rangle\overline{\langle \mathbf{X}, \mathbf{e}_k\rangle} \\ &= \frac{1}{n}\left(\sum_{t=1}^{n} X_t \exp(-it\omega_k)\right)\left(\sum_{s=1}^{n}\overline{X}_s \exp(is\omega_k)\right) \\ &= \frac{1}{n}\left(\sum_{t=1}^{n}(X_t - m)\exp(-it\omega_k)\right)\left(\sum_{s=1}^{n}(\overline{X}_s - \overline{m})\exp(is\omega_k)\right) \\ &= \frac{1}{n}\sum_{t=1}^{n}\sum_{s=1}^{n}(X_t - m)(\overline{X}_s - \overline{m})\exp(-i(t-s)\omega_k). \end{aligned}$$

Collecting the terms $t - s = h$ finishes the proof for $k \neq 0$.

The result for $k = 0$ is trivial. □

7.2 Testing for hidden periodicities

Let us assume the following *linear model*:

$$\mathbf{X} = S\boldsymbol{\alpha} + \mathbf{Z}, \quad \mathbf{Z} \sim \mathcal{N}_n(0, \sigma^2 I_n) \tag{7.2}$$

where S is the matrix

$$S = (\mathbf{1}, \mathbf{s}_1, \mathbf{s}_2) = \begin{pmatrix} 1 & \cos(\omega) & \sin(\omega) \\ 1 & \cos(2\omega) & \sin(2\omega) \\ \vdots & \vdots & \vdots \\ 1 & \cos(n\omega) & \sin(n\omega) \end{pmatrix}$$

In other words

$$X_t = \alpha_0 + \alpha_1\cos(\omega t) + \alpha_2\sin(\omega t) + Z_t.$$

For the moment the frequency ω is assumed to be fixed and known. E.g. for a conjectured weekly pattern we would choose $\omega = 2\pi/7$, as then every seventh observation has the same mean. We wish to test

$$H_0 : \alpha_1 = \alpha_2 = 0 \quad \text{v.s.} \quad H_1 : \text{not both, } \alpha_1 \text{ and } \alpha_2 \text{ are zero.}$$

Let $L_1 = \overline{\text{sp}}\{\mathbf{1}, \mathbf{s}_1, \mathbf{s}_2\}$ and $L_0 = \overline{\text{sp}}\{\mathbf{1}\}$. Then we test the linear hypothesis $\boldsymbol{\mu} = E\mathbf{X} \in L_0$ v.s. $\boldsymbol{\mu} = E\mathbf{X} \in L_1$. The classical test to use, is the F-test. Under H_0

$$F(\mathbf{X}) = \frac{\|P_{L_1}(\mathbf{X}) - P_{L_0}(\mathbf{X})\|^2/(k_1 - k_0)}{\|\mathbf{X} - P_{L_1}(\mathbf{X})\|^2/(n - k_1)} \sim F_{k_1-k_0, n-k_1},$$

where $k_0 = \dim(L_0)$ and $k_1 = \dim(L_1)$. If ω is of the form $\omega = \frac{2\pi k}{n}$, $k \in \mathcal{F}_n$, then the vectors $\mathbf{1}$, $\mathbf{s}_1$ and $\mathbf{s}_2$ are mutually orthogonal and $\|\mathbf{1}\|^2 = n$ and $\|\mathbf{s}_1\|^2 = \|\mathbf{s}_2\|^2 = n/2$. This implies that (exercise)

$$\begin{aligned}\|P_{L_1}(\mathbf{X}) - P_{L_0}(\mathbf{X})\|^2 &= \left\|P_{\overline{\text{sp}}\{\mathbf{s}_1,\mathbf{s}_2\}}(\mathbf{X})\right\|^2 \\ &= \frac{|\langle \mathbf{s}_1, \mathbf{X}\rangle|^2}{\|\mathbf{s}_1\|^2} + \frac{|\langle \mathbf{s}_2, \mathbf{X}\rangle|^2}{\|\mathbf{s}_2\|^2} = 2I(\omega_k).\end{aligned}$$

Furthermore we have

$$\begin{aligned}\|\mathbf{X} - P_{L_1}(\mathbf{X})\|^2 &= \langle \mathbf{X} - P_{L_1}(\mathbf{X}), \mathbf{X}\rangle \\ &= \|\mathbf{X}\|^2 - \langle P_{L_1}(\mathbf{X}), \mathbf{X}\rangle \\ &= \|\mathbf{X}\|^2 - \Bigg[\underbrace{\frac{|\langle \mathbf{1}, \mathbf{X}\rangle|^2}{\|\mathbf{1}\|^2}}_{=I(0)} + \underbrace{\frac{|\langle \mathbf{s}_1, \mathbf{X}\rangle|^2}{\|\mathbf{s}_1\|^2} + \frac{|\langle \mathbf{s}_2, \mathbf{X}\rangle|^2}{\|\mathbf{s}_2\|^2}}_{=2I(\omega_k)}\Bigg].\end{aligned}$$

Hence under H_0 we have

$$\frac{(n-3)I(\omega_k)}{\|\mathbf{X}\|^2 - I(0) - 2I(\omega_k)} \sim F_{2,n-3}.$$

Noting that

$$\frac{1}{n}\left(\|\mathbf{X}\|^2 - I(0)\right) = \frac{1}{n}\sum_{j=1}^n X_j^2 - m^2 = \cdots = \frac{1}{n}\sum_{j=1}^n (X_j - m)^2 =: S^2,$$

we have that

$$\frac{(1-3/n)I(\omega_k)}{S^2 - 2I(\omega_k)/n} \sim \frac{\chi_2^2/2}{\chi_{n-3}^2/(n-3)}.$$

For large n the left hand side approaches $\frac{I(\omega_k)}{S^2}$, while the right hand side converges to (exercise) $\chi_2^2/2$. So we have under the above assumptions that

$$\frac{I(\omega_k)}{S^2} \xrightarrow{d} \chi_2^2/2 \stackrel{d}{=} \text{Exp}(1),$$

where the last relation is left as an exercise as well.

In fact, if we standardize by σ^2 rather than by its estimator S^2 and if $\mathbf{X} \sim \mathcal{N}(\alpha_0\mathbf{1}, \sigma^2 I_n)$, then one can show that (exercise) $V_k := \frac{I(\omega_k)}{\sigma^2}$, $k = 1, \ldots, [(n-1)/2] =: q$ are i.i.d. exponential random variables. This allows us to develop a test for hidden periodicities with unknown frequencies, which is based on $\max_{1\le k\le q} V_k$ (respectively $\max_{1\le k\le q} \frac{I(\omega_k)}{S^2}$ if σ^2 is unknown). (Exercise.)

Using a trick, one can slightly modify the test, to obtain an explicit distribution which avoids estimation of σ^2. It is a well known result in probability that

$$Y_i := \frac{\sum_{k=1}^{i} V_k}{\sum_{k=1}^{q} V_k}, \quad i \in \{1, \ldots, q-1\},$$

are distributed as order statistics of a sample of $q-1$ independent $U(0,1)$ random variables. Thus, if we define $Y_0 = 0$ and $Y_q = 1$, then

$$M_q = \max_{1 \le i \le q} (Y_i - Y_{i-1}) = \max_{1 \le i \le q} \frac{I(\omega_i)}{\sum_{j=1}^{q} I(\omega_j)},$$

has the same distribution as the length of the largest of the intervals in $[0,1]$, which are obtained by "throwing" $q-1$ i.i.d. uniform random variables into $[0,1]$. Again, this distribution is know:

$$G_q(v) = P(M_q \le v) = \sum_{j=0}^{q} (-1)^j \binom{q}{j} (1 - jv)_+^{q-1}.$$

We reject H_0, if $M_q > G_q^{-1}(1-\alpha)$. This is known as *Fisher's test*.

Example. We generate in `R` 420 i.i.d. $N(0, 0.5)$ observations and superimpose a weekly and monthly pattern.

```
## preliminary definitions

t=seq(1:420)
n=length(t)

## generate time series, with monthly and weekly pattern

season=0.2*cos(2*pi*t/30)+0.2*sin(2*pi*t/7)
noise=rnorm(n,0,0.5)
x=season+noise

season=ts(season)
noise=ts(noise)
x=ts(x)
```

To compute the periodogram we use the built-in function `fft(x)`, which computes the unscaled Fourier transform of a sequence $x = (x_1, \ldots, x_n)'$ as

$$\text{fft}(x) = \left\{ \sum_{t=1}^{n} x_t \exp(-i(t-1)2\pi k/n),\ k = 0, \ldots, n-1 \right\}.$$

To obtain the periodogram we need to multiply by the factor $\frac{1}{\sqrt{n}}\exp(-2\pi ik/n)$ and take the squared modulus. Since the modulus of the factor $\frac{1}{\sqrt{n}}\exp(-2\pi ik/n) = 1/n$, we may proceed as follows:

```
## estimate periodogram

q=floor((n-1)/2)
period=abs(fft(x))^2/n
f=2*pi*0:q/n

## plots

par(mfrow=c(2,2))
plot(season,ylim=c(-1.5,1.5))
plot(noise,ylim=c(-1.5,1.5))
plot(x,ylim=c(-1.5,1.5))
plot(f,period[1:210],type="o",ylab="I(w)",xlab="frequency")
```

Figure 7.4 shows, that the seasonal component is quite small in comparison to the noise, and its presence in $\mathbf{X}$ is not evident in the graphs. However, from visual inspection of the periodogram we see very well those 2 dominating frequencies. If we want to apply a test, we can proceed in two ways: (1) if a weekly pattern was conjectured a priory, we reject H_0 if

$$2I(\omega_{60})/s^2 > \chi^2_{2;0.95} = 5.99.$$

If a monthly pattern was conjectured, we reject H_0 if

$$2I(\omega_{14})/s^2 > \chi^2_{2;0.95} = 5.99.$$

In our example we had $I(\omega_{14}) = 2.363$, $I(\omega_{60}) = 3.465$ and $s^2 = 0.263$. Thus in both cases, we clearly reject. If a priory it is not clear which frequency might be prevailing, we can use a test that is based on $\max_{1\leq k\leq 210} I(\omega_k)/s^2$ or we can use Fisher's test (exercise).

7.3 Asymptotics of the periodogram

Recall that the periodogram can be represented as

$$I(\omega_k) = \sum_{|h|<n} \hat{\gamma}(h)\exp(-ih\omega_k), \quad \text{for } k \in \mathcal{F}_n\backslash\{0\}. \tag{7.3}$$

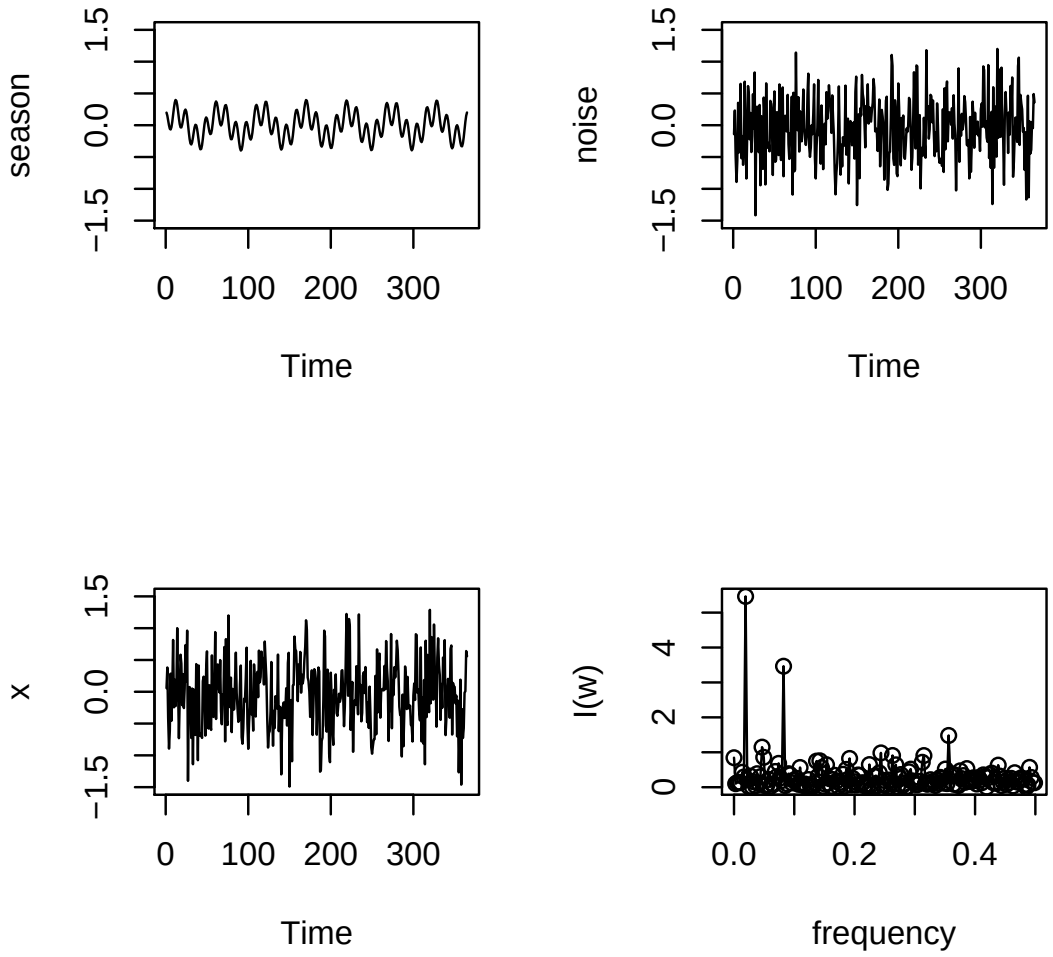


Figure 7.4: Seasonal component (upper left), noise (upper right) and resulting observations (lower left). The periodogram (lower right) shows two conspicuously large values $I(2\pi/7) \equiv I(\omega_{60})$ and $I(2\pi/30) \equiv I(\omega_{14})$.

This suggests, that the population analogue for the periodogram is the function

$$2\pi f(\omega) := \sum_{h\in\mathbb{Z}} \gamma_X(h) \exp(-ih\omega), \quad \omega \in (-\pi, \pi]\backslash\{0\}.$$

If $\sum_{h\in\mathbb{Z}} |\gamma_X(h)| < \infty$ (which is true for all ARMA models), then $f(\omega)$ exists and it is called *the spectral density of* $\{X_t\}$. It is easy to see, that in this case $f(\omega)$ is a

uniformly continuous function (exercise). The spectral density is the discrete-time Fourier transform (DTFT) of the sequence $\{\gamma_X(h),\, h \in \mathbb{Z}\}$. Notice that

$$\gamma_X(h) = \int_{-\pi}^{\pi} f(\omega) \exp(ih\omega) d\omega,$$

and consequently there is a 1:1 correspondence between the ACF γ_X and the spectral density $f(\omega)$. Hence, whatever analysis of a time series is based on studying its ACF (*analysis in the time domain*), can be equally done on its spectral density (*analysis in the frequency domain*). For certain purposes this can be very convenient.

Before we analyze the spectral density in more detail (see next section), let us first study the relation of the periodogram and the spectral density function. *For the rest of this section we assume to work with a real time series* $\{X_t\}$. If we see the periodogram $I(\omega_k) = I_n(\omega_k)$ as an estimator of $2\pi f(\omega_k)$, then we should check if the most basic properties of an estimator hold: for example (a) (asymptotic) unbiasedness and (b) consistency. Since the periodogram so far is only defined at the fundamental frequencies ω_k, we first extend its domain to $[-\pi, \pi]$. This can be done in several ways. One possibility is to just replace ω_k in (7.3) by ω. Following Brockwell and Davis (1991) we use here the following extension:

Definition 7.3.1 (Extension of the Periodogram)**.** For any $\omega \in [-\pi, \pi]$, the periodogram is defined as

$$I_n(\omega) = \begin{cases} I_n(\omega_k) & \text{if } \omega_k - \pi/n < \omega \leq \omega_k + \pi/n \text{ and } 0 \leq \omega \leq \pi, \\ I_n(-\omega) & \text{if } \omega \in [-\pi, 0). \end{cases}$$

Also, we can extend the definition of the periodogram to the entire line, by letting $I_n(\omega + 2k\pi) = I_n(\omega)$ for any $k \in \mathbb{Z}$. It holds that

Proposition 7.3.1. *If* $\{X_t\}$ *is stationary with mean* μ *and absolutely summable ACF* $\gamma_X(\cdot)$*, then*

$$\begin{aligned} &(i) \quad EI_n(0) - n\mu^2 \to 2\pi f(0); \\ &(ii) \quad EI_n(\omega) \to 2\pi f(\omega) \quad \textit{if} \quad \omega \neq 0. \end{aligned}$$

(If $\mu = 0$ *then* $EI_n(\omega)$ *converges uniformly to* $2\pi f(\omega)$ *on* $[-\pi, \pi]$*.)*

Proof. We first assume $\omega \neq 0$. For ω_k (a fundamental frequency) with $k \neq 0$

$$I(\omega_k) = \sum_{|h|<n} \tilde{\gamma}(h) \exp(-ih\omega_k),$$

with $\tilde{\gamma}(h) = \frac{1}{n}\sum_{t=1}^{n-h}(X_{t+h} - \mu)(\overline{X}_t - \overline{\mu})$ for $h \geq 0$ and $\tilde{\gamma}(-h) = \overline{\tilde{\gamma}(h)}$. Here $\mu = EX_1$. (Check the proof of Proposition 7.1.1.) Hence

$$EI_n(\omega_k) = \sum_{|h|<n} \left(1 - \frac{|h|}{n}\right) \gamma(h) \exp(-ih\omega_k).$$

Let ω^n be the element in $\mathcal{F}^n$ which is closest to ω. For n large enough, $\omega^n \neq 0$ and hence

$$\begin{aligned} |EI_n(\omega) - 2\pi f(\omega)| = &\sum_{|h|<n} \left(1 - \frac{|h|}{n}\right) \gamma(h) \big[\exp(-ih\omega^n) - \exp(-ih\omega)\big] \\ &- \sum_{|h|>n} \gamma(h) \exp(-ih\omega) \\ &+ \sum_{|h|<n} \frac{|h|}{n} \gamma(h) \exp(-ih\omega). \end{aligned}$$

Each series converges to zero. (Exercise.)

For (i) we note that

$$EI_n(0) = nE|\bar{X}|^2 = nE|\bar{X} - \mu|^2 + n\mu^2.$$

We saw in Chapter 1 that $nE|\bar{X} - \mu|^2 \to \sum_{h\in\mathbb{Z}} \gamma(h) = 2\pi f(0)$. □

While (a) is more or less true (if we exclude the frequency zero for non-centered time series) we will show in the following example that (b) cannot hold.

Example 7.3.1. Let us compute the spectral density of $f(\omega)$ of an i.i.d. $N(0,1)$ sequence $\{X_t\}$. Here $\gamma_X(0) = 1$ and $\gamma_X(h) = 0$ for all $h \neq 0$. Consequently $2\pi f(\omega) = 1$ for all $\omega \in [-\pi, \pi]$. On the other hand, we have already seen that $I(\omega_k)$ are i.i.d. Exp(1) in this case.

The example shows, that not even for an i.i.d. Gaussian sample the periodogram can be used to estimate the spectral density, as the limit is still random. In fact this is a special case of a more general phenomenon. One can show the following:

Theorem 7.3.1. *Let $\{X_t\}$ be the (real) linear process*

$$X_t = \sum_{j\in\mathbb{Z}} \psi_j Z_{t-j}, \quad \{Z_t\} \sim \mathrm{IID}(0, \sigma^2),$$

where $\sum_{j\in\mathbb{Z}} |\psi_j| < \infty$. Let $I_n(\lambda)$ denote the periodogram of $\mathbf{X}$ and let $f(\lambda)$ be the spectral density of $\{X_t\}$.

(i) If $f(\lambda) > 0$ for all $\lambda \in [-\pi, \pi]$ and if $0 < \lambda_1 < \lambda_2 < \cdots < \lambda_m < \pi$, then the random vector $(I_n(\lambda_1), \ldots, I_n(\lambda_m))'$ converges in distribution to a vector of independent and exponentially distributed random variables with mean $2\pi \times (f(\lambda_1), \ldots, f(\lambda_m))'$.

(ii) If $\sum_{j\in\mathbb{Z}} |\psi_j||j|^{1/2} < \infty$ and $EZ_1^4 < \infty$ then for the fundamental frequencies ω_j, ω_k we have constants C and D independent of j and k such that

$$|\mathrm{Cov}\,(I_n(\omega_j), I_n(\omega_k))\,| \le \begin{cases} C/n & \text{if } j \neq k, \\ D & \text{if } j = k. \end{cases}$$

Proof. Not here. It is not difficult but requires very lengthy and tedious calculations. □

How can we overcome the problem of non-consistency? One possibility is suggested by Theorem 7.3.1. If we pick a fixed ω we may hope that by asymptotic independence the "law of large number effect" implies for example that

$$\frac{1}{2m+1} \sum_{|j|\le m} I_n(\omega + 2\pi j/n)$$

converges to $2\pi f(\omega)$, provided $m = m_n$ is chosen properly.

More generally, we consider estimators of the form

$$\hat{f}(\omega) = (2\pi)^{-1} \sum_{j\in\mathbb{Z}} W_n(j) I_n(\omega + \omega_j),$$

where $\{W_n(\cdot)\}$ is a sequence of weight functions. We are thus *smoothing the periodogram*. Under some technical assumptions on the weight function and the sequence $\{X_t\}$ one can obtain the *uniform consistency* of such estimators, in the sense

$$\sup_{-\pi\le\omega\le\pi} |E\hat{f}(\omega) - f(\omega)| \to 0 \quad \text{and} \quad \sup_{-\pi\le\omega\le\pi} \mathrm{Var}\left(\hat{f}(\omega)\right) \to 0.$$

We will work under the following assumptions for $W_n(\cdot)$.

Assumption W.

$$\begin{aligned} &W_n(k) \ge 0, \quad \text{for all } k, \\ &W_n(k) = 0 \quad \text{if } |k| > m = m_n \text{ where } m \to \infty \text{ and } m/n \to 0, \\ &\sum_{k\in\mathbb{Z}} W_n(k) = 1 \quad \text{and} \quad \sum_{k\in\mathbb{Z}} W_n^2(k) \to 0 \quad \text{as } n \to \infty. \end{aligned}$$

Theorem 7.3.2 (Asymptotic unbiasedness of the smoothed periodogram)**.** *Let Assumption W and the conditions of Theorem 7.3.1 hold. Then we have that*

$$\sup_{\omega\in[-\pi,\pi]} |E\hat{f}(\omega) - f(\omega)| \to 0.$$

Proof. By Proposition 7.3.1 there is for any ε an $n_0 \geq 1$, such that for $n \geq n_0$

$$|EI_n(\omega) - 2\pi f(\omega)| \leq \varepsilon.$$

(Note that our assumptions imply that $EX_t = 0$.) As the function $f(\omega)$ is uniformly continuous, it follows that for small enough $h > 0$ we have

$$\sup_{\omega\in[-\pi,\pi]} \sup_{|s|\leq h} |f(\omega + s) - f(\omega)| \leq \varepsilon.$$

Assume now that n is large enough in order that $|\omega_j| \leq h$ for $j \leq m$. Then

$$\begin{aligned}
|E\hat{f}(\omega) - f(\omega)| &= \left| \frac{1}{2\pi} \sum_{j\in\mathbb{Z}} W_n(j) E\left(I_n(\omega+\omega_j) - 2\pi f(\omega)\right) \right| \\
&\leq \frac{1}{2\pi} \sum_{j=-m}^{m} W_n(j) \left| E\left(I_n(\omega+\omega_j) - 2\pi f(\omega)\right) \right| \\
&\leq \frac{1}{2\pi} \sum_{j=-m}^{m} W_n(j) \left| E\left(I_n(\omega+\omega_j) - 2\pi f(\omega+\omega_j)\right) \right| \\
&\quad + \frac{1}{2\pi} \sum_{j=-m}^{m} W_n(j) \left| \left(2\pi f(\omega+\omega_j) - 2\pi f(\omega)\right) \right| \\
&\leq \frac{1}{2\pi} \sum_{j=-m}^{m} W_n(j) \left(\varepsilon + 2\pi\varepsilon\right) < 2\varepsilon.
\end{aligned}$$

□

Theorem 7.3.3 (Consistency of the smoothed periodogram)**.** *Let Assumption W and the conditions of Theorem 7.3.1 (ii) hold. Then we have that* $\sup_{\omega\in[-\pi,\pi]} \mathrm{Var}(\hat{f}(\omega)) \to 0$.

Proof. By the definition of the extended periodogram $I(\omega) = I(\omega_k)$, where ω_k is a fundamental frequency. Then since $\omega_k + \omega_j$ is again a fundamental frequency we

can apply Theorem 7.3.1 to obtain

$$\begin{aligned}
\mathrm{Var}\left(2\pi\hat{f}(\omega)\right) &= \mathrm{Var}\left(\sum_{|j|<m} W_n(j) I_n(\omega_k+\omega_j)\right) \\
&= \sum_{|j|<m} W_n^2(j)\mathrm{Var}\left(I_n(\omega_k+\omega_j)\right) \\
&\quad + \sum_{|j|<m}\sum_{\substack{|\ell|<m \\ \ell\neq j}} W_n(j)W_n(\ell)\mathrm{Cov}\left(I_n(\omega_k+\omega_j), I_n(\omega_k+\omega_\ell)\right) \\
&\leq D\sum_{|j|<m} W_n^2(j) + \frac{C}{n}\sum_{|j|<m}\sum_{|\ell|<m} W_n(j)W_n(\ell) \\
&\leq D\sum_{|j|<m} W_n^2(j) + \frac{C}{n}\underbrace{\left(\sum_{j\in\mathbb{Z}} W_n(j)\right)^2}_{=1} \to 0.
\end{aligned}$$

□

Example 7.3.2. We generate 200 observations of an MA(1) process $X_t = Z_t - 0.8Z_{t-1}$ and calculate the periodgram as well as some smoothed versions using the simple weights $W_n(j) = \frac{1}{2m+1} I\{|j| \leq m\}$. This is shown in Figure 7.5. We see that the low frequencies don't play an important role in the Fourier representation of the observed sequence $\mathbf{x}$.

Lag window estimators. An alternative way to obtained a smoothed periodgram, is to apply so-called lag window estimators. The idea is to argue that

$$I_n(\omega_k) = \frac{1}{2\pi}\sum_{|h|<n} \hat{\gamma}(h)\exp(-ih\omega_k),$$

is not a good estimator of $f(\omega_k)$, because the estimators $\hat{\gamma}(h)$ are very bad when h is close to n. We should weight them and hence may set

$$\hat{f}_L(\omega) = \frac{1}{2\pi}\sum_{|h|<n} w(h/r)\hat{\gamma}(h)\exp(-ih\omega_k),$$

where w is an even, (piecewise) continuous function of x satisfying the conditions:

$$\begin{aligned}
&w(0) = 1, \\
&|w(x)| \leq 1, \quad \forall\, x, \\
&w(x) = 0 \quad \text{for } |x| > 1.
\end{aligned}$$

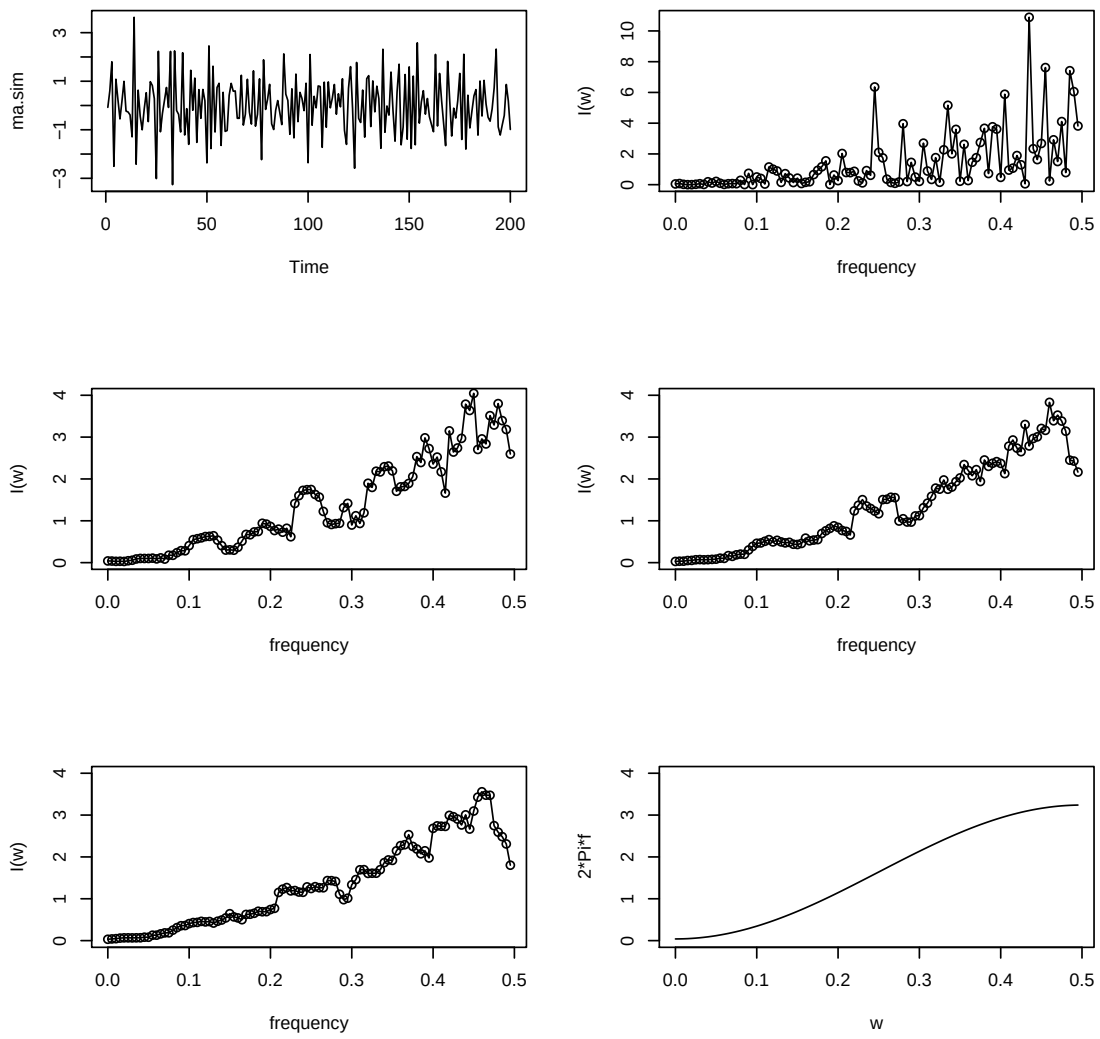


Figure 7.5: 200 observations of an MA(1) process $X_t = Z_t - 0.8Z_{t-1}$ (upper left) and raw periodgram (upper right). Smoothed periodograms with $W_n(j) = \frac{1}{2m+1} I\{|j| \leq m\}$, $m = 3, 5, 7$ and the spectral density multiplied by 2π (lower right).

Then $\hat{f}_L$ is called *lag window spectral density estimator*. Again one can show consistency and certain assumptions. One example of particular importance is the *Bartlett window*, defined by $w(x) = \begin{cases} 1 - |x| & \text{if } |x| \leq 1, \\ 0 & \text{else.} \end{cases}$

7.4 Herglotz's theorem

As before we will work with complex valued processes, therefore we will introduce some further simple generalizations.

Definition 7.4.1 (Stationary complex process)**.** A process $\{X_t\}$ with values in $\mathbb{C}$ is called stationary, if

$E|X_t|^2 < \infty$ for all t,

$EX_t = \mu$ doesn't depend on t,

$\gamma(h) := E[X_{t+h}\overline{X}_t] - EX_{t+h}E\overline{X}_t$ is independent of t.

The inner product $\langle X, Y\rangle = EX\overline{Y}$. The collection of complex variables X with $E|X|^2 < \infty$ forms a Hilbert space. Similar as in the real case one can show that if $\gamma(h)$ is an ACF of a stationary process, then $\gamma(0) \geq |\gamma(h)|$ and $\gamma(h) = \overline{\gamma(-h)}$ (the function is *Hermitian*). Furthermore, one can characterize ACFs like in the real case:

Theorem 7.4.1 (Characterization of ACF for stationary complex processes)**.** *A function $\gamma(h)$ defined on the integers is ACF of a stationary process, iff (i) it is Hermitian and (ii) it is non-negative definite.*[1]

We give now another characterization:

Theorem 7.4.2 (Herglotz's theorem)**.** *A function $\gamma(h)$ defined on the integers is ACF of a stationary process, iff it can be written as*

$$\gamma(h) = \int_{[-\pi,\pi]} \exp(ih\lambda)dF(\lambda), \tag{7.4}$$

where F is a non-decreasing, right-continuous function on $[-\pi, \pi]$ with $F(-\pi) = 0$ and $F(\pi) = \gamma(0)$. The function F is called the spectral-distribution function*. If $F(\lambda)$ is absolutely continuous with respect to the Lebesgue measure, then $F(\lambda) = \int_{-\pi}^{\lambda} f(\omega)d\omega$, and $f(\omega)$ is the spectral density function of the process. In this case*

$$\gamma(h) = \int_{-\pi}^{\pi} \exp(ih\lambda)f(\lambda)d\lambda.$$

Proof. First of all it is clear that if γ has representation (7.4), then it is Hermitian. Hence it remains to show that γ is non-negative definite, if and only if it has representation (7.4).

[1] This means here that $\sum_{i=1}^{n}\sum_{j=1}^{n} a_i\gamma(i-j)\overline{a}_j \geq 0$, for all $(a_1, \ldots, a_n)' \in \mathbb{C}^n$.

If it has representation (7.4), then

$$\sum_{r,s=1}^{n} a_r \gamma(r-s)\overline{a}_s = \sum_{r,s=1}^{n} a_r \overline{a}_s \int_{[-\pi,\pi]} \exp(i(r-s)\lambda) dF(\lambda)$$
$$= \int_{[-\pi,\pi]} \left| \sum_{r=1}^{n} a_r \exp(ir\lambda) \right|^2 dF(\lambda) \geq 0.$$

On the other hand, if γ is n.n.d. then

$$f_n(\lambda) = \frac{1}{2\pi n} \sum_{r,s=1}^{n} \exp(-ir\lambda)\gamma(r-s)\exp(is\lambda) \geq 0 \quad \forall\, n \geq 1,$$

and

$$F_n(\lambda) = \int_{-\pi}^{\lambda} f_n(\omega) d\omega, \quad -\pi \leq \lambda \leq \pi,$$

defines a distribution function[2] on $[-\pi, \pi]$. Note that $F_n(-\pi) = 0$ and $F_n(\pi) = \gamma(0)$ for all n. By collecting the terms $r - s = k$ in $f_n(\lambda)$ we obtain

$$\begin{aligned}
\int_{[-\pi,\pi]} \exp(ih\lambda) dF_n(\lambda) &= \int_{-\pi}^{\pi} \exp(ih\lambda) f_n(\lambda) d\lambda \\
&= \int_{-\pi}^{\pi} \exp(ih\lambda) \frac{1}{2\pi n} \sum_{|k|<n} (n - |k|)\gamma(k)\exp(-ik\lambda) d\lambda \\
&= \frac{1}{2\pi} \int_{-\pi}^{\pi} \exp(ih\lambda) \sum_{|k|<n} \left(1 - \frac{|k|}{n}\right) \gamma(k)\exp(-ik\lambda) d\lambda \\
&= \begin{cases} \left(1 - \frac{|h|}{n}\right)\gamma(h) & \text{if } |h| < n, \\ 0 \quad \text{else.} \end{cases}
\end{aligned}$$

If we let n go to infinity, then clearly the right hand side converges to $\gamma(h)$. For the left hand side we need to take a bit more care. We need to be sure that we can (i) take the limit inside the integral, and that (ii) the limit of F_n is again a distribution function.

We first show (ii). To this end we need Helly's theorem, which states that for every sequence $\{F_n\}$ of distribution functions, there exists a subsequence $\{F_{n_k}\}$

[2]Distribution function is here used in a slightly more general sense. As usual we set $F(x) = \mu\{(-\infty, x]\}$ but we allow for finite measures and not just for probabilty measures. Since the scale has no influence on almost all results shown for probability distribution functions, we can safely ignore this fact.

and a non-decreasing, right continuous function F, such that $\lim_k F_{n_k}(x) = F(x)$ at all continuity points of F. Since all the mass is concentrated on this interval, the corresponding sequence of measures is *tight* and thus $F(-\pi) = 0$ and $F(\pi) = \gamma(0)$. Consequently, by the Portmanteau theorem

$$\int_{[-\pi,\pi]} g(\lambda) dF_{n_k}(\lambda) \to \int_{[-\pi,\pi]} g(\lambda) dF(\lambda),$$

for every bounded and continuous function g. □

Remark. One can show that F is unique.

Corollary 7.4.1. *An absolutely summable, complex valued function γ defined on the integers is the ACF of a stationary process $\{X_t\}$, iff*

$$f(\lambda) = \frac{1}{2\pi} \sum_{h\in\mathbb{Z}} \gamma(h) \exp(-ih\lambda) \geq 0 \quad \forall\, \lambda \in [-\pi, \pi].$$

In this case f is the spectral density of $\{X_t\}$.

Proof. First assume that γ is an ACF of a stationary process. Since γ is n.n.d. we have as in the previous proof

$$\begin{aligned}
0 \leq f_n(\lambda) &= \frac{1}{2\pi n} \sum_{r,s=1}^{n} \exp(-ir\lambda)\gamma(r-s)\exp(is\lambda) \\
&= \frac{1}{2\pi} \sum_{|h|<n} \left(1 - \frac{|h|}{n}\right) \gamma(h) \exp(-ih\lambda) \\
&= \frac{1}{2\pi} \sum_{h\in\mathbb{Z}} \left(1 - \frac{|h|}{n}\right) \gamma(h) \exp(-ih\lambda) I\{|h| < n\} \\
&\to \frac{1}{2\pi} \sum_{h\in\mathbb{Z}} \gamma(h) \exp(-ih\lambda) = f(\lambda) \quad \forall \lambda \in [-\pi, \pi],
\end{aligned}$$

where we used the dominated convergence theorem (requires the absolute summability of the ACF!).

Next assume that $f(\lambda) \geq 0$ for all $\lambda \in [-\pi, \pi]$. Then $F(\lambda) = \int_{-\pi}^{\lambda} f(t)dt$ defines a distribution function and

$$\gamma(h) = \int_{-\pi}^{\pi} \exp(ih\lambda) f(\lambda) d\lambda = \int_{-\pi}^{\pi} \exp(ih\lambda) dF(\lambda).$$

The proof follows by application of Herglotz's theorem. □

Example 7.4.1. We have seen in Chapter 1 that it is quite tricky to check for which values of ρ the function

$$\kappa(h) = \begin{cases} 1 & \text{if } h = 0, \\ \rho & \text{if } |h| = 1, \\ 0 & \text{else,} \end{cases}$$

is an ACF of a stationary process. Now obviously $\kappa(h)$ is absolutely summable and it easy to see that

$$f(\lambda) = \frac{1}{2\pi}\left(1 + 2\rho\cos(\lambda)\right).$$

This is non-negative on $[-\pi, \pi]$, iff $|\rho| \leq 1/2$.

7.5 The spectral density of ARMA processes

The following lemma shows, how to compute the spectral density of linear processes.

Proposition 7.5.1. *Consider a real, zero-mean linear process*

$$X_t = \sum_{k\in\mathbb{Z}} \psi_k Z_{t-k} = \psi(B)Z_t, \quad Z_t \sim \mathrm{WN}(0, \sigma^2),$$

with $\sum_{k\in\mathbb{Z}} |\psi_k| < \infty$. *Then* $f_X(\lambda) = \frac{\sigma^2}{2\pi}|\psi(e^{-i\lambda})|^2, \quad -\pi \leq \lambda \leq \pi.$

Proof. We have learned that $\gamma_X(h) = \sigma^2 \sum_{k\in\mathbb{Z}} \psi_k\psi_{k+|h|}$. In the sequel we will repeatedly make use of the simple fact that $\sum_{k\in\mathbb{Z}} a_k = \sum_{k\in\mathbb{Z}} a_{k+m}$. It implies amoung others, that $\gamma_X(h) = \sigma^2 \sum_{k\in\mathbb{Z}} \psi_k\psi_{k+h}$. Hence, due to

$$\sum_{h\in\mathbb{Z}} |\gamma_X(h)| \leq \sigma^2 \sum_{h\in\mathbb{Z}}\sum_{k\in\mathbb{Z}} |\psi_k||\psi_{k+h}| = \sigma^2 \left(\sum_{k\in\mathbb{Z}} |\psi_k|\right)^2 < \infty,$$

the spectral density exists, and thus we have

$$\begin{aligned} f_X(\lambda) &= \frac{1}{2\pi}\sum_{h\in\mathbb{Z}} \gamma_X(h)\exp(-ih\lambda) \\ &= \frac{\sigma^2}{2\pi} \underbrace{\sum_{k\in\mathbb{Z}} \psi_k \exp(ik\lambda)}_{\psi(e^{i\lambda}) = \overline{\psi(e^{-i\lambda})}} \underbrace{\sum_{h\in\mathbb{Z}} \psi_{k+h}\exp(-i(h+k)\lambda)}_{\psi(e^{-i\lambda})}. \end{aligned}$$

□

Corollary 7.5.1. *Let $\{X_t\}$ be an ARMA(p,q) process given by*

$$\phi(B)X_t = \theta(B)Z_t, \quad \{Z_t\} \sim \mathrm{WN}(0,\sigma^2).$$

Then if $\phi(z)$ and $\theta(z)$ have no common roots, and if $\phi(z) \neq 0$ for all z with modulus 1, then $f_X(\lambda) = \frac{\sigma^2}{2\pi}\frac{|\theta(e^{-i\lambda})|^2}{|\phi(e^{-i\lambda})|^2}, \quad -\pi \leq \lambda \leq \pi.$

Proof. The proof is immediate from Proposition 7.5.1 and the characterization of stationary causal solutions for ARMA equations. (The result can be extended to non-causal processes.) □

Example 7.5.1. If $X_t = Z_t + \theta Z_{t-1}$, with $\{Z_t\} \sim \mathrm{WN}(0,\sigma^2)$ is an MA(1) process, then

$$f_X(\lambda) = \frac{\sigma^2}{2\pi}(1+\theta e^{-i\lambda})(1+\theta e^{i\lambda}) = \frac{\sigma^2}{2\pi}(1+2\theta\cos(\lambda)+\theta^2).$$

If $X_t = \phi X_{t-1} + Z_t$, with $\{Z_t\} \sim \mathrm{WN}(0,\sigma^2)$ is an AR(1) process, then

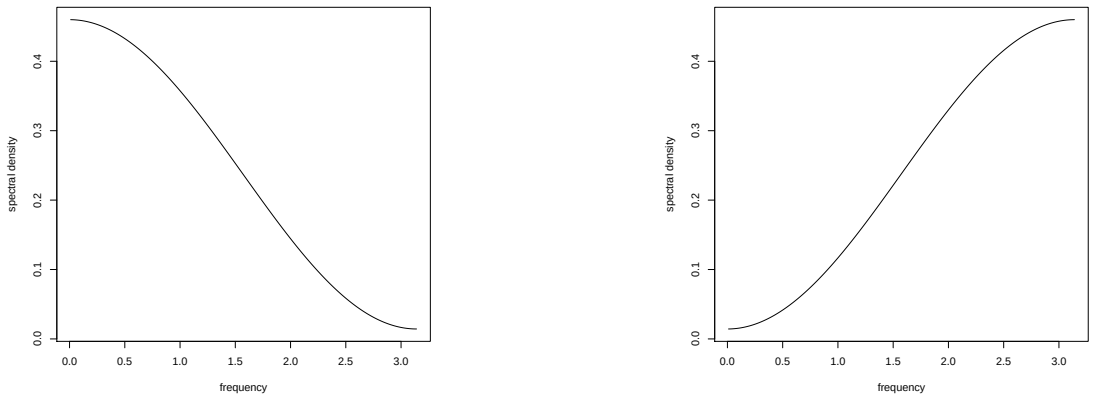


Figure 7.6: The spectral densities of the MA(1) processes $X_t = Z_t + \theta Z_{t-1}$, with $\{Z_t\} \sim \mathrm{WN}(0,\sigma^2)$, with $\theta = 0.7$ (left) and $\theta = -0.7$ (right).

$$f_X(\lambda) = \frac{\sigma^2}{2\pi}\frac{1}{(1-\phi e^{-i\lambda})(1-\phi e^{i\lambda})} = \frac{\sigma^2}{2\pi}(1-2\phi\cos(\lambda)+\phi^2)^{-1}.$$

To illustrate the usefulness of the frequency-domain approach we consider the problem of finding a causal and invertible representation of some ARMA(p,q) process.

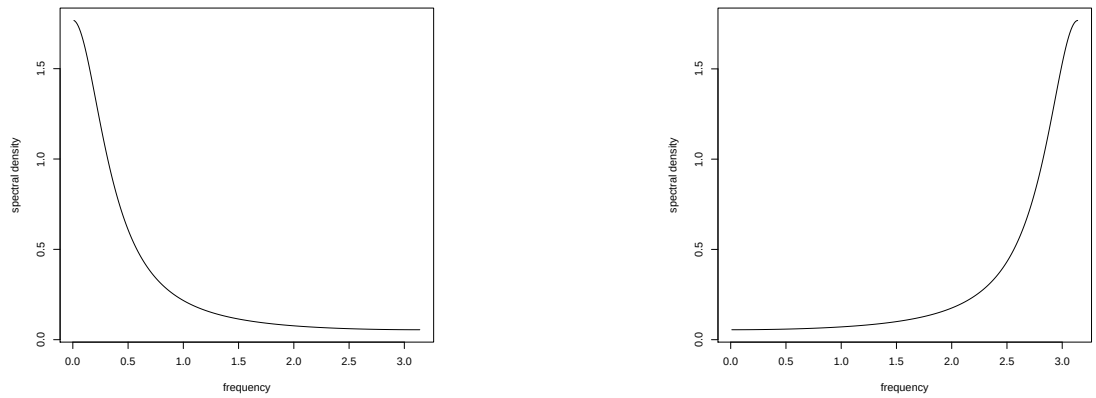


Figure 7.7: The spectral densities of the AR(1) processes $X_t = \phi X_{t-1} + Z_t$, with $\{Z_t\} \sim \mathrm{WN}(0, \sigma^2)$, with $\phi = 0.7$ (left) and $\phi = -0.7$ (right).

We assume that X_t satisfies $\phi(B)X_t = \theta(B)Z_t$ where $\phi(z)\theta(z) \neq 0$ for all z with modulus 1. We factorize now both, the MA and the AR polynomial and in each case we separate the roots inside and outside the unit circle:

$$\phi(z) = \prod_{k=1}^{p}(1 - a_k^{-1}z) \quad \text{and} \quad \theta(z) = \prod_{j=1}^{q}(1 - b_j^{-1}z),$$

where

$$\begin{aligned} &|a_k| > 1, \quad 1 \leq k \leq r \quad \text{and} \quad |a_k| < 1, \quad r+1 \leq k \leq p, \\ &|b_j| > 1, \quad 1 \leq j \leq s \quad \text{and} \quad |b_j| < 1, \quad s+1 \leq j \leq q. \end{aligned}$$

By Corollary 7.5.1 we have

$$f_X(\lambda) = \frac{\sigma^2}{2\pi} \frac{\prod_{j=1}^{q} |1 - b_j^{-1}e^{-i\lambda}|^2}{\prod_{k=1}^{p} |1 - a_k^{-1}e^{-i\lambda}|^2}.$$

Now

$$|1 - \overline{c}e^{-i\lambda}| = |1 - ce^{i\lambda}| = |c||c^{-1} - e^{i\lambda}| = |c| \underbrace{|e^{-i\lambda}|}_{=1} |1 - c^{-1}e^{-i\lambda}|,$$

showing that

$$f_X(\lambda) = \frac{\sigma^2}{2\pi} \frac{\prod_{j=1}^{s} |1 - b_j^{-1}e^{-i\lambda}|^2 \prod_{j=s+1}^{q} |1 - \overline{b_j}e^{-i\lambda}|^2 \prod_{k=r+1}^{p} |a_k|^2}{\prod_{k=1}^{r} |1 - a_k^{-1}e^{-i\lambda}|^2 \prod_{k=r+1}^{p} |1 - \overline{a_k}e^{-i\lambda}|^2 \prod_{j=s+1}^{p} |b_j|^2}.$$

Letting $\tau^2 = \sigma^2 \frac{\prod_{k=r+1}^{p} |a_k|^2}{\prod_{j=s+1}^{p} |b_j|^2}$, we obtain that $f_X(\lambda)$ is also the spectral density of the process

$$\tilde{\phi}(B) X_t^* = \tilde{\theta}(B) Z_t^*,$$

where $\{Z_t^*\} \sim \mathrm{WN}(0, \tau^2)$ and

$$\tilde{\phi}(z) = \prod_{k=1}^{r} (1 - a_k^{-1} z) \prod_{k=r+1}^{p} (1 - \overline{a_k} z),$$
$$\tilde{\theta}(z) = \prod_{j=1}^{s} (1 - b_j^{-1} z) \prod_{j=s+1}^{q} (1 - \overline{b_j} z).$$

Now both, the MA and the AR polynomial have all roots outside the unit-circle. Since both processes have the same spectral density, they have the same ACF.

Remark. Take the equation

$$\phi(B) X_t = \theta(B) Z_t,$$

and apply the operator $\tilde{\phi}(B)/\phi(B)$ on both sides. This gives

$$\tilde{\phi}(B) X_t = \tilde{\theta}(B) \Bigg[\underbrace{\left(\frac{\tilde{\phi}(B)}{\tilde{\theta}(B)} \frac{\theta(B)}{\phi(B)} Z_t \right)}_{=: Z_t^*} \Bigg].$$

Indeed, one can show that $\{Z_t^*\} \sim \mathrm{WN}(0, \tau^2)$ (exercise), and so we obtain a causal ARMA representation involving the original process $\{X_t\}$.

7.6 Exercises

Ex 2.1 Let $\mathbf{Z} \sim \mathcal{N}_n(0, \sigma^2 I_n)$ and $L_1, \ldots, L_r$ be orthogonal subspaces of $\mathbb{R}^n$, such that $\mathbb{R}^n = L_1 \oplus \cdots \oplus L_r$. Let $k_i = \dim(L_i)$. Then

(a) $P_{L_1}(\mathbf{Z}), \ldots, P_{L_r}(\mathbf{Z})$ are mutually independent.

(b) $\|P_{L_i}(\mathbf{Z})\|^2 \sim \sigma^2 \chi^2(k_i)$, $1 \le i \le r$.

Ex 7.2 Use the previous result to derive again the F-test in Section 7.2.

Ex 7.3 If $\mathbf{X} \sim \mathcal{N}(\mathbf{0}, \sigma^2 I_n)$, then $V_k := \frac{I(\omega_k)}{\sigma^2}$, $k = 1, \ldots, [(n-1)/2] =: q$ are i.i.d. Exp(1) variables.

Ex 7.4 Assume the setup of Section 7.2. Develop a test for hidden periodicities with unknown frequencies, which is based on the test statistic $\max_{1\le k\le q} V_k$ (respectively $\max_{1\le k\le q} \frac{I(\omega_k)}{S^2}$ if σ^2 is unknown). Here q is defined as in the previous exercise.

Ex 7.5 What is the result of Fisher's test, when applied to the example of Section 7.2? Compare it to your test of Ex 2.4.
Hint: In this numerical example we had $\|\mathbf{X}\|^2 = 110.38$.

Ex 7.6 Make a simulation study, which compares the power of your test in Ex 2.4 to Fisher's test.

Ex 7.7 Consider the AR(1) process $X_t = \phi X_{t-1} + Z_t$, where $\{Z_t\} \sim \mathrm{WN}(0, \sigma^2)$.

(a) Show that the spectral density of X is given by $\frac{\sigma^2}{2\pi}(1 - 2\phi\cos(\lambda) + \phi^2)^{-1}$.

(b) Use (a) to verify that for any $h \in \mathbb{Z}$ we have $\gamma_X(h) = \frac{\sigma^2\phi^{|h|}}{1-\phi^2}$.

Ex 7.8 Determine the autocovariance function of the process with spectral density $f(\lambda) = \frac{(\pi - |\lambda|)}{\pi^2}$.

Ex 7.9 Establish whether or not the following function is the ACF of a stationary process:

$$\gamma(h) = \begin{cases} 1 & \text{if } h = 0, \\ -1/2 & \text{if } h = \pm 2, \\ -1/4 & \text{if } h = \pm 3, \\ 0 & \text{otherwise.} \end{cases}$$

Ex 7.10 What is the spectral density of $\{Z_t\} \sim \mathrm{WN}(0, \sigma^2)$?

Ex 7.11 Suppose $\{X_t\}$ and $\{Y_t\}$ are uncorrelated stationary processes with spectral distribution $F_X(\lambda)$ and $F_Y(\lambda)$. Determine the spectral distribution function of $\{X_t + Y_t\}$.

Ex 7.12 Suppose that $\{X_t\}$ is a zero mean stationary process with spectral density $f_X(\lambda)$. What is the spectral density of $Y_t = \sum_{k\in Z} a_k X_{t-k}$ for some absolutely summable sequence (a_k)?

Ex 7.13 The process $X_t - 2X_{t-1} = Z_t + 4Z_{t-1}$, $\{Z_t\} \sim \mathrm{WN}(0, \sigma^2)$ is neither causal nor invertible. Find an ARMA(1,1) representation of the process X_t which is causal and invertible.

Ex 7.14 Show that the 3 series in the proof of Proposition 7.3.1 tend to zero.

Ex 7.15 Referring to the end of Section 7.5, show that $\{Z_t^*\} \sim \text{WN}(0, \tau^2)$.
Hint: Compute the spectral density of $\{Z_t^*\}$.